Logic

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Chapter 1

Atomes

1.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 1.1.1. ***Urelements*** *are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.*

In this book, we will denote individual urelements using lowercase letters u, v, w,...

Undefined 1.1.2. ***Type*** *parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).*

1.	u		
2.		v	
3.			w

Rule 1.1.1. ***fusion*** *it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)*

$$\begin{array}{r|l|l}
m. & u_0 & \\
 & \vdots & \\
m+n. & u_n & \\
o. & & \{u_0, u_1 ... u_n\}
\end{array}$$

Undefined 1.1.3. ***Set*** *we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).*

definition 1.1.1. ***Object*** *urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).*

Rule 1.1.2. ***fission*** *it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)*

$$\begin{array}{r|l|l}
o. & & \{a_0, a_1 ... a_n\} \\
m. & a_0 & \\
 & \vdots & \\
m+n. & a_n &
\end{array}$$

Example 1.1.1. $\{a, b, c\} / \therefore \{a, b\}$ *(prove if* $\{a, b, c\}$ *exists then* $\{a, c\}$ *exists)*

$$\begin{array}{r|l|ll}
1. & & \{a, b, c\} & \\
2. & a & & fi1 \\
3. & b & & fi1 \\
4. & c & & fi1 \\
5. & & \{a, c\} & fu2,3,4
\end{array}$$

Example 1.1.2. $\{a, b\} / \therefore \{b, a\}$

1.		$\{a, b\}$	
2.	a		*fi1*
3.	b		*fi1*
4.		$\{b, a\}$	*fu2,3*

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Chapter 2

propositional logic (zeroth-order logic)

2.1 What is proposition?

definition 2.1.1. ***predicate*** *"set()" and identity "=" and membership "$\in$" symbols are predicates. "set()" is unary and identity and membership are binary.*

capital letter "(P, Q, R)" showes **predicates**.
definition 2.1.2. ***Atomic proposition*** *we define atomic propositions like this.*

1. *if "a" is an obj then "set(a)" is an atomic proposition.*
2. *if "a, b" are obj then "$a = b$" is an atomic proposition.*
3. *if "a, b" are obj then "$a \in b$" is an atomic proposition.*

we denote atomic propositions with "p_i".
Undefined 2.1.1. ***logical operators*** *($\neg, \rightarrow, \wedge, \vee$) tool that used between atoms to make more complex propositions (all of the opratores are binary except "$\neg$" that is unary).*

you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.
definition 2.1.3. ***Propositions*** *these are propositions.*

1. *atomic propositions "(p_i)".*
2. *if "p" is a formula then "$\neg p$" is a formula too.*
3. *if "p, q" are two propositions then "$p \bigcirc q$" is a formula too.($\bigcirc$ can be "$\rightarrow, \wedge, \vee$").*

for example "$(p_0 \rightarrow p_1, \neg p_3)$"
we denote propositions with "$p, q, r, ...$".

2.2 Rules of propositions.

definition 2.2.1. ***proposition schema(variable proposition)*** *a proposition that is unknown for us and it works as variable.(we show them with "$\varphi, \psi, \theta, \ldots$")*
Undefined 2.2.1. ***Assumption for conditional proof(acp)*** *is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shoter than left one.(it can't fall out). and every hypothesis that opens must be closed.)*

1	$\vert$		
2	$\vert$	$\vert$	
3	$\vert$	$\vert$	$\vert$
4	$\vert$	$\vert$	$\vert$
5	$\vert$	$\vert$	
6	$\vert$		

Rule 2.2.1. ***rules of logical operations*** $(\to I, \to E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \perp I, \perp E)$ *this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)*

$\vert\ \varphi$ $\vert\ \vdots$ $\vert\ \psi$ $\therefore \varphi \to \psi \qquad \to I$	$\varphi \to \psi$ φ $\therefore \psi \qquad \to E$

Example 2.2.1. *for "p, q, r" we can prove these.*

1. $p \to q, q \to r / \therefore p \to r$
2. $q / \therefore p \to q$

1	$p \to q$		ass
2	$q \to r$		ass
3		$\vert\ p$	acp
4		$\vert\ q$	$\to E1,3$
5		$\vert\ r$	$\to E2,4$
6	$p \to r$		$\to I$

1	q		ass
2		$\vert\ p$	acp
3		$\vert\ q$	ass
4	$p \to q$		$\to I$

φ ψ $\therefore \varphi \wedge \psi \qquad \wedge I$	$\varphi \wedge \psi$ $\therefore \varphi \qquad \wedge E$	$\varphi \wedge \psi$ $\therefore \psi \qquad \wedge E$

Example 2.2.2. *for "p, q, r" we can prove these.*

1. $p \wedge q / \therefore q \wedge p$
2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$
3. $p \to q / \therefore (p \wedge r) \to q$
4. $(p \wedge q) \to r / \therefore p \to (q \to r)$
5. $p \to (q \to r) / \therefore (p \wedge r) \to q$

1	$p \wedge q$	ass
2	p	$\wedge E, 1$
3	q	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$

1	$(p \wedge q) \wedge r$	ass
2	$p \wedge q$	$\wedge E, 1$
3	r	$\wedge E, 1$
4	p	$\wedge E, 2$
5	q	$\wedge E, 2$
6	$q \wedge r$	$\wedge I, 3, 5$
7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \to q$	ass
2	$\mid\ p \wedge r$	acp
3	$\mid\ p$	$\wedge E, 2$
4	$\mid\ q$	$\to E, 1, 3$
5	$(p \wedge r) \to q$	$\to I$

1	$(p \wedge q) \to r$	ass
2	$\mid\ p$	acp
3	$\mid\ \mid\ q$	acp
4	$\mid\ \mid\ p \wedge q$	$\wedge I2, 3$
5	$\mid\ \mid\ r$	$\to E1, 4$
6	$\mid\ q \to r$	$\to I$
7	$p \to (q \to r)$	$\to I$

1	$p \to (q \to r)$	ass
2	$\mid\ p \wedge q$	acp
3	$\mid\ p$	$\wedge E, 2$
4	$\mid\ q \to r$	$\to E, 1, 3$
5	$\mid\ q$	$\wedge E, 2$
6	$\mid\ r$	$\to E, 4, 5$
7	$(p \wedge q) \to r$	$\to I$

$$\frac{\varphi}{\therefore \varphi \vee \psi} \quad \vee I \qquad\qquad \frac{\varphi}{\therefore \psi \vee \varphi} \quad \vee I$$

$\varphi \vee \psi$

$\mid\ \varphi$
$\mid\ \vdots$
$\mid\ \theta$

$\mid\ \psi$
$\mid\ \vdots$
$\mid\ \theta$

$\therefore \theta \qquad \vee E$

Example 2.2.3. *for "p, q, r" we can prove these.*

1. $p \vee q / \therefore q \vee p$
2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$
3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$

4. $(p \to q) \wedge (r \to s) / \therefore (p \vee r) \to (q \vee s)$

1	$p \vee q$	ass
2	$\mid p$	acp
3	$\mid p \vee q$	$\vee I, 2$
4	$\mid q$	acp
5	$\mid q \vee p$	$\vee I, 4$
6	$q \vee p$	$\vee E, 1$

1	$(p \vee q) \vee r$	ass
2	$\mid p \vee q$	acp
3	$\mid \mid p$	acp
4	$\mid \mid p \vee (q \vee r)$	$\vee I, 3$
5	$\mid \mid q$	acp
6	$\mid \mid q \vee r$	$\vee I, 5$
7	$\mid \mid p \vee (q \vee r)$	$\vee I, 6$
8	$\mid p \vee (q \vee r)$	$\vee E, 2$
9	$\mid r$	acp
10	$\mid q \vee r$	$\vee I, 9$
11	$\mid p \vee (q \vee r)$	$\vee I, 10$
12	$p \vee (q \vee r)$	$\vee E, 1$

1	$p \wedge (q \vee r)$	ass
2	$\mid p$	$\wedge E1$
3	$\mid q \vee r$	$\wedge E1$
4	$\mid \mid q$	acp
5	$\mid \mid p \wedge q$	$\wedge E2, 4$
6	$\mid \mid (p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	$\mid \mid r$	acp
8	$\mid \mid p \wedge r$	*I*$\wedge$ *2,7*
9	$\mid \mid (p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$\mid (p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \to q) \wedge (r \to s)$	ass
2	$(p \to q)$	$\wedge E1$
3	$(r \to s)$	$\wedge E1$
4	$\mid p \vee r$	acp
5	$\mid \mid p$	acp
6	$\mid \mid q$	$\to E2, 5$
7	$\mid \mid q \vee s$	$\vee I, 6$
8	$\mid \mid r$	acp
9	$\mid \mid s$	$\to E3, 8$
10	$\mid \mid q \vee s$	$\vee I9$
11	$\mid q \vee s$	$\vee E, 4$
12	$(p \vee r) \to (q \vee s)$	$\to I$

definition 2.2.2. $\perp$ *:* $\varphi \wedge \neg\varphi$

$\mid \varphi$ $\mid \vdots$ $\mid \perp$ $\therefore \neg\varphi \quad \neg I$	$\mid \neg\varphi$ $\mid \vdots$ $\mid \perp$ $\therefore \varphi \quad \neg E$

definition 2.2.3. ***schema of prove*** *you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form.you can't prove it directly. here we use a technique*

calls ***"schema of prove"***. *it let us prove for all situations at the same time.*

Theorem 2.2.1. (*schema*) $(\perp I, \perp E)$

$\dfrac{\varphi \qquad \neg\varphi}{\therefore \perp} \quad \perp I$	$\dfrac{\perp}{\therefore \varphi} \quad \perp E$

1	φ	ass
2	$\neg\varphi$	ass
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\perp$	$def.3$

1	$\perp$	ass
2	$\quad \mid \neg\varphi$	acp
3	$\quad \mid \vdots$	
4	$\quad \mid \perp$	$rp1$
5	φ	$\neg E$

Attention 2.2.1. *Intuitionists can skip the "$\neg E$" rule that is the "RAA" rule in old notation and use "$\perp E$" as a rule instead.*

Example 2.2.4. *for "p, q, r" we can prove these.*

1. $p \to q, \neg q / \therefore \neg p$

2. $p \to q / \therefore \neg p \vee q$

1	$p \to q$	ass
2	$\neg q$	ass
3	$\quad \mid p$	acp
4	$\quad \mid q$	$\to E, 1, 3$
5	$\quad \mid \perp$	$\perp I, 2, 4$
6	$\neg p$	$\neg I$

1	$p \to q$	ass
2	$\quad \mid \neg(\neg p \vee q)$	acp
3	$\quad \mid \quad \mid p$	acp
4	$\quad \mid \quad \mid q$	$\to E, 1, 3$
5	$\quad \mid \quad \mid \neg p \vee q$	$\vee I, 4$
6	$\quad \mid \quad \mid \perp$	$\perp I, 2, 5$
7	$\quad \mid \neg p$	$\neg I$
8	$\quad \mid \neg p \vee q$	$\vee I, 7$
9	$\quad \mid \perp$	$\perp I, 2, 8$
10	$\neg p \vee q$	$\neg I$

definition 2.2.4. $\varphi \leftrightarrow \psi : (\varphi \to \psi) \wedge (\psi \to \varphi)$

Theorem 2.2.2. (*schema*) $(\leftrightarrow I, \leftrightarrow E)$

$\begin{array}{ll} \left\lvert \begin{array}{l} \varphi \\ \vdots \\ \psi \end{array} \right. & \\ \left\lvert \begin{array}{l} \psi \\ \vdots \\ \varphi \end{array} \right. & \\ \therefore \varphi \leftrightarrow \psi & \leftrightarrow I \end{array}$	$\begin{array}{ll} \varphi \leftrightarrow \psi & \\ \underline{\varphi} & \\ \therefore \psi & \leftrightarrow E \end{array} \qquad \begin{array}{ll} \varphi \leftrightarrow \psi & \\ \underline{\psi} & \\ \therefore \varphi & \leftrightarrow E \end{array}$

1	$\mid\ \varphi$	acp
	$\mid\ \vdots$	
n	$\mid\ \psi$	
$n+1$	$\varphi \rightarrow \psi$	$\rightarrow, I$
$n+2$	$\mid\ \psi$	acp
	$\mid\ \vdots$	
m	$\mid\ \varphi$	
$m+1$	$\psi \rightarrow \varphi$	$\rightarrow, I$
$m+2$	$(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$	$\wedge I$
$m+3$	$\varphi \leftrightarrow \psi$	$def., m+3$

1	$\varphi \leftrightarrow \psi$	ass
2	φ	ass
3	$(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$	$def.1$
4	$\varphi \rightarrow \psi$	$\wedge E3$
5	ψ	$\rightarrow E1, 4$

Exercise 2.2.1. *for "p, q, r" prove.*

1. $p / \therefore p$
2. $/ \therefore p \vee \neg p$
3. $\neg p, p \vee q / \therefore q$
4. $\neg p / \therefore p \rightarrow q$
5. $p \rightarrow q \therefore / \therefore \neg q \rightarrow \neg p$
6. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$
7. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$ *(De morgan)*
8. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$ *(De morgan)*
9. $(\neg p \rightarrow q) / \therefore ((p \rightarrow q) \rightarrow q)$
10. $/ \therefore (p \wedge q \rightarrow r) \wedge (p \wedge s \rightarrow r) \rightarrow (p \wedge (q \vee s) \rightarrow r)$
11. $/ \therefore (p \wedge q \rightarrow r) \vee (p \wedge s \rightarrow r) \rightarrow (p \wedge (q \wedge s) \rightarrow r)$
12. $/ \therefore (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
13. $/ \therefore (p \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (p \rightarrow r))$

Attention 2.2.2. *we can rewrite* ***Example*** *and* ***Exercise*** *propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

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Chapter 3

first-order logic

3.1 what is formula?

obj are two types.we know them and we know what exactly are we talking about and unknowns are mystery.all the obj we have used until now were known ones we call them **"constants"**.
Undefined 3.1.1. ***variables*** *A variable is a symbol (usually a letter like $x, y, or z$) that represents an unknown obj.*
definition 3.1.1. ***Atomic formula*** *we define atomic formulas like this.*

1. *if "x" is an obj(constant or variable) then "$Set(x)$" is an atomic formula.*
2. *if "x, y" are obj(constant or variable) then "$x = y$" is an atomic formula.*
3. *if "x, y" are obj(constant or variable) then "$x \in y$" is an atomic formula.*

we denote atomic formulas with "A_i".
definition 3.1.2. ***well formed formula(WWF)*** *these are formulas.*

1. *atomic formulas (A_i).*
2. *if A is a formula then $\neg A$ is a formula too.*
3. *if A, B are two formulas then $A \bigcirc B$ is a formula too.($\bigcirc$ can be $\rightarrow, \wedge, \vee$).*

for example $(A_0 \rightarrow A_1, \neg A_3)$
we easily call WWF "formula" and also we denote WWF with $A, B, C, ...$
definition 3.1.3. ***open formula*** *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call **"closed formulas"** or **"sentence"**. Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.
definition 3.1.4. ***Substitution for obj*** *we define Substitution an obj with variable like this.*

1. *for variables if $x = y$ then $y[a/x] := a$ and if $x \neq y$ then $y[a/x] := y$.*
2. *for constants $c[a/x] := c$.*

definition 3.1.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like $x = y$ or $x \in y$ (x, y can be both constants and variables). we define it like this.*

$$Set(y)[a/x] := Set(y[a/x])$$

$$(x = y)[a/z] := (x[a/z] = y[a/z])$$

$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ can be $\to, \wedge, \vee$).

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x.we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 3.1.6. ***Open proposition*** *assume formula "A" with variables "$x_0, x_1, .., x_n$" in it and a set "U". we say formula "A"is a "open proposition in U" when for every obj "$a_0, a_1, .., a_n$" in "U"*

$$A([a_0/x_0][a_1/x_1]...[a_n/x_n])$$

is a proposition we call it open proposition and denote it with "$(P(x), Q(x, y), ..)$"

definition 3.1.7. ***Big "AND" and "OR"*** *assume a set like "$\{a_0, a_1, a_2, ..., a_n\}$" we define **big "AND"** like this.*

$$\bigwedge_{i=0}^{n} P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge ... \wedge P(a_n)$$

*and **big "OR"***

$$\bigvee_{i=0}^{n} P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee ... \vee P(a_n)$$

definition 3.1.8. ***Universe of discourse*** *we specify a set like "$\{a_0, a_1, a_2, ..., a_n\}$" as universe of a Logic world.*

Rule 3.1.1. ***use of open propositions*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$"*

$c \quad P(x)$

means :

$c_0 \quad P[a_0/x]$

$c_1 \quad P[a_1/x]$

$c_2 \quad P[a_2/x]$

$\quad \vdots$

$c_n \quad P[a_n/x]$

if Px was been assemed then like this

$$\begin{array}{l|l} c & P(x) \\ & \vdots \end{array}$$

means :

$$\begin{array}{l|l} c_0 & P[a_0/x] \\ & \vdots \\ \hline c_1 & P[a_1/x] \\ & \vdots \\ \hline c_2 & P[a_2/x] \\ & \vdots \\ \hline c_n & P[a_n/x] \\ & \vdots \end{array}$$

definition 3.1.9. ***predicate schema(variable predicate)*** *a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, ...$)*
definition 3.1.10. ***Universal quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\forall x P(x)$"*

$$\forall x \phi(x) : \bigwedge_{i=0}^{n} \phi[a_i/x]$$

definition 3.1.11. ***Existential quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\exists x Px$"*

$$\exists x \phi(x) : \bigvee_{i=0}^{n} \phi[a_i/x]$$

as you can see "$\forall x \phi(x)$" and "$\exists x \phi(x)$" are sentences.
definition 3.1.12. ***Free variables*** *variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sentece is not necessary a proposition)*

3.2 Rules of first order logic.

Theorem 3.2.1. (*schema*) ($\forall I, \forall E, \exists I, \exists E$) *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$"*

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x\phi(x)} \quad \forall I$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed.*
$\dfrac{\forall x\phi(x)}{\therefore \phi[a/x]} \quad \forall E$	*1. a is an obj in universe of discourse.(it can be constant or variable.)*
$\dfrac{\phi[a/x]}{\therefore \exists x\phi(x)} \quad \exists I$	*1. a is an obj in universe of discourse.(it can be conatant or variable.)*
$\exists x\phi(x)$ $\left\vert \begin{array}{l} \phi(x) \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \quad \exists E$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed in previous lines.(except ones that are closed)* *3. x is not free in θ*

$$
\begin{array}{lll}
1 & \phi(x) & ass \\
2 & \phi[a_0/x] & def.1 \\
 & \vdots & \\
n & \phi[a_n/x] & def.1 \\
n+1 & \bigwedge_{i=0}^{n} \phi[a_i/x] & \wedge I, 2, 3, ..., n+2 \\
n+2 & \forall x\phi(x) & def.n+3
\end{array}
$$

$$
\begin{array}{lll}
1 & \forall x\phi(x) & ass \\
2 & \bigwedge_{i=0}^{n} \phi[a_i/x] & def.1 \\
3 & \phi[a/x] & \wedge E2
\end{array}
$$

$$
\begin{array}{lll}
1 & \phi[a/x] & \\
2 & \bigvee_{i=0}^{n} \phi[a_i/x] & \vee I1 \\
3 & \exists x\phi(x) & def2
\end{array}
$$

$$
\begin{array}{llll}
1 & \exists x\phi(x) & & \\
2 & \bigvee_{i=0}^{n} \phi[a_i/x] & & \\
3 & \quad \vert & \phi[a_0/x] & acp \\
 & \quad \vert & \vdots & \\
n & \quad \vert & \theta & \\
n+1 & \quad \vert & \phi[a_1/x] & acp \\
 & \quad \vert & \vdots & \\
m & \quad \vert & \theta & \\
 & \vdots & & \\
l & \quad \vert & \phi[a_2/x] & acp \\
 & \quad \vert & \vdots & \\
r & \quad \vert & \theta & \\
r+1 & \theta & & \exists E, 3
\end{array}
$$

Rule 3.2.1. (*schema*) $(= I, = E)$

$\therefore a = a \qquad = I$	$\dfrac{a = b \quad \phi(a)}{\therefore \phi(b)} \quad = E \qquad \dfrac{a = b \quad \phi(a)}{\therefore \phi(a)} \quad = E$

definition 3.2.1. ***Uniqueness*** *we define "$\exists! x\phi(x)$"*

$$\exists! x\phi(x) : \exists x(\phi(x) \wedge \forall y(\phi(y) \to x = y))$$

Example 3.2.1. *for P,Q,R we can prove these.Proof of last ones Is Left As An Exercise To The Reader.*😀

1. $\forall x(P(x) \wedge Q(x)) \therefore / \therefore \forall xP(x) \wedge \forall xQ(x)$ *(D F $\forall\wedge$)(Distribution and Factorization of $\forall\wedge$)*
2. $\exists x(P(x) \vee Q(x)) \therefore / \therefore \exists xP(x) \vee \exists xQ(x)$ *(D F $\exists\vee$)*
3. $\forall xP(x) \vee \forall xQ(x) / \therefore \forall x(P(x) \vee Q(x))$ *(F $\forall\vee$)*
4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists xP(x) \wedge \exists xQ(x)$ *(D $\exists\wedge$)*
5. $\forall x(P(x) \to Q(x)) / \therefore \forall xP(x) \to \forall xQ(x)$ *(D $\forall \to$)*
6. $\exists xP(x) \to \exists xQ(x) / \therefore \exists x(P(x) \to Q(x))$ *(F $\exists \to$)*

1	$\forall x(P(x) \wedge Q(x))$	*ass*	1	$\forall xP(x) \wedge \forall xQ(x)$	*ass*
2	$P(x) \wedge Q(x)$	*acp*	2	$\forall xP(x)$	$\wedge E$
3	$P(x)$	$\wedge E$	3	$\forall xQ(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$	4	$P(x)$	$\forall E$
5	$\forall xP(x)$	$\forall I$	5	$Q(x)$	$\forall E$
6	$\forall xQ(x)$	$\forall I$	6	$P(x) \wedge Q(x)$	$\wedge I$
7	$\forall xP(x) \wedge \forall xQ(x)$	$\wedge I$	7	$\forall x(P(x) \wedge Q(x))$	$\forall I$

1	$\exists x(P(x) \vee Q(x))$	*ass*
2	$\quad P(x) \vee Q(x)$	*acp*
3	$\quad\quad P(x)$	*acp*
4	$\quad\quad \exists x P(x)$	$\exists I$
5	$\quad\quad \exists x P(x) \vee \exists x Q(x)$	$\vee I$
6	$\quad\quad Q(x)$	*acp*
7	$\quad\quad \exists x Q(x)$	$\exists I$
8	$\quad\quad \exists x P(x) \vee \exists x Q(x)$	$\vee I$
9	$\quad \exists x P(x) \vee \exists x Q(x)$	$\vee E$
10	$\exists x P(x) \vee \exists x Q(x)$	$\exists E$

1	$\exists x P(x) \vee \exists x Q(x)$	*ass*
2	$\quad \exists x P(x)$	*acp*
3	$\quad\quad P(x)$	*acp*
4	$\quad\quad P(x) \vee Q(x)$	$\vee I$
5	$\quad\quad \exists x(P(x) \vee Q(x))$	$\exists I$
6	$\quad \exists x(P(x) \vee Q(x))$	$\exists E$
7	$\quad \exists x Q(x)$	*acp*
8	$\quad\quad Q(x)$	*acp*
9	$\quad\quad P(x) \vee Q(x)$	$\vee I$
10	$\quad\quad \exists x(P(x) \vee Q(x))$	$\exists I$
11	$\quad \exists x(P(x) \vee Q(x))$	$\exists E$
12	$\exists x(P(x) \vee Q(x))$	$\vee E$

1	$\forall x P(x) \vee \forall x Q(x)$	*ass*
2	$\quad \forall x P(x)$	*acp*
3	$\quad P(x)$	$\forall E$
4	$\quad P(x) \vee Q(x)$	$\vee I$
5	$\quad \forall x(P(x) \vee Q(x))$	$\forall I$
6	$\quad \forall x Q(x)$	*acp*
7	$\quad Q(x)$	$\forall E$
8	$\quad P(x) \vee Q(x)$	$\vee I$
9	$\quad \forall x(P(x) \vee Q(x))$	$\forall I$
10	$\forall x(P(x) \vee Q(x))$	$\vee E$

1	$\exists x(P(x) \wedge Q(x))$	*ass*
2	$\quad P(x) \wedge Q(x)$	*acp*
3	$\quad P(x)$	$\wedge E$
4	$\quad Q(x)$	$\wedge E$
5	$\quad \exists x P(x)$	$\exists I$
6	$\quad \exists x Q(x)$	$\exists I$
7	$\exists x P(x)$	$\exists E$
8	$\exists x Q(x)$	$\exists E$
9	$\exists x P(x) \wedge \exists x Q(x)$	$\wedge I$

1	$\forall x(P(x) \to Q(x))$	*ass*
2	$\forall x P(x)$	*acp*
3	$P(x)$	$\forall E$
4	$P(x) \to Q(x)$	$\forall E$
5	$Q(x)$	$\to E$
6	$\forall x Q(x)$	$\forall I$
7	$\forall x P(x) \to \forall x Q(x)$	$\to I$

1	$\exists x P(x) \to \exists x Q(x)$	$\to I$
2	$\exists x P(x) \vee \neg(\exists x P(x))$	*ex.*
3	$\exists x P(x)$	*acp*
4	$\exists Q(x)$	$\to I$
5	$Q(x)$	*acp*
6	$P(x) \to Q(x)$	*ex.*
7	$\exists x(P(x) \to Q(x))$	$\exists I$
8	$\exists x(P(x) \to Q(x))$	$\exists E$
9	$\neg(\exists x P(x))$	*acp*
10	$\forall x \neg P(x)$	*De morgan*
11	$\neg P(x)$	$\forall E$
12	$P(x) \to Q(x)$	*ex.*
13	$\exists x(P(x) \to Q(x))$	$\exists I$
14	$\exists x(P(x) \to Q(x))$	$\vee E$

Exercise 3.2.1. *for P,Q,R prove.*

1. $\forall x P(x) \wedge Q \therefore / \therefore \forall x(P(x) \wedge Q)$
2. $\forall x P(x) \vee Q \therefore / \therefore \forall x(P(x) \vee Q)$
3. $\exists x P(x) \wedge Q \therefore / \therefore \exists x(P(x) \wedge Q)$
4. $\exists x P(x) \vee Q \therefore / \therefore \exists x(P(x) \vee Q)$
5. $Q \to \forall x P(x) \therefore / \therefore \forall x(Q \to P(x))$
6. $Q \to \exists x P(x) \therefore / \therefore \exists x(Q \to P(x))$
7. $\forall x P(x) \to Q \therefore / \therefore \exists x(P(x) \to Q)$
8. $\exists x P(x) \to Q \therefore / \therefore \forall x(P(x) \to Q)$
9. $\exists x \neg P(x) \therefore / \therefore \neg \forall x P(x)$ *(De morgan)*
10. $\forall x \neg P(x) \therefore / \therefore \neg \exists x P(x)$ *(De morgan)*
11. $\exists x\Big(P(x) \wedge \exists y Q(y) \to \forall z R(z)\Big) / \therefore \forall y \forall z\Big(\forall x P(x) \wedge Q(y) \to R(z)\Big)$
12. $\forall x \exists y\Big(P(x) \wedge Q(y) \to \forall z R(z)\Big) / \therefore \forall z\Big(\exists x P(x) \wedge \forall y Q(y) \to R(z)\Big)$
13. $\forall x P(x) \to \exists y Q(y) / \therefore \exists x \exists y\Big(P(x) \to Q(y)\Big)$
14. $\forall x \forall z \exists y \exists w\Big(P(y,z) \vee Q(z) \to R(x,w)\Big) / \therefore \Big(\exists z \forall y P(y,z) \vee \exists z Q(z)\Big) \to \forall x \exists w R(x,w)$
15. $\exists x P(x) \vee \exists x\Big(Q(x) \wedge R(x)\Big) / \therefore \exists x\Big(P(x) \vee Q(x)\Big) \wedge \exists x\Big(P(x) \vee R(x)\Big)$

16. $\exists x \exists z \forall y \Big(P(x,y) \to Q(z) \vee R(x,y) \Big) / \therefore \forall x \exists y P(x,y) \to \Big(\exists z Q(z) \vee \exists x \exists z R(x,y) \Big)$

17. $\forall x \exists y \Big(\exists z P(x,y,z) \wedge Q(x,y) \Big) \vee \forall x \exists y \exists z \Big(P(x,y,z) \wedge R(x,y) \Big) / \therefore \forall x \exists y \exists z \Big(P(x,y,z) \wedge (Q(x,y) \vee R(x,y)) \Big)$

Attention 3.2.1. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 4

The beginning of set theory

In language of set theory if we are talking about "$\forall X(P(X))$" it means "$\forall x(Set(x) \to P(x))$" and when we are talking about "$\exists X(P(X))$" it means "$\exists x(Set(x) \land P(x))$". similarly if we are talking about "$\forall x \in A \quad P(x)$" it means "$\forall x(x \in A \to P(x))$" and when we are talking about "$\exists x \in A \quad P(x)$" it means "$\exists x(x \in A \land P(x))$".

and we have a definitions.

$$Type_0(X) : \exists Y \quad X \in Y$$

$$Type_1(X) : \neg\exists Y \quad X \in Y$$

4.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 4.1.1. ***Existence of universe of set theory*** *There exists a set V_2 (we denote it as V_2.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition "ϕ" in "set theory" we mean proposition "ϕ" in "V_2").*

definition 4.1.1. ***Subset*** *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \to x \in B)$$

or i.e.

$$\forall x \in A \quad x \in B$$

Theorem 4.1.1. (*schema*) ($\subseteq I, \subseteq E$)

$x \in A$ $\vdots$ $x \in B$ $\therefore A \subseteq B \qquad \subseteq I$	$A \subseteq B$ $a \in A$ $\therefore a \in B \qquad \subseteq E$

1	$x \in A$	acp
	$\vdots$	
n	$x \in B$	
$n+1$	$x \in A \to x \in B$	$I \to, 1, n$
$n+2$	$\forall x(x \in A \to x \in B)$	$\forall I, n+1$
$n+3$	$A \subseteq B$	$def.n+2$

1	$A \subseteq B$	ass
2	$a \in A$	ass
3	$\forall x(x \in A \to x \in A)$	$def.1$
4	$a \in A \to a \in B$	$\forall E, 4$
5	$a \in B$	$E \to 3, 5$

Example 4.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	ass
2	$B \subseteq C$	ass
3	$x \in A$	acp
4	$x \in B$	$\subseteq E, 1, 3$
5	$x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 4.1.2. ***Extensionality****Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \to X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$$

Theorem 4.1.2. (*schema*) $(= I, = E)$

$x \in A$ $\vdots$ $x \in B$ $x \in B$ $\vdots$ $x \in A$ $\therefore A = B \qquad = I$	$A = B$ $a \in A$ $\therefore a \in B \qquad = E$ $A = B$ $a \in B$ $\therefore a \in A \qquad = E$

1	$\quad x \in A$	*acp*
	$\quad \vdots$	
n	$\quad x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$\quad x \in B$	*acp*
	$\quad \vdots$	
m	$\quad x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$	*Extensionality*
$m+3$	$(A \subseteq B \wedge B \subseteq A) \rightarrow A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \rightarrow, m+3, m+4$

Axiom 4.1.3. ***Weak comprehension*** (*schema*) *this axioms says for every open proposition like "$\phi(x, x_0, x_1, ..., x_n)$" there is a set like "$X$" such that every "$x$" with "$Type_0$" is a member of that set if and only if it holds "$\phi(x, x_0, x_1, ..., x_n)$".*

$$\forall x_0, x_1, ..., x_n \exists X \forall x(Type_0(x) \wedge \phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as "$\{x \mid \phi(x, x_0, x_1, ..., x_n)\}$". sometimes instead of "$\{x \mid x \in A \wedge \phi(x, x_0, x_1, ..., x_n)\}$" we write "$\{x \in A \mid \phi(x, x_0, x_1, ..., x_n)\}$".

Let's talk about some axioms in set theoty.These daxioms will come in handy later.

Axiom 4.1.4. ***Existence*** *there is a "Set" with no member i.e..*

$$\exists x \quad Type_0(x)$$

definition 4.1.2. ***Empty set*** *also known as the null set, is the unique mathematical set containing no elements at all. (uniqueness can be proven.)*

$$\varnothing := \{x \mid \bot\}$$

therefore.

$$\forall x(x \in \varnothing \leftrightarrow \bot \wedge Type_0(x))$$

therefore.

$$\forall x(x \notin \varnothing)$$

Example 4.1.2. $/ \therefore \forall X(\varnothing \subseteq X)$

1	$\quad \mid x \in \varnothing$	*acp*
2	$\quad \mid \forall x(x \notin \varnothing)$	*def*
3	$\quad \mid x \notin \varnothing$	$\forall E, 2$
4	$\quad \mid \bot$	$\bot I, 1, 3$
5	$\quad \mid x \in X$	$\bot E, 4$
6	$\varnothing \subseteq X$	$\subseteq I, 1, 5$
7	$\forall X(\varnothing \subseteq X)$	$\forall I, 6$

Lemma 4.1.1. $A = \varnothing \therefore / \therefore \forall y(y \notin A)$

1	$\forall y(y \notin A)$	*ass*
2	$\quad \mid y \in A$	*acp*
3	$\quad \mid y \notin A$	$\forall E, 1$
4	$\quad \mid \bot$	$\bot I, 2, 3$
5	$\quad \mid y \in \varnothing$	$\bot E, 4$
6	$\quad \mid y \in \varnothing$	*acp*
7	$\quad \mid \forall y(y \notin \varnothing)$	*def.*
8	$\quad \mid y \notin \varnothing$	$\forall E, 7$
9	$\quad \mid \bot$	$\bot I, 6, 8$
10	$\quad \mid y \in A$	$\bot E, 9$
11	$A = \varnothing$	$I =$

1	$A = \varnothing$	*ass*
2	$\forall y(y \notin \varnothing)$	*def.*
3	$\forall y(y \notin A)$	$= E, 1, 2$

Theorem 4.1.3. (*schema*) $(\varnothing \neq I, \varnothing \neq E, \varnothing = I, \varnothing = E)$

$\dfrac{a \in A}{\therefore A \neq \varnothing} \qquad \varnothing \neq I$	$\begin{array}{l} A \neq \varnothing \\ \quad \left\lvert \begin{array}{l} x \in A \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \qquad\qquad \varnothing \neq E \end{array}$
$\begin{array}{l} \quad \left\lvert \begin{array}{l} x \in A \\ \vdots \\ \bot \end{array} \right. \\ \therefore A = \varnothing \qquad\qquad \varnothing = I \end{array}$	$\dfrac{A = \varnothing}{\therefore a \notin A} \qquad \varnothing = E$

1	$a \in A$	*ass*
2	$\exists x(x \in A)$	$\exists I, 1$
3	$\neg(\forall x(x \notin A))$	*De Morgan* 2
4	$A \neq \varnothing$	*lemma* 4.4.1, 3

1	$\lvert\ x \in A$	*acp*
	$\lvert\ \vdots$	
n	$\lvert\ \bot$	
$n+1$	$x \notin A$	$\neg I$
$n+2$	$\forall x(x \notin A)$	$\forall I, n+1$
$n+3$	$A = \varnothing$	*lemma* 4.4.1, 1

1	$A = \varnothing$	*ass*
2	$\forall x(x \notin A)$	*lemma* 4.4.1, 1
3	$a \notin A$	$\forall E2$

Axiom 4.1.5. ***Existence***

$$Type_0(\varnothing)$$

definition 4.1.3. ***Pairing set***

$$\{a_0, a_1, a_2, ..., a_n\} := \{x \mid x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, ..., a_n\} \leftrightarrow (Type_0(x) \wedge x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n))$$

Theorem 4.1.4. (*schema*) $(\{\}I, \{\}E)$

$\dfrac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, ..., a_n\}} \quad \{\}I$	$a \in \{a_0, a_1, a_2, ..., a_n\}$ $\quad\vert\ a = a_0$ $\quad\vert\ \vdots$ $\quad\vert\ \theta$ $\vdots$ $\quad\vert\ a = a_n$ $\quad\vert\ \vdots$ $\quad\vert\ \theta$ $\therefore \theta \qquad \{\}E$

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.6. ***Pairing*** (*schema*)

$$Type_0(\{a_0, a_1, a_2, ..., a_n\})$$

definition 4.1.4. ***Union and Intersection***
Intersection:

$$A \cap B := \{x \mid x \in A \land x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \land x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \lor x \in B\}$$

therefore.

$$\forall x(x \in A \cup B \leftrightarrow (x \in A \lor x \in B))$$

Theorem 4.1.5. (*schema*) $(\cap I, \cap E, \cup I, \cup E)$

$a \in A$ $a \in B$ $\overline{\therefore a \in A \cap B}$ $\quad \cap I$	$\dfrac{a \in A \cap B}{\therefore a \in A}$ $\quad \cap E$ $\qquad \dfrac{a \in A \cap B}{\therefore a \in B}$ $\quad \cap E$
$\dfrac{a \in A}{\therefore a \in A \cup B}$ $\quad \cup I$ $\dfrac{a \in A}{\therefore a \in B \cup A}$ $\quad \cup I$	$a \in A \cup B$ $\quad \vert\ a \in A$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\quad \vert\ a \in B$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta$ $\quad \cup E$

Proof Is Left As An Exercise To The Reader.😀

somtimes all members of a set is also set we call it "set of sets".
definition 4.1.5. ***Big Union and Intersection*** *if "A" is "set of sets" and "$A \neq \varnothing$" we define Big Intersection:*

$$\bigcap A := \{x \mid \forall X(X \in A \to x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcap A \leftrightarrow (Type_0(x) \wedge \forall X(X \in A \to x \in X))\Big)$$

Big Union:

$$\bigcup A := \{x \mid \exists X(X \in A \wedge x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcup A \leftrightarrow (Type_0(x) \wedge \exists X(X \in A \wedge x \in X))\Big)$$

Theorem 4.1.6. (*schema*) ($\bigcap I, \bigcap E, \bigcup I, \bigcup E$)

$\begin{array}{\|l} X \in A \\ \vdots \\ a \in X \end{array}$ $\therefore a \in \bigcap A \qquad \bigcap I$	$a \in \bigcap A$ $\underline{B \in A}$ $\therefore a \in B \qquad \bigcap E$
$a \in B$ $\underline{B \in A}$ $\therefore a \in \bigcup A \qquad \bigcup I$	$a \in \bigcup A$ $\begin{array}{\|l} a \in X \\ X \in A \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad \bigcup E$

Proof Is Left As An Exercise To The Reader.😃

definition 4.1.6. ***Complement***

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow Type_0(x) \land x \notin A)$$

Theorem 4.1.7. *(schema)* $(()^c I, ()^c E)$

$\begin{array}{\|l} a \in A \\ \vdots \\ \bot \end{array}$ $\therefore a \notin A \qquad ()^c I$	$\begin{array}{\|l} a \notin A \\ \vdots \\ \bot \end{array}$ $\therefore a \in A \qquad ()^c E$

Proof Is Left As An Exercise To The Reader.😃

Observation 4.1.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow Type_0(x) \land \top)$$

therefore.

$$\forall x(x \in V)$$

definition 4.1.7. ***Subtraction***

$$A - B := \{x \mid x \in A \land x \notin B\}$$

therefore.

$$\forall x\Big(x \in A - B \leftrightarrow (Type_0(x) \land x \in A \land x \notin B)\Big)$$

Theorem 4.1.8. *(schema)* $(-I, -E)$

$a \in A$ $a \notin B$ $\therefore a \in A - B \qquad -I$	$\dfrac{a \in A - B}{\therefore a \in A} \quad -E \qquad \dfrac{a \in A - B}{\therefore a \notin B} \quad -E$

Proof Is Left As An Exercise To The Reader.😃

Lemma 4.1.2. $/ \therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.7. ***Seperation***

$$\forall X, Y \quad Type_0(X) \to Type_0(X \cap Y)$$

therefore.

$$Type_0(A) \to Type_0(A \cap B)$$

therefore.

$$Type_0(\bigcap A)$$

therefore.

$$Type_0(A) \to Type_0(A - B)$$

Axiom 4.1.8. ***union***

$$\forall X \quad Type_0(X) \to Type_0(\bigcup X)$$

therefore.

$$Type_0(A) \land Type_0(B) \to Type_0(A \cup B)$$

definition 4.1.8. ***Powerset***

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X \Big(X \in \mathcal{P}(A) \leftrightarrow (Type_0(X) \land X \subseteq A) \Big)$$

Theorem 4.1.9. *(schema)* $(\mathcal{P}I, \mathcal{P}E)$

$\dfrac{A \subseteq B}{\therefore A \in \mathcal{P}(B)} \quad \mathcal{P}I$	$\dfrac{A \in \mathcal{P}(B)}{\therefore A \subseteq B} \quad \mathcal{P}E$

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.9. ***Powerset***

$$\forall X \quad Type_0(X) \to Type_0(P(X))$$

from now on we don't get in to all details of proofs because it will be too long and hard to read. so we will take it easy from now and don't write all details like reasons of lines and all lines we left it to reader to undrestand what happened there.😃

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Chapter 5

Relations and Functions

5.1 Orderd pair and Cartesian product

definition 5.1.1. ***Ordered pair***

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

so $\forall x, y \quad Type_0((x, y))$

definition 5.1.2. ***Tuples****(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)*

$$(x_1, x_2, ..., x_n) := ((x_1, x_2, ..., x_{n-1}), x_n)$$

Lemma 5.1.1. $/ \therefore (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader.😀

definition 5.1.3. ***Cartesian product***

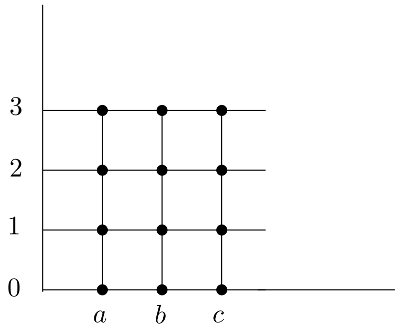
$$A \times B := \{u \mid \exists x, y\Big((x, y) = u \wedge x \in A \wedge y \in B\Big)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 5.1.1. *if* $A = \{a, b, c\}$ *and* $B = \{0, 1, 2, 3\}$ *then*

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



so $\forall X, Y \quad Type_0(X) \wedge Type_0(Y) \rightarrow Type_0(X \times Y)$
Theorem 5.1.1. (*schema*) $(\times I, \times E)$

$t \in A$ $s \in B$ $\therefore (t,s) \in A \times B \qquad \times I$	$\therefore t \in A \times B$ $\quad t = (x,y)$ $\quad x \in A$ $\quad y \in B$ $\quad \vdots$ $\quad \theta$ $\theta \qquad \times E$

Proof Is Left As An Exercise To The Reader.😃
Theorem 5.1.2. (*schema*)

$(t,s) \in A \times B$ $\therefore t \in A$	$(t,s) \in A \times B$ $\therefore s \in B$

Proof Is Left As An Exercise To The Reader.😃
Exercise 5.1.1. *for sets A, B, C, prove.*

1. $/ \therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$
2. $/ \therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $/ \therefore A \times (B - C) = (A \times B) - (A \times C)$

5.2 Relations

definition 5.2.1. ***Relation*** *we call a set like* ***R*** *a* ***Relation*** *when there are two sets like X and Y s.t. $R \subseteq X \times Y$.*

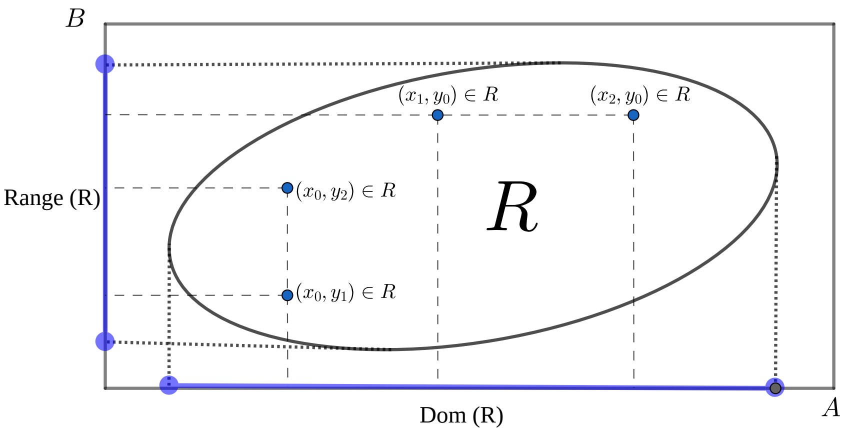
$$Rel(R) : \exists X, Y (R \subseteq X \times Y).$$

somtimes we use different notations in this book. you have to be smart and recognize it by yourself. sometimes instead of "$x \in R$" we denote it by "$R(x)$". and instead of "$(x_0, x_1, ..., x_n) \in R$" we denote it by "$R(x_0, x_1, ..., x_n)$". if tuple is binary, instead of "$(x,y) \in R$" we denote it by "xRy". or "$R(x,y)$" instead of "$R\big((x,y)\big)$"
so if we have a definition like "$R(x) : \phi(x)$" s.t. "$\phi(x)$" is a first order proposition. "weak comprehension" show us there is a straight relatioon between "propositions" and "sets" so you can see it as "$R := \{x \mid \phi(x)\}$". and somtimes we might show a formula in natural language for example below instead of "$Rel(X)$" we can say "X is relation". if it is confusing so forget it.😃
definition 5.2.2. ***Domain and Range(also called Image)***
Domain of a set "R" is all "x" that there exists a "y" s.t. "xRy"

$$Dom(R) := \{x \mid \exists y \quad xRy\}$$

Range(Image) of a set "R" is all "y" that there exists a "x" s.t. "xRy"

$$Range(R) := \{y \mid \exists x \quad xRy\}.$$



Theorem 5.2.1. $(schema)$ $(DomI, DomE)$

$\dfrac{tRs}{\therefore t \in Dom(R)}$ $\quad DomI$	$t \in Dom(R)$ $\quad tRy$ $\quad \vdots$ $\quad \theta$ $\therefore \theta \qquad DomE$

Proof Is Left As An Exercise To The Reader.😀

Theorem 5.2.2. $(schema)$ $(RangeI, RangeE)$

$\dfrac{tRs}{\therefore s \in Range(R)}$ $\quad RangeI$	$s \in Range(R)$ $\quad xRs$ $\quad \vdots$ $\quad \theta$ $\therefore \theta \qquad RangeE$

Proof Is Left As An Exercise To The Reader.😀

definition 5.2.3. ***Inverse of a Relation***

$$R^{-1} = \{u \mid \exists x, y\Big(u = (x, y) \wedge yRx\Big)\}$$

or i.e.

$$R^{-1} = \{(x, y) \mid yRx\}$$

therefore.

$$\forall x, y\Big(xR^{-1}y \leftrightarrow yRx\Big)$$

Theorem 5.2.3. (*schema*) $(()^{-1}I, ()^{-1}E)$

tRs $\therefore sR^{-1}t \qquad ()^{-1}I$	$t \in R^{-1}$ $\quad \vert\ t = (x, y)$ $\quad \vert\ yR^{-1}x$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \qquad ()^{-1}E$

Proof Is Left As An Exercise To The Reader.😃

definition 5.2.4. ***Combination of a Relations***

$$S \circ R = \{u \mid \exists x, y\Big(u = (x, y) \wedge \exists z\big(xRz \wedge zSy\big)\Big)\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z\Big(xRz \wedge zSy\Big)\}$$

therefore.

$$\forall x, y\Big((x, y) \in S \circ R \leftrightarrow \exists z\big(xRz \wedge zSy\big)\Big)$$

Theorem 5.2.4. (*schema*) $(\circ I, \circ E)$

tSs sRr $\therefore sR \circ Sr \qquad \circ I$	$t \in R \circ S$ $\quad \vert\ t = (x, y)$ $\quad \vert\ xSz$ $\quad \vert\ zRy$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \qquad \circ E$

Proof Is Left As An Exercise To The Reader.😃

definition 5.2.5. ***Image***

$$R[A] := \{y \mid \exists x \in A \quad xRy\}$$

you can olso see by definition of "inverse of a Relation" and "image" this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad xRy\}$$

*we cal this "**Pre-image**"*
Theorem 5.2.5. (*schema*) ($R[]I, R[]E, R^{-1}[]I, R^{-1}[]E$)

$t \in A$ $\underline{tRs}$ $\therefore s \in R[A] \qquad R[]I$	$s \in R[A]$ $\begin{array}{\|l} x \in A \\ xRs \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad R[]E$
$s \in B$ $\underline{tRs}$ $\therefore t \in R^{-1}[B] \qquad R^{-1}[]I$	$t \in R^{-1}[B]$ $\begin{array}{\|l} y \in B \\ tRy \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad R^{-1}[]E$

Proof Is Left As An Exercise To The Reader.😀

5.3 Functions

definition 5.3.1. ***Function*** *we call a relation "**f**" a "**function**" when every input has exactly one output i.e.*

$$Func(f) : Rel(f) \wedge \forall x, y_1, y_2 \Big((x, y_1) \in f \wedge (x, y_2) \in f \rightarrow y_1 = y_2 \Big)$$

Theorem 5.3.1. (*schema*) ($FuncI, FuncE$)

$\begin{array}{\|l} xfy \\ xfz \\ \vdots \\ y = z \end{array}$ $\therefore Func(f) \qquad FuncI$	$Func(f)$ tfs $\underline{tfr}$ $\therefore s = r \qquad FuncE$

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.2. ***Restriction of a function*** *to a set "X" (usually a subset of "dom(f)") is the function*

$$f \restriction X := \{(x, y) \in f \mid x \in X\}$$

definition 5.3.3. ***Output of a function*** *assume a function* ***f*** *and* $(x, y) \in f$*, we define "*$f(x) := y$*" and we call "*$f(x)$*" a "****output of a function****". and the reverse is also true, if we use "*$f(x)$*", we have a function "****f****" and we can say "*$x \in Dom(f)$*". "output of a function" is object too.*

Theorem 5.3.2. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$

$\dfrac{\begin{array}{l} Func(f) \\ (x,y) \in f \end{array}}{\therefore f(x) = y} \quad f()E$	$\dfrac{\begin{array}{l} f(x) = y \\ Func(f) \end{array}}{\therefore (x,y) \in f} \quad f()E$

Attention 5.3.1. *input of a function can be a ordered pair itself, so we can have* $f((x, y))$ *and it is a "****output of a function****". we define "*$f(x, y) := f((x, y))$*" for simplicity. this is true for tuples as well, so we can have "*$f((x_1, x_2, ..., x_n))$*" and we define "*$f(x_1, x_2, ..., x_n) := f((x_1, x_2, ..., x_n))$*" for simplicity.*

definition 5.3.4. ***Term*** *we define term like this.*

1. *every obj is a term.*
2. *if* $t_0, ...t_n$ *are terms and "*$Func(f)$*" and "*$(t_0, ...t_n) \in Dom(f)$*". then "*$f(t_0, ...t_n)$*" is also a term.*

we denote terms with small letters like $t, s, r, ...$

definition 5.3.5. ***Substitution for term*** *we define Substitution an term with variable like this.*

1. *for variables if* $x = y$ *then* $y[t/x] := t$ *and if* $x \neq y$ *then* $y[t/x] := y$*.*
2. *for constants* $c[t/x] := c$*.*
3. *for terms like* $f(t_0, ...t_n)$

$$f(t_0, ...t_n)[t/x] := f(t_0[t/x], ...t_n[t/x])$$

definition 5.3.6. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single term we do it as we explain in "****Substitution for term****". if it's like* $t_0 = t_1$ *or* $t_0 \in t_1$ *(*t_0, t_1 *are terms). we define it like this.*

$$Set(y)[t/x] := Set(y[t/x])$$
$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$
$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[t/x] := (B[t/x] \bigcirc C[t/x])$$

($\bigcirc$ *can be* $\rightarrow, \wedge, \vee$*).*

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 5.3.3. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x \phi(x)}$ $\forall I$	1. x is a variable. 2. "$\phi(x)$" must not be assumed.
$\dfrac{\forall x \phi(x)}{\therefore \phi[t/x]}$ $\forall E$	1. t is a term.
$\dfrac{\phi[t/x]}{\therefore \exists x \phi(x)}$ $\exists I$	1. t is a term.
$\exists x \phi(x)$ $\quad \vert\ \phi(x)$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta$ $\exists E$	1. x is a variable. 2. "$\phi(x)$" must not be assumed in previous lines.(except ones that are closed) 3. x is not free in θ

$(\forall I, \exists E)$ is the same as $(\forall I, \exists E)$ in chapter three so we have to prove $(\forall E, \exists I)$. so assume $t = f(x)$ so to it's easy to see $Func(f)$ and $x \in Dom(f)$.

1	$\forall x \phi(x)$	ass
2	$Func(f)$	1
3	$x \in Dom(f)$	1
4	$\quad \vert\ (x, y) \in f$	acp
5	$\quad \vert\ f(x) = y$	def
6	$\quad \vert\ \phi(y)$	$\forall E$
7	$\quad \vert\ \phi(f(x))$	$= E$
8	$\phi(f(x))$	$\exists E$

1	$\phi(f(x))$	ass
2	$Func(f)$	ass
3	$x \in Dom(f)$	ass
4	$\quad \vert\ (x, y) \in f$	acp
5	$\quad \vert\ f(x) = y$	def
6	$\quad \vert\ \phi(y)$	$= E$
7	$\quad \vert\ \exists x \phi(x)$	$\exists I$
8	$\exists x \phi(x)$	$\exists E$

Attention 5.3.2. *in all rules and theorems you can replace all constants like $a, b, c, \ldots$ with terms like $t, s, r, \ldots.$ and rewrite them.*

definition 5.3.7. ***Compatible Function***

1. *Compatible*$(f, g) : \forall x \in dom(f) \cap dom(g) \quad f(x) = g(x)$.
2. *Compatible*$(F) : \forall f, g \in F \quad$ *Compatible*(f, g).

Lemma 5.3.1. .

1. *Compatible*$(f, g) \therefore / \therefore Func(f \cup g)$.
2. *Compatible*$(f, g) \therefore / \therefore f \upharpoonright (dom(f) \cap dom(g)) = g \upharpoonright (dom(f) \cap dom(g))$.

Proof Is Left As An Exercise To The Reader.😃

Theorem 5.3.4. *Compatible*$(F) / \therefore (1), (2), (3)$

1. $Func(\bigcup F)$.
2. $dom(\bigcup F) = \bigcup\{dom(f) \mid f \in F\}$.
3. $\forall f \in F \quad Compatible(f, \bigcup F)$.

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.8. ***Function from set to set*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is subset of **B** i.e.*

$$(f : A \to B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 5.3.9. ***Set of functions from set to set*** *we denote set of all "$f : A \to B$" with "B^A" i.e.*

$$B^A : \{f \mid f : A \to B\}$$

definition 5.3.10. ***Injective function(Relation)*** *we call a function(Relation) **f** injective when every output has at most one input i.e.*

$$Injective(f) : \forall y, x_1, x_2 \Big((x_1, y) \in f \wedge (x_2, y) \in f \to x_1 = x_2 \Big).$$

*we call a function **f** injective from **A** to **B** when it's function from **A** to **B** and it's injective i.e.*

$$(f : A \xrightarrow{inj} B) : (f : A \to B) \wedge Injective(f)$$

Theorem 5.3.5. *(schema)* $(InjectiveI, InjectiveE)$

$\quad\vert\; xfz$ $\quad\vert\; yfz$ $\quad\vert\; \vdots$ $\quad\vert\; x = y$ $\therefore Injective(f) \qquad InjectiveI$	$Injective(f)$ tfr sfr ――― $\therefore s = r \qquad InjectiveE$

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.11. ***Surjective function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 5.3.12. ***Bijection function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{inj} B) : (f : A \xrightarrow{inj} B) \wedge (f : A \xrightarrow[surj]{} B)$$

5.4 Family of sets

Attention 5.4.1. ***Indexes*** *are objects (like numbers) allows us to give a unique "tag" to every object. we use* $i, j, k, ...$ *for variable indexes and we use* $m, n, l, ...$ *for constant indexes.*
definition 5.4.1. ***Family of sets*** *assume* $A : I \to \mathcal{F}$ *is a function from a nonempty set of indices* I *to a family of sets* $\mathcal{F}$*, then we call* $(Range[A])$ *an indexed family of sets. we can write* A_i *instead of* $A(i)$ *and we can write* $(A_i)_{i\in I}$ *instead of "*A*" and* $\{A_i\}_{i\in I}$ *instead of "*$Range[A]$*" or "*$A[I]$*".*

$$(A_i)_{i\in I} := A$$

and

$$\{A_i\}_{i\in I} := A[I]$$

definition 5.4.2. ***Intersrction and union of an family of sets***

$$\bigcap_{i\in I} A_i := \bigcap\{A_i\}_{i\in I}$$

$$\bigcup_{i\in I} A_i := \bigcup\{A_i\}_{i\in I}$$

Attention 5.4.2. *we can write* $\forall i \in I \quad P(i)$ *instead of* $\forall i(i \in I \to P(i))$ *and we can write* $\exists i \in I \quad P(i)$ *instead of* $\exists i(i \in I \wedge P(i))$*.*
Lemma 5.4.1.

$$/ \therefore \bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i \in I} A_i$	acp
2	$x \in \bigcap \{A_i\}_{i \in I}$	$def.1$
3	$x \in \bigcap A[I]$	$def.1$
4	$\forall Y \in A[I] \quad x \in Y$	$def \bigcap$
5	$\forall Y((\exists i \in I \quad A_i = Y) \rightarrow x \in Y)$	$def image$
6	$(\exists i \in I \quad A_i = A_{i'}) \rightarrow x \in A_{i'}$	$\forall E$
7	$i' \in I$	acp
8	$A_{i'} = A_{i'}$	$= I$
9	$i' \in I \wedge A_{i'} = A_{i'}$	$\wedge I$
10	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
11	$x \in A_{i'}$	$\rightarrow E$
12	$i' \in I \rightarrow x \in A_{i'}$	$\rightarrow I$
13	$\forall i \in I \quad x \in A_i$	$\forall I$
14	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	def
15	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	acp
16	$\forall i \in I \quad x \in A_i$	def
17	$\exists i \in I \quad A_i = Y$	acp
18	$i \in I$	acp
19	$x \in A_i$	$\forall E$
20	$A_i = Y$	$\wedge E$
21	$x \in Y$	$= E$
22	$x \in Y$	$\exists E$
23	$(\exists i \in I \quad A_i = Y) \rightarrow x \in Y$	$\rightarrow I$
24	$\forall Y((\exists i \in I \quad A_i = Y) \rightarrow x \in Y)$	$\forall I$
25	$\forall Y \in A[I] \quad x \in Y$	def
26	$x \in \bigcap A[I]$	def
27	$x \in \bigcap \{A_i\}_{i \in I}$	def
28	$x \in \bigcap_{i \in I} A_i$	def
29	$\bigcap_{i \in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/ \therefore \bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$x \in \bigcup_{i \in I} A_i$	*acp*
2	$x \in \bigcup \{A_i\}_{i \in I}$	$def.1$
3	$x \in \bigcup A[I]$	$def.1$
4	$\exists Y \in A[I] \quad x \in Y$	$def \bigcup$
5	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$def.image$
6	$x \in Y$	*acp*
7	$\exists i \in I \quad A_i = Y$	*acp*
8	$i \in I$	*acp*
9	$A_i = Y$	*acp*
10	$x \in A_i$	$= E$
11	$\exists i \in I \quad x \in A_i$	$\exists I$
12	$\exists i \in I \quad x \in A_i$	$\exists E$
13	$\exists i \in I \quad x \in A_i$	$\exists E$
14	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	def
15	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	*acp*
16	$\exists i \in I \quad x \in A_i$	def
17	$i' \in I$	*acp*
18	$x \in A_{i'}$	*acp*
19	$A_{i'} = A_{i'}$	$= I$
20	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
21	$(\exists i \in I \quad A_i = A_{i'}) \wedge x \in A_{i'}$	$\wedge I$
22	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists I$
23	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists E$
24	$\exists Y \in A[I] \quad x \in Y$	def
25	$x \in \bigcup A[I]$	def
26	$x \in \bigcup \{A_i\}_{i \in I}$	def
27	$x \in \bigcup_{i \in I} A_i$	def
28	$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$= I$

Theorem 5.4.1. (*schema*) ($\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E$)

$\quad i \in I$ $\quad \vdots$ $\quad t \in A_i$ $\therefore t \in \bigcap_{i\in I} A_i \qquad \bigcap_I I$	$t \in \bigcap_{i\in I} A_i$ $n \in I$ $\therefore t \in A_n \qquad \bigcap_I E$
$t \in A_n$ $n \in I$ $\therefore t \in \bigcup_{i\in I} A_i \qquad \bigcup_I I$	$t \in \bigcup_{i\in I} A_i$ $\quad i \in I$ $\quad t \in A_i$ $\quad \vdots$ $\quad \theta$ $\therefore \theta \qquad \bigcup_I E$

Proof Is Left As An Exercise To The Reader.😀

5.5 Structures and morphisms

definition 5.5.1. ***Structure*** *"$\mathfrak{M}$" is a tuple defined like this.*

$$\mathfrak{M} := (M, f_0, f_1, ..., f_n, R_0, R_1, ..., R_m, c_0, c_1, ..., c_k)$$

"M" is a ***nonempty*** *set shows our universe "$\forall i \in \{0, ..., n\} \quad f_i : M^{n'} \to M$" are funcions and " $\forall i \in \{0, ..., m\} \quad R_i \subseteq M^{m'}$" are relations and "$\forall i \in \{0, ..., k\} \quad c_i \in M$" are constants. ($M^n$ means $\overbrace{M \times M \times ... \times M}^{n\ times}$ we will define it later) and for the case we have arbitary family of functions and relations and constants.*

$$\mathfrak{M} := (M, \{f_i\}_{i\in I_f}, \{R_i\}_{i\in I_R}, \{c_i\}_{i\in I})$$

definition 5.5.2. ***Homorphism*** *if "$\mathfrak{M} := (M, \{f_{i,m}\}_{i\in I_f}, \{R_{i,m}\}_{i\in I_R}, \{c_{i,m}\}_{i\in I})$" and "$\mathfrak{N} := (N, \{f_{i,n}\}_{i\in I_f}, \{R_{i,n}\}_{i\in I_R}, \{c_{i,n}\}_{i\in I})$" are two structures. function "$h : M \to N$"is called "(weak) Homorphism" when this conditions holds*

1. *for eveery relation "$R_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad R_{i,m}(x_0, x_1, x_2, ..., x_k) \to R_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

2. *for eveery function "$f_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad h(f_{i,m}(x_0, x_1, x_2, ..., x_k)) = f_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

3. *for eveery constant "$c_{i,m}$"*

$$h(c_{i,m}) = c_{i,n}$$

.

in condition (1) if "$\leftrightarrow$" be instead of "$\rightarrow$" then it's called "(strong) Homorphism"

few notices: when we are talking about "Homorphism" we usually talk about "(strong) Homorphism" .
in the case (in definition "(strong) Homorphism") we denote it by "$Hom(h, \mathfrak{M}, \mathfrak{N})$". if exists function "$h$" "$Hom(h, \mathfrak{M}, \mathfrak{N})$" we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are homomorphic. we denote it by "$Hom(\mathfrak{M}, \mathfrak{N})$".
if "$h : M \xrightarrow{inj} N$" we call it "Monomorphism" or "Embedding".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are momomorphic.
if "$h : M \xrightarrow[surj]{} N$" we call it "Epimorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Eomomorphic.
if "$h : M \xrightarrow[surj]{inj} N$" we call it "Isomorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Isomomorphic. and denote it by "$\mathfrak{M} \cong \mathfrak{N}$".
if "$k : M \to M$" is a function and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Indomorphism".
if "$k : M \xrightarrow[surj]{inj} M$" is a bijection and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Automorphism".

5.6 Equivalences and Partitions

definition 5.6.1. ***Equivalence relation*** *assume a structure "$\mathfrak{A} := (A, \sim)$"("$\sim$" is a relation). we say "$\sim$" is a "Equivalence relation" in "$\mathfrak{A}$" if it's reflexive,symmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \sim x$$

$$(symmetric) : \forall x, y \in A \quad x \sim y \to y \sim x$$

$$(transitive) : \forall x, y, z \in A \quad x \sim y \wedge y \sim z \to x \sim z$$

Lemma 5.6.1. *"$\cong$" is a equivalence relation in "$(Structure, \cong)$"*
Proof Is Left As An Exercise To The Reader.😃
definition 5.6.2. ***Equivalence class of "x"*** *Let "$\sim$" be an equivalence relation on "$\mathfrak{A}$". For every "$x \in A$"*

$$[x]_\sim := \{y \in A \mid y \sim x\}$$

you can also see it as a function $[\_]_\sim : A \to V$.
Theorem 5.6.1. $(schema)$ $([]_\sim I, []_\sim E)$

$\dfrac{t \sim s}{\therefore t \in [s]_\sim} \quad []_\sim I$	$\dfrac{t \in [s]_\sim}{\therefore t \sim s} \quad []_\sim E$

Proof Is Left As An Exercise To The Reader.😃
Lemma 5.6.2. *For every "$a, b \in A$"*

1. $a \sim b \therefore / \therefore [a]_\sim = [b]_\sim$
2. $a \nsim b \therefore / \therefore [a]_\sim \cap [b]_\sim = \varnothing$

prove.1

	$/ \therefore$	
1	$a \sim b$	*ass*
2	$\mid\ x \in [a]_\sim$	*acp*
3	$\mid\ x \sim a$	$[]_\sim E$
4	$\mid\ x \sim b$	*trans*
5	$\mid\ x \in [b]_\sim$	$[]_\sim I$
6	$\mid\ x \in [b]_\sim$	*acp*
7	$\mid\ x \sim b$	$[]_\sim E$
8	$\mid\ x \sim a$	*trans*
9	$\mid\ x \in [a]_\sim$	$[]_\sim I$
10	$[a]_\sim = [b]_\sim$	

	$\therefore /$	
1	$[a]_\sim = [b]_\sim$	*ass*
2	$a \sim a$	*reflex*
3	$a \in [a]_\sim$	$[]_\sim I$
4	$a \in [b]_\sim$	$= E$
5	$a \sim b$	$[]_\sim E$

prove.2

	$/ \therefore$	
1	$a \nsim b$	*ass*
2	$\mid\ [a]_\sim \cap [b]_\sim \neq \varnothing$	*acp*
3	$\mid\ \mid\ x \in [a]_\sim \cap [b]_\sim$	*acp*
4	$\mid\ \mid\ x \in [a]_\sim$	$\cap E$
5	$\mid\ \mid\ x \in [b]_\sim$	$\cap E$
6	$\mid\ \mid\ x \sim a$	$[]_\sim E$
7	$\mid\ \mid\ x \sim b$	$[]_\sim E$
8	$\mid\ \mid\ a \sim b$	*trans*
9	$\mid\ \mid\ \perp$	
10	$\mid\ \perp$	
11	$[a]_\sim \cap [b]_\sim = \varnothing$	

	$\therefore /$	
1	$[a]_\sim \cap [b]_\sim = \varnothing$	
2	$\mid\ a \sim b$	*acp*
3	$\mid\ a \sim a$	*reflex*
4	$\mid\ a \in [a]_\sim$	$[]_\sim I$
5	$\mid\ a \in [b]_\sim$	$[]_\sim I$
6	$\mid\ a \in [a]_\sim \cap [b]_\sim$	$\cap I$
7	$\mid\ \perp$	
8	$a \nsim b$	

definition 5.6.3. ***Partition*** *is a "set of sets" like "P" of a set like "A" with this conditions.*

1. $\forall X \in P \quad X \neq \varnothing$
2. $\forall X, Y \in P \quad X \neq Y \rightarrow X \cap Y = \varnothing$
3. $\bigcup P = A$
   if these condition holds then we say "$Part(P, A)$*".*

definition 5.6.4. ***Set of Equivalence classes***

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

Theorem 5.6.2. *if "$\sim$" is a equivalence class in "$(A, \sim)$" then "$A/\sim$" is a partition for "A" i.e. "$Part(A/\sim, A)$".*

1	$\quad [x]_\sim \in A/\sim$	acp
2	$\quad x \sim x$	$reflex$
3	$\quad x \in [x]_\sim$	$[]_\sim I$
4	$\quad [x]_\sim \neq \varnothing$	$\neq \varnothing I$
5	$\forall [x]_\sim \in A/\sim \quad [x]_\sim \neq \varnothing$	

1	$\quad [x]_\sim \neq [y]_\sim$	acp
2	$\quad x \nsim y$	$Lemma 5.6.2.1$
3	$\quad [x]_\sim \cap [y]_\sim = \varnothing$	$Lemma 5.6.2.2$
4	$\forall [x]_\sim, [y]_\sim \in A/\sim \quad [x]_\sim \neq [y]_\sim \rightarrow [x]_\sim \cap [y]_\sim = \varnothing$	

1	$\quad x \in \bigcup(A/\sim)$	acp
2	$\qquad [y]_\sim \in A/\sim$	acp
3	$\qquad x \in [y]_\sim$	acp
4	$\qquad [y]_\sim \subseteq A$	
5	$\qquad x \in A$	$\subseteq E$
6	$\quad x \in A$	$\bigcup E$
7	$\quad x \in A$	
8	$\quad x \in [x]_\sim$	
9	$\quad [x]_\sim \in (A/\sim)$	
10	$\quad x \in \bigcup(A/\sim)$	$\bigcup I$
11	$A = \bigcup(A/\sim)$	$= I$

Theorem 5.6.3. *if "P" is a partition for "A" we define*

$$\sim_P := \{(x, y) \mid \exists Y \in P \quad (x \in Y \wedge y \in Y)\}$$

then "$\sim_P$" is a equivalence class in "$(A, \sim_P)$".

1	$x \in A$	*acp*
2	$x \in \bigcup P$	*Def*
3	$x \in Y$	*acp*
4	$Y \in P$	*acp*
5	$\exists Y \in P \quad (x \in Y \wedge x \in Y)$	
6	$x \sim_P x$	
7	$x \sim_P x$	
8	$\forall x \in A \quad x \sim_P x$	

1	$x, y \in A$	*acp*
2	$x \sim_P y$	*acp*
3	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$	*Def*
4	$Y \in P \wedge x \in Y \wedge y \in Y$	*acp*
5	$Y \in P \wedge y \in Y \wedge x \in Y$	*commutative*$\wedge$
6	$\exists Y \in P \quad (y \in Y \wedge x \in Y)$	
7	$y \sim_P x$	
8	$y \sim_P x$	
9	$x \sim_P y \rightarrow y \sim_P x$	
10	$\forall x, y \in A \quad x \sim_P y \rightarrow y \sim_P x$	

1	$x, y, z \in A$	*acp*
2	$x \sim_P y$	*acp*
3	$y \sim_P z$	*acp*
4	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$	
5	$\exists Y \in P \quad (y \in Y \wedge z \in Y)$	
6	$Y \in P \wedge x \in Y \wedge y \in Y$	
7	$Y' \in P \wedge y \in Y' \wedge z \in Y'$	
8	$Y \cap Y' \neq \varnothing$	
9	$Y = Y'$	
10	$Y \in P \wedge x \in Y \wedge z \in Y$	
11	$\exists Y \in P \quad (x \in Y \wedge z \in Y)$	
12	$x \sim_P z$	
13	$x \sim_P z$	
14	$x \sim_P z$	
15	$x \sim_P y \wedge y \sim_p z \to x \sim_P z$	
16	$\forall x, y, z \in A \quad x \sim_P y \wedge y \sim_p z \to x \sim_P z$	

5.7 Orderings

definition 5.7.1. ***(partial) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a relation). we say "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" if it's reflexive,antisymmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \preceq x$$

$$(antisymmetric) : \forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y$$

$$(transitive) : \forall x, y, z \in A \quad x \preceq y \wedge y \preceq z \to x \preceq z$$

definition 5.7.2. ***(partial) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a relation). we say "$\prec$" is a "(partial) Strict order relation" in "$\mathfrak{A}$" if it's irreflexive, and transitive.*

$$(irreflexive) : \forall x \in A \quad x \nprec x$$

$$(transitive) : \forall x, y, z \in A \quad x \prec y \wedge y \prec z \to x \prec z$$

by two conditions bellow we can proof asymmetric too.

$$(asymmetric) : \forall x, y \in A \quad x \prec y \to y \nprec x$$

now we will show you relation between "(partial) Order relation" and "(partial) Strict order relation".

Theorem 5.7.1. .

1. *if "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(partial) Strict order relation".

1	│ $x \in A$	*acp*
2	│ │ $x \prec x$	*acp*
3	│ │ $x \neq x$	$\wedge E$
4	│ │ $x = x$	$= I$
5	│ │ $\perp$	$\perp I$
6	│ $x \not\prec x$	$\neg I$
7	$\forall x \in A \quad x \not\prec x$	$\rightarrow I, \forall I$

1	│ $x, y, z \in A$	*acp*
2	│ │ $x \prec y$	*acp*
3	│ │ $y \prec z$	*acp*
4	│ │ $x \preceq y$	$\wedge E$
5	│ │ $y \preceq z$	$\wedge E$
6	│ │ $x \preceq z$	*trans*
7	│ │ $x \neq y$	$\wedge E$
8	│ │ $y \neq z$	$\wedge E$
9	│ │ │ $x = z$	*acp*
10	│ │ │ $y \preceq x$	$= E$
11	│ │ │ $x = y$	*antisym*
12	│ │ │ $\perp$	$\perp I$
13	│ │ $x \neq z$	$\neg I$
14	│ │ $x \prec z$	$\wedge I$
15	│ $x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I$
16	$\forall x, y, z \in A \quad x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I, \forall I$

2. *if "$\prec$" is a "(partial) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(partial) order relation".

1	$x \in A$	*acp*
2	$x = x$	$= I$
3	$x \preceq x$	$\vee I$
4	$\forall x \in A \quad x \preceq x$	$\to I, \forall I$

1	$x, y \in A$	*acp*
2	$x \preceq y \wedge y \preceq x$	*acp*
3	$x \prec y \wedge y \prec x$	*acp*
4	$\perp$	*asym*
5	$x = y$	
6	$x \prec y \wedge y = x$	*acp*
7	$x = y$	
8	$x = y \wedge y \prec x$	*acp*
9	$x = y$	
10	$x = y \wedge y = x$	*acp*
11	$x = y$	
12	$x = y$	$\vee E$
13	$x \preceq y \wedge y \preceq x \to x = y$	$\to I$
14	$\forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y$	$\to I, \forall I$

1	$x, y, z \in A$	*acp*
2	$\quad x \preceq y \wedge y \preceq z$	*acp*
3	$\quad\quad x \prec y \wedge y \prec z$	*acp*
4	$\quad\quad x \prec z$	*trans*
5	$\quad\quad x \preceq z$	$\vee I$
6	$\quad\quad x \prec y \wedge y = z$	*acp*
7	$\quad\quad x \prec z$	$= E$
8	$\quad\quad x \preceq z$	$\vee I$
9	$\quad\quad x = y \wedge y \prec z$	*acp*
10	$\quad\quad x \prec z$	$= E$
11	$\quad\quad x \preceq z$	$\vee I$
12	$\quad\quad x = y \wedge y = z$	*acp*
13	$\quad\quad x = z$	$= E$
14	$\quad\quad x \preceq z$	$\vee I$
15	$\quad x \preceq z$	$\vee E$
16	$x \preceq y \wedge y \prec z \to x \preceq z$	$\to I$
17	$\forall x, y, z \in A \quad x \preceq y \wedge y \prec z \to x \preceq z$	$\to I, \forall I$

definition 5.7.3. ***(Total-Linear) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \preceq y \vee y \preceq x$$

definition 5.7.4. ***(Total-Linear) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a Strict order relation). we say "$\prec$" is a "(Total-Linear) Strict order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$$

Theorem 5.7.2. *.*

1. *if "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(Total-Linear) Strict order relation".*
2. *if "$\prec$" is a "(Total-Linear) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(Total-Linear) order relation".*

1	$\quad x, y \in A$	*acp*
2	$\quad x \preceq y \vee y \preceq x$	*linearity of $\preceq$ (1)*
3	$\quad\quad x = y$	*acp*
4	$\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ *(3)*
5	$\quad\quad x \neq y$	*acp*
6	$\quad\quad\quad x \preceq y$	*acp*
7	$\quad\quad\quad x \preceq y \wedge x \neq y$	$\wedge I$ *(7, 6)*
8	$\quad\quad\quad x \prec y$	*def $\prec$ (8)*
9	$\quad\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ *(9)*
10	$\quad\quad\quad y \preceq x$	*acp*
11	$\quad\quad\quad y \neq x$	*symmetry of $\neq$ (6)*
12	$\quad\quad\quad y \preceq x \wedge y \neq x$	$\wedge I$ *(12, 13)*
13	$\quad\quad\quad y \prec x$	*def $\prec$ (14)*
14	$\quad\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ *(15)*
15	$\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee E$ *(2, 11, 17)*
16	$\quad x \prec y \vee x = y \vee y \prec x$	$\vee E$ *(20, 5, 19)*
17	$\forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$	$\forall I$ *(1-21)*

1	$\quad x, y \in A$	*acp*
2	$\quad x \prec y \vee x = y \vee y \prec x$	*linearity of $\prec$*
3	$\quad\quad x \prec y \vee x = y$	*acp (Case 1 from 2)*
4	$\quad\quad x \preceq y$	*def $\preceq$ (3)*
5	$\quad\quad x \preceq y \vee y \preceq x$	$\vee I$ *(4)*
6	$\quad\quad y \prec x$	*acp (Case 2 from 2)*
7	$\quad\quad y \prec x \vee y = x$	$\vee I$ *(6)*
8	$\quad\quad y \preceq x$	*def $\preceq$ (7)*
9	$\quad\quad x \preceq y \vee y \preceq x$	$\vee I$ *(8)*
10	$\quad x \preceq y \vee y \preceq x$	$\vee E$ *(2, 3-5, 6-9)*
11	$\forall x, y \in A \quad x \preceq y \vee y \preceq x$	$\forall I$ *(1-10)*

Lemma 5.7.1. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B, C \subseteq A$ prove*

1. *$\exists b \in B \quad \forall x \in B \quad b \preceq x / \therefore \exists! b \in B \quad \forall x \in B \quad b \preceq x$ in this case we say "B" has least element or Minimum or "$HaveLeast(B)$"*

2. *$\exists c \in C \quad \forall x \in C \quad x \preceq c / \therefore \exists! c \in C \quad \forall x \in C \quad x \preceq c$ in this case we say "C" has greatest element or has Maximum or "$HaveGreatest(C)$"*

1	$\exists b \in B \ \forall x \in B (b \preceq x)$	*ass*
2	$b_1, b_2 \in B$	*acp (assumption for uniqueness)*
3	$(\forall x \in B \quad b_1 \preceq x) \wedge (\forall x \in B \ b_2 \preceq x)$	*acp*
4	$\forall x \in B \quad b_1 \preceq x$	*$\wedge E$ (3)*
5	$\forall x \in B \quad b_2 \preceq x$	*$\wedge E$ (3)*
6	$b_1 \preceq b_2$	*$\forall E$ (4, $x \to b_2$)*
7	$b_2 \preceq b_1$	*$\forall E$ (5, $x \to b_1$)*
8	$b_1 \preceq b_2 \wedge b_2 \preceq b_1$	*$\wedge I$ (6, 7)*
9	$b_1 = b_2$	*antisym of $\preceq$ (8)*
10	$((\forall x \in B \quad b_1 \preceq x) \wedge (\forall x \in B \ b_2 \preceq x)) \to b_1 = b_2$	*$\to I$ (3-9)*
11	$\exists! b \in B \quad \forall x \in B (b \preceq x)$	*$\exists!$ def (1, 10)*

1	$\exists c \in C \ \forall x \in C (x \preceq c)$	*ass*
2	$c_1, c_2 \in C$	*acp (assumption for uniqueness)*
3	$(\forall x \in C \ x \preceq c_1) \wedge (\forall x \in C \ x \preceq c_2)$	*acp*
4	$\forall x \in C \ x \preceq c_1$	*$\wedge E$ (3)*
5	$\forall x \in C \ x \preceq c_2$	*$\wedge E$ (3)*
6	$c_2 \preceq c_1$	*$\forall E$ (4, $x \to c_2$)*
7	$c_1 \preceq c_2$	*$\forall E$ (5, $x \to c_1$)*
8	$c_1 \preceq c_2 \wedge c_2 \preceq c_1$	*$\wedge I$ (7, 6)*
9	$c_1 = c_2$	*antisym of $\preceq$ (8)*
10	$((\forall x \in C \ x \preceq c_1) \wedge (\forall x \in C \ x \preceq c_2)) \to c_1 = c_2$	*$\to I$ (3-9)*
11	$\exists! c \in C \ \forall x \in C (x \preceq c)$	*$\exists!$ def (1, 10)*

Theorem 5.7.3. ***Minimum and Maximum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid HaveLeast(B)\}$ and $\mathcal{C} := \{C \subseteq A \mid HaveGreatest(C)\}$ we define like this*

$$min_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times V \mid \forall x \in B \quad b \preceq x\}$$

$$max_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times V \mid \forall x \in C \quad x \preceq c\}$$

we must prove these two relations are two functions "$min_{\mathfrak{A}} : \mathcal{B} \to V$" and "$max_{\mathfrak{A}} : \mathcal{C} \to V$".

1	$B \in \mathcal{B}$	*acp*
2	$\exists b \in B \quad \forall x \in B(b \preceq x)$	*def $\mathcal{B}$ (1)*
3	$b_0 \in B \wedge \forall x \in B(b_0 \preceq x)$	*acp (assumption for $\exists E$)*
4	$(B, b_0) \in min_A$	*def min_A (3)*
5	$\exists b \in V((B, b) \in min_A)$	*$\exists I$ (4)*
6	$\exists b \in V((B, b) \in min_A)$	*$\exists E$ (2, 3-5)*
7	$b_1, b_2 \in V$	*acp*
8	$(B, b_1) \in min_A \wedge (B, b_2) \in min_A$	*acp*
9	$(\forall x \in B \quad b_1 \preceq x) \wedge (\forall x \in B \; b_2 \preceq x)$	*def min_A, $\wedge E$ (8)*
10	$b_1 = b_2$	*Lemma 5.7.1 (uniqueness)*
11	$((B, b_1) \in min_A \wedge (B, b_2) \in min_A) \to b_1 = b_2$	*$\to I$ (8-10)*
12	$\exists! b \in V((B, b) \in min_A)$	*def $\exists!$ (6, 11)*
13	$\forall B \in \mathcal{B} \quad \exists! b \in V((B, b) \in min_A)$	*$\forall I$ (1-12)*

1	$C \in \mathcal{C}$	*acp*
2	$\exists c \in C \; \forall x \in C(x \preceq c)$	*def $\mathcal{C}$ (1)*
3	$c_0 \in C \wedge \forall x \in C(x \preceq c_0)$	*acp (assumption for $\exists E$)*
4	$(C, c_0) \in max_A$	*def max_A (3)*
5	$\exists c \in V((C, c) \in max_A)$	*$\exists I$ (4)*
6	$\exists c \in V((C, c) \in max_A)$	*$\exists E$ (2, 3-5)*
7	$c_1, c_2 \in V$	*acp*
8	$(C, c_1) \in max_A \wedge (C, c_2) \in max_A$	*acp*
9	$(\forall x \in C \quad x \preceq c_1) \wedge (\forall x \in C \; x \preceq c_2)$	*def max_A, $\wedge E$ (8)*
10	$c_1 = c_2$	*Lemma 5.7.1 (uniqueness)*
11	$((C, c_1) \in max_A \wedge (C, c_2) \in max_A) \to c_1 = c_2$	*$\to I$ (8-10)*
12	$\exists! c \in V((C, c) \in max_A)$	*def $\exists!$ (6, 11)*
13	$\forall C \in \mathcal{C} \quad \exists! c \in V((C, c) \in max_A)$	*$\forall I$ (1-12)*

definition 5.7.5. ***Minimal and Maximal*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we define "Minimal and Maximal" like this*

$$Minimal_{\mathfrak{A}}(b) : \forall x \in A \quad x \not\prec b$$

i.e. "b" is the minimal in "$\mathfrak{A}$".

$$Maximal_{\mathfrak{A}}(c) : \forall x \in A \quad c \not\prec x$$

i.e. "c" is the maximal in "$\mathfrak{A}$".

definition 5.7.6. ***Well ordering*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\mathfrak{A}$ " is "Well ordered" when every nonempty subset of "A " has a "Minimum " i.e.*

$$WellOrdered(\mathfrak{A}) : \forall X \subseteq A \quad X \neq \varnothing \to HaveLeast(X)$$

definition 5.7.7. ***Lower and Upper Bound*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). $B, C \subseteq A$. we define "Lower and Upper Bound" like this*

$$Lowerb_{\mathfrak{A}}(b, B) : \forall x \in B \quad b \preceq x$$

i.e. "b" is the lowerbound of B in "$\mathfrak{A}$".

$$Upperb_{\mathfrak{A}}(c, C) : \forall x \in C \quad x \preceq c$$

i.e. "C" is the upperbound of B in "$\mathfrak{A}$".

if the stucture clear to us we don't write stucture index "$min(\_), Minimal(\_), Lowerb(\_, \_)$". and we don't mension it.

Theorem 5.7.4. ***Infimum and Supremum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid HaveGreatest\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}\}$ and $\mathcal{C} := \{C \subseteq A \mid HaveLeast\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}\}$ we define "$inf_{\mathfrak{A}} : \mathcal{B} \to V$" and "$sup_{\mathfrak{A}} : \mathcal{C} \to V$" like this*

$$inf_{\mathfrak{A}}(B) := max_{\mathfrak{A}}\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}$$

$$sup_{\mathfrak{A}}(B) := min_{\mathfrak{A}}\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}$$

Theorem 5.7.5. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B \subseteq A$ prove*

1. $min_{\mathfrak{A}}(B) = b / \therefore inf_{\mathfrak{A}}(B) = b$.

1	$min_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B \wedge \forall y \in B(b \preceq y)$	*def* $min_{\mathfrak{A}}$ *(1)*
3	$b \in B$	$\wedge E$ *(2)*
4	$\forall y \in B(b \preceq y)$	$\wedge E$ *(2)*
5	$Lowerb_{\mathfrak{A}}(b, B)$	*def* $Lowerb_{\mathfrak{A}}$ *(4)*
6	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*set construction (5)*
7	$\quad z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*acp*
8	$\quad Lowerb_{\mathfrak{A}}(z, B)$	*set membership (7)*
9	$\quad \forall y \in B(z \preceq y)$	*def* $Lowerb_{\mathfrak{A}}$ *(8)*
10	$\quad z \preceq b$	$\forall E$ *(9,* $y \to b$ *using 3)*
11	$\forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	$\forall I$ *(7-10)*
12	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\} \wedge \forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	$\wedge I$ *(6, 11)*
13	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $max_{\mathfrak{A}}$ *(12)*
14	$inf_{\mathfrak{A}}(B) = b$	*def* $inf_{\mathfrak{A}}$ *(13)*

2. $inf_{\mathfrak{A}}(B) = b, b \in B / \therefore min_{\mathfrak{A}}(B) = b.$

1	$inf_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B$	*ass*
3	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $inf_{\mathfrak{A}}$ *(2)*
4	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\} \wedge \forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	*def* $max_{\mathfrak{A}}$ *(4)*
5	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	$\wedge E$ *(5)*
6	$Lowerb_{\mathfrak{A}}(b, B)$	*set membership (6)*
7	$\forall y \in B(b \preceq y)$	*def* $Lowerb_{\mathfrak{A}}$ *(7)*
8	$b \in B \wedge \forall y \in B(b \preceq y)$	$\wedge I$ *(3, 8)*
9	$min_{\mathfrak{A}}(B) = b$	*def* $min_{\mathfrak{A}}$ *(9)*

this theorem holds for "max, sup" too.

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Chapter 6

The beginning of Abstract Algebra

we want to introduce some functions "$\bigcirc : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" to you. we use "$m \bigcirc n$" instead of "$\bigcirc(m, n)$".

6.1 Group theory

definition 6.1.1. *A structure $\mathfrak{G} := (G, \cdot)$*

1. *If $\forall x, y, z \in G \quad (xy)z = x(yz)$ (assosiative $\times$), $\mathfrak{G}$ is a* ***semigroup****.*
2. *If $\mathfrak{G}$ is a* ***semigroup*** *and $\forall x \in G \quad \exists y \in G \quad \forall x \in G \quad xy = yx = x$ (has identity $\times$), $\mathfrak{G}$ is a* ***monoid****. you can prove for every* ***monoid*** *$\mathfrak{G}$ identity element is unique we denote it by "e"(in $\times$ with "1" and in $+$ with 0).*
3. *If $\mathfrak{G}$ is a* ***monoid*** *and $\forall x \in G \quad \exists y \in G \quad xy = yx = e$ (has inverse $\times$), $\mathfrak{G}$ is a group.*
4. *If $\mathfrak{G}$ is a* ***semigroup/monoid*** *and $\forall x, y, z \in G \quad xz = yz \to x = y$ (cancellative $\times$), $\mathfrak{G}$ is a* ***cancellative semigroup/monoid****. (you can prove every group is cancellative as an exercise.😀)*
5. *If $\mathfrak{G}$ is a* ***semigroup/monoid/group*** *and $\forall x, y \in G \quad xy = yx$ (commutative(abelian) $\times$), $\mathfrak{G}$ is a* ***commutative(abelian) semigroup/monoid/group****.*

Theorem 6.1.1 (Grothendieck Group Construction)**.** *Let $(S, +, 0)$ be a cancellative commutative monoid.*

Then there exists an abelian group $G(S)$ together with a monoid mbedding

$$\eta : S \longrightarrow G(S)$$

such that for every abelian group A and every monoid mbedding

$$f : S \to A,$$

there exists a unique group mbedding

$$\tilde{f} : G(S) \to A$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$	
2	$a + b = b + a$	
3	$(a, b) \sim (a, b)$	

Symmetry

1	$(a, b) \sim (c, d)$	
2	$a + d = b + c$	
3	$c + b = d + a$	
4	$(c, d) \sim (a, b)$	

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

Step 1: Define Grothendieck group

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operation

$$[(a,b)] + [(c,d)] := [(a+c, b+d)]$$

$$-[(a,b)] := [(b,a)]$$

—

Step 3: Well-definedness

(Addition and inverse are well-defined by compatibility of $\sim$ with $+$.)

—

Step 4: Group structure

(a,b)] + [(b,a)] = [(a+b, a+b)][(b,a)] = [(a+b, a+b)] (a+b,a+b)] = 01	+
0 2	=
3	every element has inverse
4	$G(S)$ is an abelian group

—

Step 5: Define extension

Let $f : S \to A$ be a monoid mbedding into an abelian group A.

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

—

Step 6: Well-definedness

1	$(a,b) \sim (c,d)$
2	$a + d = b + c$
3	$f(a) + f(d) = f(b) + f(c)$
4	$f(a) - f(b) = f(c) - f(d)$

—

Step 7: Homomorphism property

(Straightforward from group operations in A.)

—

Step 8: Factorization

$$\tilde{f}([(a,0)]) = f(a)$$

—

Step 9: Uniqueness

Any mbedding is forced by:

$$[(a,b)] = [(a,0)] - [(b,0)].$$

□

6.2 Ring theory

definition 6.2.1. *A structure $\mathfrak{R} := (R, +, \cdot)$*

1. *If $(R, +)$ is a abelian **monoid/group** and $\forall x, y, z \in R \quad (xy)z = x(yz)$ (assosiative $\times$), $\forall x, y, z \in R \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ (distributive $\times$), $\mathfrak{R}$ is a **Semiring/Ring**.*

2. *If $\mathfrak{R}$ is a **Semiring/Ring** and $\exists y \in R \quad \forall x \in R \quad xy = yx = x$ (has identity $\times$), $\mathfrak{R}$ is a **Semiring/Ring with identity**. you can prove for every **Semiring/Ring with identity** $\mathfrak{R}$ identity element is unique we denote it by "1".*

3. *If $\mathfrak{R}$ is a **Semiring/Ring** and $\forall x, y \in R \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (zero divisor free $\times$), $\mathfrak{R}$ is a **zero divisor free Semiring/Ring**. (you can prove every **division Semiring/Ring** is zero divisor free as an exercise.😃)*

4. *If $\mathfrak{R}$ is a **Semiring/Ring with identity** and $\forall x \in R, \neq 0 \quad \exists y \in R \quad xy = yx = 1$ (has inverse $\times$), $\mathfrak{R}$ is a **division Semiring/Ring** with.*

5. *If $\mathfrak{R}$ is a **Semiring/Ring** and $\forall x, y \in R \quad xy = yx$ (commutative(abelian) $\times$), $\mathfrak{R}$ is a **commutative(abelian) Semiring/Ring**.*

6. *If $\mathfrak{R}$ is a **zero divisor free Semiring/Ring** and **commutative(abelian) Semiring/Ring**. $\mathfrak{R}$ is a **Semidomain/Domain** .*

7. *If $\mathfrak{R}$ is a **division Semiring/Ring** and **commutative(abelian) Semiring/Ring**. $\mathfrak{R}$ is a **SemiField/Field** .*

Theorem 6.2.1 (Grothendieck Construction for Semirings)**.** *Let $(S, +, \cdot)$ be a semiring whose additive structure $(S, +, 0)$ is a cancellative commutative monoid.*

Then there exists a ring $G(S)$ together with a semiring mbedding

$$\eta : S \longrightarrow G(S)$$

such that for every ring T and every semiring mbedding

$$f : S \to T,$$

there exists a unique ring mbedding

$$\tilde{f} : G(S) \to T$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define the underlying relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$	
2	$a + b = b + a$	
3	$(a, b) \sim (a, b)$	

Symmetry

1	$(a, b) \sim (c, d)$	
2	$a + d = b + c$	
3	$c + b = d + a$	
4	$(c, d) \sim (a, b)$	

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

(additive commutative monoid cancellation NOT required; we instead justify via rearrangement in monoid equality)

Step 1: Define the carrier set.

Define

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operations.

Define

$$[(a,b)] + [(c,d)] := [(a+c,\ b+d)],$$
$$[(a,b)] \cdot [(c,d)] := [(ac+bd,\ ad+bc)].$$

Define

$$\eta(a) := [(a,0)].$$

Step 3: Well-definedness of addition.

1 $(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$

2 $a+b' = a'+b,\ c+d' = c'+d$

3 $(a+c)+(b'+d') = (a'+c')+(b+d)$

4 $[(a+c,b+d)] = [(a'+c',b'+d')]$

5 $+$ is well-defined

Step 4: Well-definedness of multiplication.

1 $(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$

2 $a+b' = a'+b,\ c+d' = c'+d$

3 $ac+bd+(a'd'+b'c') = a'c'+bd+ac'+b'd$

4 $[(ac+bd,ad+bc)] = [(a'c'+b'd',a'd'+b'c')]$

5 $\cdot$ is well-defined

Step 5: Ring structure.

1 $[(a,b)] + [(c,d)] = [(a+c,b+d)]$

2 associativity inherited from S

3 $[(a,b)] \cdot [(c,d)]$ associative

4 $distributivityfollowsfromsemiringlaws$

5 $(0,0)$ is additive identity

6 $[(a,a)] = 0$ in $G(S)$

7 $G(S)$ is a ring

Step 6: Define extension of a mbedding.

Let

$$f : S \to T$$

be a semiring mbedding into a ring T.

Define

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

Claim 6: $\tilde{f}$ is well-defined.

1 $(a,b) \sim (c,d)$

2 $a + d = b + c$

3 $f(a) + f(d) = f(b) + f(c)$

4 $f(a) - f(b) = f(c) - f(d)$

5 $\tilde{f}([(a,b)]) = \tilde{f}([(c,d)])$

Step 7: $\tilde{f}$ is a ring mbedding.

1 $\tilde{f}([(a,b)] + [(c,d)]) = f(a+c) - f(b+d)$

2 $= (f(a) - f(b)) + (f(c) - f(d))$

3 $= \tilde{f}([(a,b)]) + \tilde{f}([(c,d)])$

4 $\tilde{f}([(a,b)] \cdot [(c,d)]) = f(ac+bd) - f(ad+bc)$

5 $= (f(a) - f(b))(f(c) - f(d))$

6 $= \tilde{f}([(a,b)])\tilde{f}([(c,d)])$

Step 8: Factorization property.

1 $\tilde{f}(\eta(a)) = \tilde{f}([(a,0)])$

2 $= f(a)$

3 $\tilde{f} \circ \eta = f$

Step 9: Uniqueness.

1 $g : G(S) \to T$ ring mbedding

2 $g \circ \eta = f$

3 $g([(a,b)]) = g([(a,0)]) - g([(b,0)])$

4 $= f(a) - f(b)$

5 $= \tilde{f}([(a,b)])$

6 $g = \tilde{f}$

Therefore $G(S)$ satisfies the universal property of the Grothendieck construction.

$\square$

Theorem 6.2.2 (Field of Fractions)**.** *Let R be a integral domain.*

Then there exists a field $\mathrm{Frac}(R)$ *together with a ring embedding*

$$\eta : R \longrightarrow \mathrm{Frac}(R)$$

such that for every field K and every ring embedding

$$f : R \to K,$$

there exists a unique field embedding

$$\tilde{f} : \mathrm{Frac}(R) \to K$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = \frac{a}{1}.$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $R \times (R \setminus \{0\})$ by

$$(a, b) \sim (c, d) \iff ad = bc.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in R \times (R \setminus \{0\})$
2	$ab = ba$
3	$(a, b) \sim (a, b)$

Symmetry

1	$(a, b) \sim (c, d)$
2	$ad = bc$
3	$cb = da$
4	$(c, d) \sim (a, b)$

Transitivity

1	$ad = bc$	
2	$cf = de$	
3	$adf = bcf$	multiply by f, 1
4	$bcf = bde$	substitute, 2
5	$adf = bde$	
6	$(af)d = (be)d$	rearrange
7	R is an integral domain	
8	$d \neq 0$	
9	$af = be$	cancellation (no zero divisors)
10	$(a, b) \sim (e, f)$	

—

Step 1: Define field of fractions

$$\mathrm{Frac}(R) := (R \times (R \setminus \{0\}))/\sim$$

$$[(a, b)] := \frac{a}{b}$$

—

Step 2: Define operations

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

—

Step 3: Define embedding

$$\eta(a) := [(a, 1)]$$

—

Step 4: Well-definedness

(Follows from cross-multiplication and compatibility of $\sim$.)

—

Step 5: Field structure

1 $[(a,b)] \cdot [(b,a)] = [(ab,ab)]$

2 $[(ab,ab)] = 1$

3 every nonzero element has inverse

4 $\mathrm{Frac}(R)$ is a field

—

Step 6: Extension map

Let $f : R \to K$ be a ring embedding into a field K.

$$\tilde{f}([(a,b)]) := \frac{f(a)}{f(b)}$$

—

Step 7: Well-definedness

1 $(a,b) \sim (c,d)$

2 $ad = bc$

3 $f(a)f(d) = f(b)f(c)$

4 $\frac{f(a)}{f(b)} = \frac{f(c)}{f(d)}$

—

Step 8: Homomorphism and factorization

$$\tilde{f}(\eta(a)) = \tilde{f}([(a,1)]) = f(a)$$

—

Step 9: Uniqueness

$$[(a,b)] = \frac{[(a,1)]}{[(b,1)]} \Rightarrow \tilde{f} \text{ is forced}$$

$\square$

Chapter 7

Natural numbers

7.1 Introduction to Natural numbers

definition 7.1.1. ***Natural numbers*** *we define "$0, 1, 2, 3, \ldots$" like this.*

$$\begin{aligned}
&0 := \varnothing \\
&1 := \{0\} = \varnothing \cup \{\varnothing\} = \{\varnothing\} \\
&2 := \{0, 1\} = \{\varnothing\} \cup \big\{\{\varnothing\}\big\} = \big\{\varnothing, \{\varnothing\}\big\} \\
&3 := \{0, 1, 2\} = \Big\{\varnothing, \{\varnothing\}, \big\{\varnothing, \{\varnothing\}\big\}\Big\} \\
&\vdots
\end{aligned}$$

definition 7.1.2. ***Successor function*** *is a function "$s : V \to V$" defined like this.*

$$s(x) := x \cup \{x\}$$

you can see $\forall x \quad Type_0(x) \to Type_0(s(x))$

definition 7.1.3. *we define a set* ***"Inductive"*** *like this.*

$$Ind(X) : \varnothing \in X \wedge \forall y (y \in X \to s(y) \in X)$$

notice that "$Ind(X)$" might be empty.

Axiom 7.1.1. ***The Axiom of Infinity*** *we define a set "Inductive" like this.*

$$\exists X \quad Type_0(X) \wedge Ind(X)$$

so "$Ind(X)$" is a nonempty set. so we can define

definition 7.1.4. ***Set of Natural numbers($\mathbb{N}$)***

$$\mathbb{N} := \bigcap Ind$$

Lemma 7.1.1. *prove.*

1. $\mathbb{N} = \{x \mid \underbrace{\forall Y\Big(\big(\underbrace{\varnothing \in Y \land \forall z(z \in Y \to s(z) \in Y)}_{Ind(Y)}\big) \to x \in Y\Big)}_{\phi(x)}\}$

2. $Ind(\mathbb{N})$
   Proof Is Left As An Exercise To The Reader.😀

notice that now "$\phi(x)$" is a proposition. so "$\{x \mid \forall Y\Big(\big(\varnothing \in Y \land \forall z(z \in Y \to s(z) \in Y)\big) \to x \in Y\Big)\}$" is a set.

7.2 Properties of Natural numbers

Theorem 7.2.1. ***Induction*** $P(0), \forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

1	$S := \{x \in \mathbb{N} \mid P(x)\}$	def
2	$P(0)$	ass
3	$\forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big)$	ass
4	$\quad n \in \mathbb{N}$	acp
5	$\quad 0 \in S$	$1, 2$
6	$\quad\quad n \in S$	acp
7	$\quad\quad P(n)$	$comprhension$
8	$\quad\quad P(n) \to P(s(n))$	$\forall E, \to E, 3, 4$
9	$\quad\quad P(s(n))$	$\to E, 7, 8$
10	$\quad\quad s(n) \in S$	$4, 9$
11	$\quad \forall n\Big(n \in S \to s(n) \in S\Big)$	$1, 3$
12	$\quad Ind(S)$	$5, 6$
13	$\quad n \in \bigcap Ind$	def
14	$\quad n \in S$	$\bigcap E$
15	$\quad P(n)$	$comprhension$
16	$\forall n \in \mathbb{N} \quad P(n)$	$\to I, \forall I$

definition 7.2.1. *A set "T" is* ***Transitive*** *if $u \in v \in T$ implies $u \in T$.*

$$Transitive(T) : \forall u, v \quad u \in v \in T \to u \in T$$

Lemma 7.2.1. *.*

1. $\forall n \in \mathbb{N} \quad Transitive(n)$
2. $Transitive(\mathbb{N})$ 😀

proof.

1	$\forall v \quad v \notin 0$	
2	$\forall u, v \quad u \in v \in 0 \to u \in 0$	
3	$Transitive(0)$	
4	$\mid\ n \in \mathbb{N}$	*acp*
5	$\mid\ \mid\ Transitive(n)$	*acp*
6	$\mid\ \mid\ \mid\ u \in v \wedge v \in s(n)$	*acp*
7	$\mid\ \mid\ \mid\ v \in n \cup \{n\}$	$\wedge E$
8	$\mid\ \mid\ \mid\ \mid\ v \in n$	$\wedge E$
9	$\mid\ \mid\ \mid\ \mid\ u \in n$	$5, 6, 8$
10	$\mid\ \mid\ \mid\ \mid\ u \in s(n)$	$\cup I$
11	$\mid\ \mid\ \mid\ \mid\ v \in \{n\}$	$\wedge E$
12	$\mid\ \mid\ \mid\ \mid\ v = n$	
13	$\mid\ \mid\ \mid\ \mid\ u \in n$	$= E$
14	$\mid\ \mid\ \mid\ \mid\ u \in s(n)$	$\cup I$
15	$\mid\ \mid\ \mid\ u \in s(n)$	
16	$\mid\ \mid\ \forall u, v \quad u \in v \in s(n) \to u \in s(n)$	
17	$\mid\ \mid\ Transitive(s(n))$	
18	$\forall n \in \mathbb{N} \quad Transitive(n) \to Transitive(s(n))$	
19	$\forall n \in \mathbb{N} \quad Transitive(n)$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

definition 7.2.2. ***Order in Natural numbers($\mathbb{N}$)***

$$<:= \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x \in y\}$$

Theorem 7.2.2. ***Order of Natural numbers($\mathbb{N}$)*** *prove*

1. $(\mathbb{N}, <)$ *is lineary ordered.* 😀
2. $\forall n \in \mathbb{N} \quad (n, <)$ *is Well ordered.*
3. $(\mathbb{N}, <)$ *is Well ordered.*

4. $\Big(\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x < y\Big)$ *(doesn't have right endpoint)*

proof.

1	$0 \not< 0$	$def\varnothing$
2	$\quad n \in \mathbb{N}$	acp
3	$\quad\quad n \not< n$	acp
4	$\quad\quad\quad s(n) < s(n)$	acp
5	$\quad\quad\quad\quad s(n) \in n$	acp
6	$\quad\quad\quad\quad n \in s(n)$	$\cup I$
7	$\quad\quad\quad\quad n \in n$	$Transitive(\mathbb{N})$
8	$\quad\quad\quad\quad \perp$	$3, 7$
9	$\quad\quad\quad\quad s(n) \in \{n\}$	acp
10	$\quad\quad\quad\quad s(n) = n$	
11	$\quad\quad\quad\quad s(n) \not< s(n)$	$= E$
12	$\quad\quad\quad\quad \perp$	$4, 11$
13	$\quad\quad\quad \perp$	$\cup E$
14	$\quad\quad s(n) \not< s(n)$	$\neg I$
15	$\forall n \in \mathbb{N} \quad n \not< n$	*Induction theorem*

1	$\forall n, m, k \in \mathbb{N} \quad n < m < k \to n < k$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

Theorem 7.2.3. ***Induction ,second vrsion*** $\Big(\forall k < n \quad P(k)\Big) \to P(n) / \therefore \forall n \in \mathbb{N} \quad P(n)$

proof.

7.3 The Recursion theorem

definition 7.3.1. ***Tuple (new definition)-Finite sequence*** *is a function "$a : n \to V$" s.t. $n \in \mathbb{N}$ we know we can show this by "$(a_i)_{i\in n}$" we can also show it by "$(a_0, a_1, a_2, ..., a_{n-1})$"*

definition 7.3.2. ***Sequence*** *is a function "$a : \mathbb{N} \to V$" we know we can show this by "$(a_i)_{i\in\mathbb{N}}$" we can also show it by "$(a_i)_{i=0}^{\infty}$"*

now we can redefine "Cartesian product". we define finite Cartesian product "$\prod_{i\in n} A_i := \{(a_i)_{i\in n} \mid a_i \in A_i\}$" and define infinite Cartesian product "$\prod_{i\in\mathbb{N}} A_i := \{(a_i)_{i\in\mathbb{N}} \mid a_i \in A_i\}$" and define Cartesian product in general way "$\prod_{i\in I} A_i := \{(a_i)_{i\in I} \mid a_i \in A_i\}$". and you can easily see that from the definition of "B^A", "A^n" is the set of all n-length finite sequences of $a \in A$ and "$A^{\mathbb{N}}$" is the set of all infinite sequences of $a \in A$.

Theorem 7.3.1 (Recursion)**.** *For any set A, any $a \in A$, and any function $g : A \times \mathbb{N} \to A$, there is a unique function $f : \mathbb{N} \to A$ such that:*

1. $f(0) = a$.

2. $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$.

Proof. Define

$$Compute_n(t_n) : t_n : s(n) \to A \ \wedge \ t_n(0) = a \ \wedge \ \forall k < n \ \big(t(s(k)) = g(t(k), k)\big).$$

$$F := \Big\{ t_n \ \Big| \ Compute_n(t_n) \Big\}.$$

Claim 0: $\forall n \in \mathbb{N} \quad \exists t_n \in F$**.**

Claim 1: $F \neq \varnothing$**.**

1	$t_0 \in F$
2	$F \neq \varnothing$

Define

$$f := \bigcup F.$$

Claim 2: f **is a function.**

1	$\vert\ t_m, t_n \in F$
2	$\vert\ \vert\ Dom(t_m) = s(m)$
3	$\vert\ \vert\ Dom(t_n) = s(n)$
4	$\vert\ \vert\ \vert\ m \leq n$
5	$\vert\ \vert\ \vert\ t_m(0) = a$
6	$\vert\ \vert\ \vert\ t_n(0) = a$
7	$\vert\ \vert\ \vert\ t_m(0) = t_n(0)$
8	$\vert\ \vert\ \vert\ \vert\ k < m$
9	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(k) = t_n(k)$
10	$\vert\ \vert\ \vert\ \vert\ \vert\ g(t_m(k), k) = g(t_n(k), k)$
11	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(s(k)) = g(t_m(k), k)$
12	$\vert\ \vert\ \vert\ \vert\ \vert\ t_n(s(k)) = g(t_n(k), k)$
13	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(s(k)) = t_n(s(k))$
14	$\vert\ \vert\ \vert\ \vert\ \forall k \leq m \quad t_m(k) = t_n(k)$
15	$\vert\ \vert\ t_m \subseteq t_n \vee t_n \subseteq t_m$
16	$\forall t_m, t_n \in F \quad (t_m \subseteq t_n \vee t_n \subseteq t_m)$
17	$Compatible(F)$
18	$Func(f)$

Claim 3: Dom(f) $= \mathbb{N}$ and Range(f) $\subseteq A$.

1	$\quad\vert\ n \in \mathbb{N}$
2	$\quad\vert\ \exists t_n \in F$
3	$\quad\vert\ n \in Dom(t_n)$
4	$\quad\vert\ n \in Dom(f)$
5	$\quad\vert\ n \in Dom(f)$
6	$\quad\vert\quad\vert\ t_m \in F$
7	$\quad\vert\quad\vert\ n \in Dom(t_m)$
8	$\quad\vert\quad\vert\ Dom(t_m) = s(m)$
9	$\quad\vert\quad\vert\ n \in s(m)$
10	$\quad\vert\quad\vert\ n \in \mathbb{N}$
11	$\quad\vert\ n \in \mathbb{N}$
12	$Dom(f) = \mathbb{N}$
13	$\quad\vert\ x \in Range(f)$
14	$\quad\vert\ (n, x) \in f$
15	$\quad\vert\quad\vert\ t_m \in F$
16	$\quad\vert\quad\vert\ (n, x) \in t$
17	$\quad\vert\quad\vert\ t_m : s(m) \to A$
18	$\quad\vert\quad\vert\ x \in A$
19	$\quad\vert\ x \in A$
20	$Range(f) \subseteq A$

Claim 4: f satisfies the recursion equations.

1	$f(0) = a$
2	$\quad\vert\ n \in \mathbb{N}$
3	$\quad\vert\ \exists t_{n+1} \in F$
4	$\quad\vert\ t_{n+1}(s(n)) = g(t_{n+1}(n), n)$
5	$\quad\vert\ t_{n+1} \subseteq f$
6	$\quad\vert\ f(s(n)) = g(f(n), n)$
7	$\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$

Claim 5 : Uniqueness.

1		$h : \mathbb{N} \to A$	
2		$h(0) = a$	
3		$\forall n \in \mathbb{N} \quad h(s(n)) = g(h(n), n)$	
4		$f(0) = h(0)$	
5			$f(n) = h(n)$
6			$g(f(n), n) = g(h(n), n)$
7			$f(s(n)) = g(f(n), n)$
8			$h(s(n)) = g(h(n), n)$
9			$f(s(n)) = h(s(n))$
10		$\forall n \in \mathbb{N} \quad f(n) = h(n)$	
11		$f = h$	
12	$\exists! f : \mathbb{N} \to A \quad \big(f(0) = a \land \forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)\big)$		

Therefore, there exists a unique function $f : \mathbb{N} \to A$ such that $f(0) = a$ and $\forall n \in \mathbb{N}\ f(s(n)) = g(f(n), n)$. $\square$

Theorem 7.3.2 (Parametric Recursion Theorem)**.** *For any sets "A, P", any function "$a : P \to A$", and any function "$g : P \times A \times \mathbb{N} \to A$", there is a unique function "$f : P \times \mathbb{N} \to A$" such that*

1.
$$\forall p \in P \quad f(p, 0) = a(p)$$

2.
$$\forall p \in P\ \forall n \in \mathbb{N} \quad f\big(p, s(n)\big) = g\big(p, f(p, n), n\big)$$

Proof. Fix $p \in P$.

Define

$$G_p(x, n) := g(p, x, n).$$

Claim 1 : $G_p : A \times \mathbb{N} \to A$.

1		$p \in P$	
2			$(x, n) \in A \times \mathbb{N}$
3			$g(p, x, n) \in A$
4			$G(p)(x, n) \in A$
5		$G(p) : A \times \mathbb{N} \to A$	
6	$\forall p \in P \quad G(p) : A \times \mathbb{N} \to A$		

For every $p \in P$, by the ordinary Recursion Theorem applied to $a(p) \in A$ and $G(p) : A \times \mathbb{N} \to A$, there exists a unique function

$$F_p : \mathbb{N} \to A$$

such that

$$F_p(0) = a(p)$$

and

$$\forall n \in \mathbb{N} \quad F_p(s(n)) = G(p)(F_p(n), n).$$

Define

$$f(p, n) := F_p(n).$$

Claim 2 : $f : P \times \mathbb{N} \to A$.

1		$(p, n) \in P \times \mathbb{N}$
2		$F_p : \mathbb{N} \to A$
3		$F_p(n) \in A$
4		$f(p, n) \in A$
5	$f : P \times \mathbb{N} \to A$	

Claim 3 : f satisfies the recursion equations.

1		$p \in P$
2		$F_p(0) = a(p)$
3		$f(p, 0) = a(p)$
4	$\forall p \in P \quad f(p, 0) = a(p)$	
5		$p \in P$
6		$n \in \mathbb{N}$
7		$F_p(s(n)) = G(p)(F_p(n), n)$
8		$G(p)(F_p(n), n) = g(p, F_p(n), n)$
9		$f(p, s(n)) = g(p, f(p, n), n)$
10	$\forall p \in P \; \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)$	

Define

$$H_p(n) := h(p, n).$$

Claim 4 : Uniqueness.

1	$\quad h : P \times \mathbb{N} \to A$
2	$\quad \forall p \in P \quad h(p, 0) = a(p)$
3	$\quad \forall p \in P\ \forall n \in \mathbb{N} \quad h(p, s(n)) = g(p, h(p, n), n)$
4	$\quad\quad p \in P$
5	$\quad\quad H_p(0) = a(p)$
6	$\quad\quad \forall n \in \mathbb{N} \quad H_p(s(n)) = g(p, H_p(n), n)$
7	$\quad\quad F_p = H_p$
8	$\quad\quad\quad n \in \mathbb{N}$
9	$\quad\quad\quad f(p, n) = F_p(n)$
10	$\quad\quad\quad H_p(n) = h(p, n)$
11	$\quad\quad\quad f(p, n) = h(p, n)$
12	$\quad\quad \forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
13	$\quad \forall p \in P\ \forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
14	$\quad f = h$
15	$\exists! f : P \times \mathbb{N} \to A \quad \Big(\forall p \in P \quad f(p, 0) = a(p) \land \forall p \in P \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)\Big)$

Therefore, there exists a unique function $f : P \times \mathbb{N} \to A$ such that

$$\forall p \in P \quad f(p, 0) = a(p)$$

and

$$\forall p \in P\ \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n).$$

$\square$

7.4 Arithmetic of Natural numbers

Theorem 7.4.1. ***Summation*** *there is a unique function "$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m + 0 = m$
2. $\forall m, n \in \mathbb{N} \quad m + s(n) = s(m + n)$

Proof Is Left As An Exercise To The Reader.
Theorem 7.4.2. ***Multiplication*** *there is a unique function "$\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m \times 0 = 0$
2. $\forall m, n \in \mathbb{N} \quad m \times s(n) = m \times n + m$

Proof Is Left As An Exercise To The Reader.

we denote also "Multiplication" with "$m \cdot n$" and "mn" too.

7.5 Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitary functions and constants and doesn't have meaning we defined before). with these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x, y \in N \quad x + s(y) = s(x + y)$
5. $\forall x \in N \quad x \cdot 0 = 0$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\phi(0) \wedge \forall y \in N \quad \Big(\phi(y) \to \phi(s(y))\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitary "ϕ")

we call "$\mathfrak{N}$" a model of "Peano arithmetic".
Lemma 7.5.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in N \quad x + 0 = x = 0 + x$ *(identity +)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative +)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) +)*

1	$\quad x, y, z \in N$	*acp*
2	$\quad (x + y) + 0 = x + y = x + (y + 0)$	*Ax. 3*
3	$\quad\quad (x + y) + z = x + (y + z)$	*acp*
4	$\quad\quad (x + y) + s(z) = s\big((x + y) + z\big) = s\big(x + (y + z)\big) = x + s(y + z) = x + (y + s(z))$	*Ax. 4* , 3
5	$\quad (x + y) + z = x + (y + z)$	*Ax. Ind*
6	$\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$	$\to I, \forall I$

1	$x \in N$	*acp*
2	$0 + 0 = 0$	*Ax. 3*
3	$\quad 0 + x = x$	*acp*
4	$\quad 0 + s(x) = s(0 + x) = s(x)$	*Ax. 3* , 4
5	$0 + x = x$	*Ax. Ind*
6	$\forall x \in N \quad x + 0 = 0 + x = x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$y + s(0) = s(y + 0) = s(y) = s(y) + 0$	*Ax. 3*
3	$\quad y + s(x) = s(y) + x$	*acp*
4	$\quad y + s(s(x)) = s(y + s(x)) = s(s(y) + x) = s(y) + s(x)$	*Ax. 4* , 3
5	$x + y = y + x$	*Ax. Ind*
6	$\forall x, y \in N \quad y + s(x) = s(y) + x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$x + 0 = 0 + x$	*Lemma 7.5.1.2*
3	$\quad x + y = y + x$	*acp*
4	$\quad x + s(y) = s(x + y) = s(y + x) = y + s(x) = s(y) + x$	*Ax. Ind*
5	$x + y = y + x$	*Ax. Ind*
6	$\forall x, y \in N \quad x + y = y + x$	$\to I, \forall I$

Proof Is Left As An Exercise To The Reader.😀

Lemma 7.5.2. *prove for "$(N, +, \cdot)$"*

1. $\forall x, y, z \in N \quad (xy)z = x(yz)$ *(assosiative $\times$)*

2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*

3. $\forall x, y \in N \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free $\times$)*

4. $\forall x, y \in N \quad xy = yx$ *(commutative(abelian) $\times$)*

5. $\forall x, y, z \in N \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😀

1	$x \in N$	*acp*
2	$0 \cdot 0 = 0$	*Ax. 5*
3	$\quad 0 \cdot x = 0$	*acp*
4	$\quad 0 \cdot s(x) = 0 \cdot x + 0 = 0$	
5	$0 \cdot x = 0$	*Ax. Ind*
6	$\forall x, y \in N \quad 0 \cdot x = 0$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$0 + y0 = s(y) \cdot 0$	*Ax. 5*
3	$\quad x + yx = s(y) \cdot x$	*acp*
4	$\quad s(x) + yx = x + s(yx) = s(x + yx) = s(s(y) \cdot x) = s(y) \cdot s(x)$	
5	$x + yx = s(y) \cdot x$	*Ax. Ind*
6	$\forall x, y \in N \quad x + yx = s(y) \cdot x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$x \cdot 0 = 0$	*Ax. 5*
3	$x \cdot 0 = 0 \cdot x$	*Lemma*
4	$\quad xy = yx$	*acp*
5	$\quad x \cdot s(y) = xy + x = yx + x = x + yx = s(y) \cdot x$	
6	$xy = yx$	*Ax. Ind*
7	$\forall x, y \in N \quad xy = yx$	$\to I, \forall I$

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall m, n \in \mathbb{N} \quad m < n \leftrightarrow \exists k \neq 0 \quad m + k = n$$

Lemma 7.5.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \land 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.😀

as you can see "$(\mathbb{N}, +, \times, s, 0)$" is a model of peano arithmetic. so properties are true for them.

Chapter 8

From Natural numbers to Complex numbers

8.1 Integers“$\mathbb{Z}$”

we learned about “$\mathbb{N}$” before and we constructed structure “$(\mathbb{N}, +, \cdot, <)$”. if you assume “$\mathbb{N} \times \mathbb{N}$” we can define an equivalence relation “$(n_1, m_1) \sim (n_2, m_2) : n_1 + m_2 = n_2 + m_1$”(check it's an equivalence relation).

definition 8.1.1. ***Integers“$\mathbb{Z}$”*** *is a set*

$$\mathbb{Z} := \mathbb{N} \times \mathbb{N} / \sim$$

we denote it's members with “$z_0, z_1, \ldots$”

in this system “$0 := [(0, 0)]$” and “$\forall n \in \mathbb{N} \quad n := [(n, 0)]$” and “$\forall n \in \mathbb{N} \quad -n := [(0, n)]$”. in general“$\forall m, n \in \mathbb{N} \quad m - n := [(m, n)]$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z}$”.

Theorem 8.1.1. ***Summation*** *there is a unique function “$+_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$” s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] +_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 + m_2, n_1 + n_2)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.1.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$” s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] \times_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 m_2 + n_1 n_2, m_1 n_2 + m_2 n_1)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.1.2. ***Order in $\mathbb{Z}$***

$$[(m_1, n_1)] <_{\mathbb{Z}} [(m_2, n_2)] : m_1 + n_2 < m_2 + n_1$$

this lemma let us use “$<$” instead of “$<_{\mathbb{Z}}$”, “$+$” instead of “$+_{\mathbb{Z}}$” and “$\times$” instead of “$\times_{\mathbb{Z}}$”.

Lemma 8.1.1. *prove.*

1. $/ \therefore \forall m, n \in \mathbb{N} \quad m <_{\mathbb{Z}} n \leftrightarrow m < n$
2. $/ \therefore \forall m, n \in \mathbb{N} \quad m +_{\mathbb{Z}} n = m + n$
3. $/ \therefore \forall m, n \in \mathbb{N} \quad m \times_{\mathbb{Z}} n = m \times n$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$z \cdot y$" and "zy" too.
Lemma 8.1.2. *prove for "$(\mathbb{Z}, +)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall z \in \mathbb{Z} \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Z} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃
Lemma 8.1.3. *prove for "$(\mathbb{Z}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (xy)z = x(yz)$ *(assosiative ×)*
2. $\forall z \in \mathbb{Z} \quad z \cdot 1 = 1 \cdot z = z$ *(identity ×)*
3. $\forall x, y \in \mathbb{Z} \quad xy = 0 \to x = 0 \lor y = 0$ *(zero divisor free ×)*
4. $\forall x, y \in \mathbb{Z} \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in \mathbb{Z} \quad \Big(x(y + z) = xy + xz\Big) \land \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃
Theorem 8.1.3. ***Order of Integers($\mathbb{Z}$)*** *prove for "$(\mathbb{Z}, +, \times, <_{\mathbb{Z}})$ "*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad x < y\Big) \land \Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z} \quad x < y \to x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z} \quad x < y \land 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.2 Rational numbers "$\mathbb{Q}$"

we learned about "$\mathbb{Z}$" before and we constructed structure "$(\mathbb{Z}, +, \cdot, <)$". if you assume "$\mathbb{Z} \times (\mathbb{Z} - \{0\})$" we can define an equivalence relation "$(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).
definition 8.2.1. ***Rational numbers"$\mathbb{Q}$"*** *is a set*

$$\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} - \{0\}) / \sim$$

we denote it's members with "$q_0, q_1, \ldots$"

in this system “$0 := [(0,1)]$” and “$1 := [(1,1)]$” and “$\forall x \in \mathbb{Z} \quad x := [(x,1)]$” and “$\forall x \in \mathbb{Z} \quad \frac{1}{x} := [(1,x)]$”. in general “$\forall x, y \in \mathbb{Z} \quad \frac{x}{y} := [(x,y)]$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$”.

Theorem 8.2.1. ***Summation*** *there is a unique function “$+_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$” s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] +_{\mathbb{Q}} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.2.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$” s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] \times_{\mathbb{Q}} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.2.2. ***Order in*** $\mathbb{Q}$.

if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use “$<$” instead of “$<_{\mathbb{Q}}$”, “$+$” instead of “$+_{\mathbb{Q}}$” and “$\times$” instead of “$\times_{\mathbb{Q}}$”.

Lemma 8.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z} \quad z <_{\mathbb{Q}} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z} \quad z +_{\mathbb{Q}} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z} \quad z \times_{\mathbb{Q}} y = z \times y$

Proof Is Left As An Exercise To The Reader.

we denote “Multiplication” with “$x \cdot y$” and “xy” too.

Lemma 8.2.2. *prove for “$(\mathbb{Q}, +)$”*

1. $\forall x, y, z \in \mathbb{Q} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in \mathbb{Q} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Q} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😀

Lemma 8.2.3. *prove for “$(\mathbb{Q}, +, \times)$”*

1. $\forall x, y, z \in \mathbb{Q} \quad (xy)z = x(yz)$ *(assosiative ×)*
2. $\forall x \in \mathbb{Q} \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x \in \mathbb{Q} - \{0\} \quad \exists y \in \mathbb{Q} \quad xy = yx = 1$ *(inverse ×)*
4. $\forall x, y \in \mathbb{Q} \quad xy = yx$ *(commutative(abelian) ×)*

5. $\forall x, y, z \in \mathbb{Q} \quad \Big(x(y+z) = xy + xz\Big) \land \Big((x+y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 8.2.3. ***Order of Rational numbers($\mathbb{Q}$)*** *prove for "$(\mathbb{Q}, <_{\mathbb{Q}})$"*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x < y\Big) \land \Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q} \quad \exists z \in \mathbb{Q} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q} \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q} \quad x < y \land 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.3 Real numbers "$\mathbb{R}$"

before we we get to lesson define function "$|\_| : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not fimiliar to this notation it means "$|\_| := \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid (0 \leq x \to y = x) \land (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}$" before and we constructed structure "$(\mathbb{Q}, +, \cdot, <)$". now assume $\mathbb{Q}^{\mathbb{N}}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}$). we don't want it all we just want a subset.

definition 8.3.1. ***Cauchy sequence*** *is a sequence like "$(a_i)_{i \in \mathbb{N}} \in \mathbb{Q}^{\mathbb{N}}$" s.t. elements of it get infinitly close to each other i.e.*

$$Cauchy(a_i)_{i \in \mathbb{N}} : \forall \epsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \epsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in \mathbb{N}} \sim (b_i)_{i \in \mathbb{N}} : \forall \epsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \epsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in \mathbb{N}}]$ with $[a_i]_{i \in \mathbb{N}}$.

1	$a_n - a_n = 0$	
2	$\lvert a_n - a_n\rvert < \epsilon$	
3	$n > N \to \lvert a_n - a_n\rvert < \epsilon$	
4	$\forall \epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - a_n\rvert < \epsilon$	$\forall I, \exists I, \forall I$
5	$(a_i)_{i \in \mathbb{N}} \sim (a_i)_{i \in \mathbb{N}}$	$def. \sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	*ass*
2	$\forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \epsilon$	$def. \sim$
3	$\forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - a_n\rvert < \epsilon$	
4	$(b_i)_{i\in\mathbb{N}} \sim (a_i)_{i\in\mathbb{N}}$	$def. \sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	*ass*
2	$(b_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	*ass*
3	$\forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \epsilon$	$def. \sim$
4	$\forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \epsilon$	$def. \sim$
5	$\exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\epsilon}{2}$	$def. \sim$
6	$\exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \frac{\epsilon}{2}$	$def. \sim$
7	$\vert\ \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\epsilon}{2}$	*acp*
8	$\vert\ \vert\ \forall n > N' \quad \lvert b_n - c_n\rvert < \frac{\epsilon}{2}$	*acp*
9	$\vert\ \vert\ \vert\ n > max\{N, N'\}$	*acp*
10	$\vert\ \vert\ \vert\ n > N \to \lvert a_n - b_n\rvert < \frac{\epsilon}{2}$	$\forall E$
11	$\vert\ \vert\ \vert\ n > N' \to \lvert b_n - c_n\rvert < \frac{\epsilon}{2}$	$\forall E$
12	$\vert\ \vert\ \vert\ \lvert a_n - b_n\rvert < \frac{\epsilon}{2}$	$\to E$
13	$\vert\ \vert\ \vert\ \lvert b_n - c_n\rvert < \frac{\epsilon}{2}$	$\to E$
14	$\vert\ \vert\ \vert\ \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert < \epsilon$	
15	$\vert\ \vert\ \vert\ \lvert a_n - c_n\rvert < \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert$	
16	$\vert\ \vert\ \vert\ \lvert a_n - c_n\rvert < \epsilon$	
17	$\vert\ \vert\ n > max\{N, N'\} \to \lvert a_n - c_n\rvert < \epsilon$	$\to I$
18	$\vert\ \vert\ \forall n > max\{N, N'\} \quad \lvert a_n - c_n\rvert < \epsilon$	$\forall I$
19	$\vert\ \vert\ \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \epsilon$	$\exists I$
20	$\exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \epsilon$	$\exists E$
21	$\forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \epsilon$	$\forall I$
22	$(a_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	$def. \sim$

definition 8.3.2. ***Real numbers "ℝ"*** *is a set*

$$\mathbb{R} := Cauchy/\sim$$

we denote it's members with "$r_0, r_1, ...$"

in this system "$0 := [0]_{i\in\mathbb{N}}$" and "$1 := [1]_{i\in\mathbb{N}}$" and "$\forall q \in \mathbb{Q} \quad q := [q]_{i\in\mathbb{N}}$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$".

Theorem 8.3.1. ***Summation*** *there is a unique function "$+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$" s.t.*

$$[a_i]_{i\in\mathbb{N}} +_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i + b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.3.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$" s.t.*

$$[a_i]_{i\in\mathbb{N}} \times_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.

definition 8.3.3. ***Order in*** $\mathbb{R}$ *.*

$$[a_i]_{i\in\mathbb{N}} <_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} : \exists\epsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \epsilon$$

this lemma let us use "$<$" instead of "$<_{\mathbb{R}}$", "$+$" instead of "$+_{\mathbb{R}}$" and "$\times$" instead of "$\times_{\mathbb{R}}$".

Lemma 8.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 8.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in \mathbb{R} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 8.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ *(assosiative ×)*
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ *(inverse ×)*
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in \mathbb{R} \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 8.3.3. ***Linear order of Real numbers(***$\mathbb{R}$***)*** *prove "*$(\mathbb{R}, <_{\mathbb{R}})$

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x\Big)$ *(doesn't have endpoints)*

3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R} \quad Uppreb(X, y)) \to (\exists y \in \mathbb{R} \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R} \quad x < y \to x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.4 Complex numbers“$\mathbb{C}$”

we learned about “$\mathbb{R}$” before and we constructed structure “$(\mathbb{R}, +, \cdot, <)$”.
definition 8.4.1. ***Complex numbers“$\mathbb{C}$”*** *is a set*

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}$$

we denote it's members with “$z_0, x, \ldots$”

in this system “$0 := (0, 0)$” and “$\forall r \in \mathbb{R} \quad r := (r, 0)$” . in general “$\forall x, y \in \mathbb{R} \quad x + iy := (x, y)$”.
so we can say “$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$”.
Theorem 8.4.1. ***Summation*** *there is a unique function “$+_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$” s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) +_{\mathbb{C}} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$” s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) \times_{\mathbb{C}} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.

this lemma let us use “$+$” instead of “$+_{\mathbb{C}}$” and “$\times$” instead of “$\times_{\mathbb{C}}$”.
Lemma 8.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R} \quad x +_{\mathbb{C}} y = x + y$
2. $/ \therefore \forall x, y \in \mathbb{R} \quad x \times_{\mathbb{C}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote “Multiplication” with “$z \cdot y$” and “zy” too.
Lemma 8.4.2. *prove for “$(\mathbb{C}, +)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (x + y) + z = x + (y + z)$ *(assosiative $+$)*
2. $\forall z \in \mathbb{C} \quad z + 0 = 0 + z = z$ *(identity $+$)*
3. $\forall z \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad z + y = y + z = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{C} \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.3. *prove for “$(\mathbb{C}, +, \times)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall z \in \mathbb{C} \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{C} \quad xy = yx$ *(commutative(abelian) $\times$)*
4. $\forall x, y, z \in \mathbb{C} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.3. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall n \in \mathbb{N} \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.4. *prove for "$(\mathbb{C}, +, \cdot, (\_)^{(\_)})$"*

1. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.4. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C} \times \mathbb{Z} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall m, n \in \mathbb{N} \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.5. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.5. ***Big Sum*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\sum_{i=0}^{(\_)}(\_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.6. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad k\sum_{i=0}^{n} a_n = \sum_{i=0}^{n} ka_n$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.6. ***Big Product*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\prod_{i=0}^{(\_)}(\_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.7. *prove for* “$(\mathbb{C}, +, \cdot)$”

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n}(a_n b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n}(a_n)^k$

Proof Is Left As An Exercise To The Reader.

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Chapter 9

Finite and Infinite sets

9.1 Equipotent and Order in sets

we talked about “Isomorphism” before now assume two Structures $\mathfrak{A} := (A)$ and $\mathfrak{B} := (B)$. because these Structures have only a set “Isomorphism” between them only means there is a bijection function $f : A \xrightarrow[surj]{inj} B$.

definition 9.1.1. ***Equipotent*** *in situation above we say “A” and “B” are “Equipotent”. we denote it with “$A \cong B$” i.e.*

$$A \cong B : \exists f(f : A \xrightarrow[surj]{inj} B)$$

Lemma 9.1.1. *“$\cong$” is a equivalence relation in “$(V, \cong)$”*

1	$id : X \xrightarrow[surj]{inj} X$	
2	$\exists f \quad f : X \xrightarrow[surj]{inj} X$	$\exists I$
3	$X \cong X$	$def.2$

1	$X \cong Y$	ass
2	$\exists f \quad f : X \xrightarrow[surj]{inj} Y$	$def.1$
3	$\quad\vert\ f : X \xrightarrow[surj]{inj} Y$	acp
4	$\quad\vert\ f^{-1} : Y \xrightarrow[surj]{inj} X$	$lemma$
5	$\quad\vert\ \exists f \quad f : Y \xrightarrow[surj]{inj} X$	$\exists I$
6	$\quad\vert\ Y \cong X$	$def.5$
7	$Y \cong X$	$\exists E$

1	$X \cong Y$	*ass*
2	$Y \cong Z$	*ass*
3	$\exists f \quad f : X \xrightarrow[surj]{inj} Y$	$def.1$
4	$\exists f \quad f : Y \xrightarrow[surj]{inj} Z$	$def.2$
5	$\quad f : X \xrightarrow[surj]{inj} Y$	*acp*
6	$\qquad g : Y \xrightarrow[surj]{inj} Z$	*acp*
7	$\qquad g \circ f : X \xrightarrow[surj]{inj} Z$	*lemma*
8	$\qquad \exists f \quad f : X \xrightarrow[surj]{inj} Z$	
9	$\qquad X \cong Z$	
10	$X \cong Z$	

so "$V/\cong$" is a partision for "V" and every partition equals to a $[A]_{\cong}$ we denote it by $[A]$.

definition 9.1.2. ***Embedding*** *if there is a function* $f : A \xrightarrow{inj} B$. *in situation above we say "A" is less than or equipotent "B". we denote it with "*$A \precsim B$*" i.e.*

$$A \precsim B : \exists f(f : A \xrightarrow{inj} B)$$

similarly we define "$[A] \preceq [B] : \forall X \in [A], Y \in [B] \quad X \precsim Y$"

Lemma 9.1.2. $A \precsim B \therefore / \therefore [A] \preceq [B]$

1	$A \precsim B$	
2	$\quad X \in [A]$	
3	$\quad Y \in [B]$	
4	$\quad X \cong A$	
5	$\quad Y \cong B$	
6	$\quad \exists f \quad f : X \xrightarrow[surj]{inj} A$	$def.4$
7	$\quad \exists f \quad f : A \xrightarrow{inj} B$	$def.1$
8	$\quad \exists f \quad f : B \xrightarrow[surj]{inj} Y$	$def.5$
9	$\quad\quad f : X \xrightarrow[surj]{inj} A$	acp
10	$\quad\quad\quad h : A \xrightarrow{inj} B$	acp
11	$\quad\quad\quad\quad g : B \xrightarrow[surj]{inj} Y$	acp
12	$\quad\quad\quad\quad g \circ h \circ f : X \xrightarrow{inj} Y$	$lemma$
13	$\quad\quad\quad\quad \exists f \quad f : X \xrightarrow{inj} Y$	$\exists I$
14	$\quad\quad\quad\quad X \precsim Y$	$def.13$
15	$\quad X \precsim Y$	$\exists E$
16	$[A] \preceq [B]$	

Theorem 9.1.1. ***Schroeder-Bernstein*** $X \precsim Y, Y \precsim X / \therefore X \cong Y$
Proof Is Left As An Exercise To The Reader.

$$h(x) := \begin{cases} f(g(y)) & \exists i \quad y \in A_i \\ y & \exists i \quad y \in B_i \end{cases}$$

$$C_0 := Y$$

$$C_{s(n)} := f \circ g[C_n]$$

$$D_0 := f[X]$$

$$D_{s(n)} := f \circ g[D_n]$$

$$A_n := C_n - D_n$$

$$B_n := D_n - C_s(n)$$

Claim 1: $\bigcup_{i \in \mathbb{N}} A_i \cap \bigcup_{i \in \mathbb{N}} B_i \neq \varnothing$

Claim 2: $Func(h)$

Claim 3: $injective(h)$

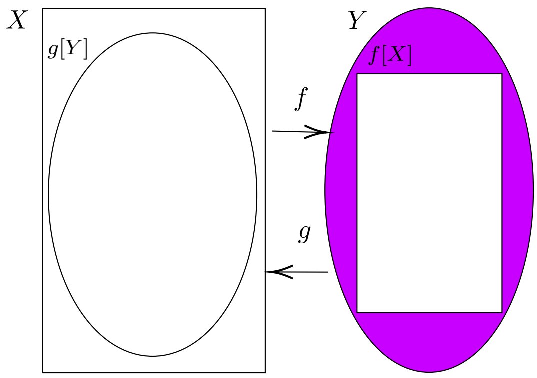
Claim 4: $h : Y \xrightarrow[surj]{inj} f[X]$

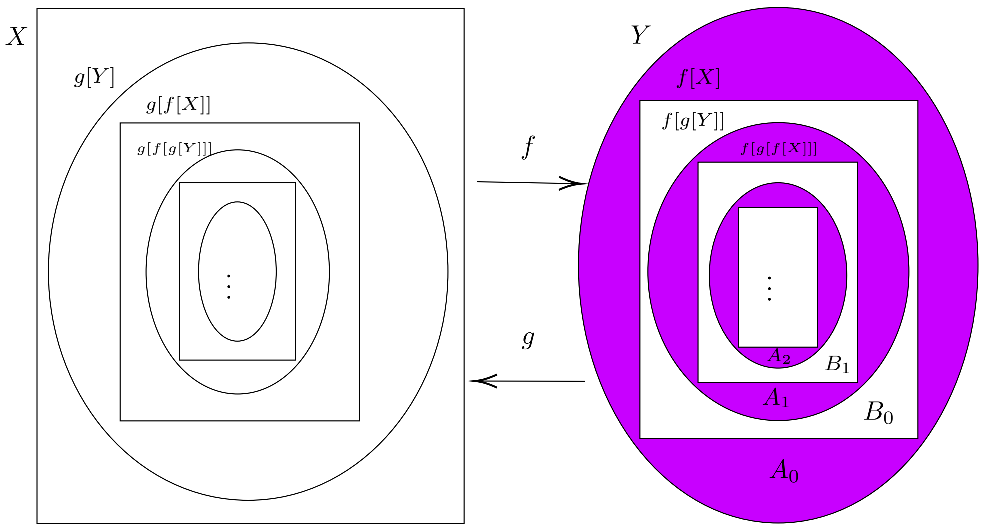
1 $h : Y \xrightarrow[surj]{inj} f[X]$

2 $Y \cong f[X]$

3 $X \cong f[X]$

4 $X \cong Y$





Lemma 9.1.3. *"$\preceq$" is a order relation in "$(V/\cong, \preceq)$"*
Proof Is Left As An Exercise To The Reader.😀

9.2 Finite sets

definition 9.2.1. *if there is a "$n \in \mathbb{N}$" s.t. $A \cong n$ we call it **"Finite"**$(Finite(A))$. if a set is not finite we call it **"Infinite"**$(Infinite(A))$.*
Lemma 9.2.1. $/ \therefore \forall n \in \mathbb{N} \quad X \subset n \to X \ncong n$

1	$\forall X \quad X \subset 0 \to X \ncong 0$	*vacuous truth*
2	$\forall X \quad X \subset n \to X \ncong n$	
3	$X \subset s(n)$	
4	$X \cong s(n)$	
5	$\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$	
6	$f : s(n) \xrightarrow[surj]{inj} X$	
7	$n \notin X$	
8	$X \subset n \cup \{n\}$	
9	$X \subseteq n$	
10	$X - \{f(n)\} \subset n$	
11	$X - \{f(n)\} \subset n \to X - \{f(n)\} \ncong n$	
12	$X - \{f(n)\} \ncong n$	
13	$f \restriction n : n \xrightarrow[surj]{inj} X - \{f(n)\}$	
14	$X - \{f(n)\} \cong n$	
15	$\perp$	
16	$n \in X$	
17	$n \in Dom(f)$	
18	$n \in Range(f)$	
19	$(k, n) \in f$	
20	$(n, m) \in f$	
21	$g := f \cup \{(k, m)\} - \{(k, n), (n, m)\}$	
22	$g : n \xrightarrow[surj]{inj} X - \{n\}$	
23	$X \subset n \cup \{n\}$	
24	$X - \{n\} \subset n$	
25	$X - \{n\} \subset n \to X - \{n\} \ncong n$	
26	$X - \{n\} \ncong n$	
27	$X - \{n\} \cong n$	
28	$\perp$	
29	$\perp$	
30	$X \ncong s(n)$	
31	$\forall X \quad X \subset s(n) \to X \ncong s(n)$	

Corollary 9.2.1. .

1. $\forall m, n \in \mathbb{N} \quad m \neq n \rightarrow m \ncong n$

2. $\forall m, n \in \mathbb{N} \quad A \cong m \wedge A \cong n \rightarrow m = n$

3. $Infinite(\mathbb{N})$

4. $\forall m, n \in \mathbb{N} \quad m \prec n \leftrightarrow m < n$

Proof Is Left As An Exercise To The Reader.😃

corollary (4) let us use “$<$” instead of “$\prec$”

Lemma 9.2.2. $Finite(X), Y \subseteq X / \therefore Finite(Y)$

1 $X \cong 0 \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

2 $X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

3 $X \cong s(n)$

4 $\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$

5 $f : s(n) \xrightarrow[surj]{inj} X$

6 $Y \subseteq X$

7 $g := f - \{(n, f(n))\}$

8 $g : n \xrightarrow[surj]{inj} X - \{f(n)\}$

9 $X - \{f(n)\} \cong n$

10 $Y - \{f(n)\} \subseteq X - \{f(n)\}$

11 $X - \{f(n)\} \cong n \wedge Y - \{f(n)\} \subseteq X - \{f(n)\} \to \exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

12 $\exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

13 $Y - \{f(n)\} \cong m$

14 $h : m \xrightarrow[surj]{inj} Y - \{f(n)\}$

15 $f(n) \in Y$

16 $Y = (Y - \{f(n)\}) \cup \{f(n)\}$

17 $h' := h \cup \{(m, f(n))\}$

18 $h' : s(m) \xrightarrow[surj]{inj} Y$

19 $s(m) \cong Y$

20 $\exists m \in \mathbb{N} \quad Y \cong m$

21 $f(n) \notin Y$

22 $Y = Y - \{f(n)\}$

23 $h : m \xrightarrow[surj]{inj} Y$

24 $m \cong Y$

25 $\exists m \in \mathbb{N} \quad Y \cong m$

26 $\exists m \in \mathbb{N} \quad Y \cong m$

27 $X \cong s(n) \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

28 $\forall n \in \mathbb{N}(X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m)$

29 $\exists n \in \mathbb{N} \quad X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

30 $Finite(X) \wedge Y \subseteq X \to Finite(Y)$

9.3 Countable sets

definition 9.3.1. *set "A" is* ***"Countable"*** *if "$A \cong \mathbb{N}$". set "A" is* ***"At most countable"*** *if "$A \precsim \mathbb{N}$"*
Lemma 9.3.1. $Countable(A), Infinite(B), B \subseteq A / \therefore Countable(B)$
Proof Is Left As An Exercise To The Reade.

Proof. Let A be countable and let $B \subseteq A$ be infinite.

Then there exists a bijective enumeration

$$f : \mathbb{N} \xrightarrow[surj]{inj} A.$$

Define recursively a sequence $(k_n)_{n\in\mathbb{N}}$ in $\mathbb{N}$:

$$k_0 := \min\{k \in \mathbb{N} : f(k) \in B\},$$
$$k_{n+1} := \min\{k \in \mathbb{N} : f(k) \in B \wedge f(k) \neq f(k_i)\ \forall i \leq n\}.$$

Define

$$b_n := f(k_n).$$

1	$b_n \in B$
2	$\quad\vert\ x \in B$
3	$\quad\vert\ \exists k \in \mathbb{N}\ f(k) = x$
4	$\quad\vert\ \exists n \in \mathbb{N}\ b_n = x$
5	$B \subseteq \{b_n : n \in \mathbb{N}\}$
6	$\{b_n : n \in \mathbb{N}\} \subseteq B$
7	$B = \{b_n : n \in \mathbb{N}\}$
8	$\forall m < n\ (b_m \neq b_n)$
9	$(b_n)_{n\in\mathbb{N}}$ injective
10	$\mathbb{N} \xrightarrow[surj]{inj} B$
11	$Countable(B)$

□

Corollary 9.3.1. *a set is at most countable iff it's finite or countable.*
Theorem 9.3.1. *The range of an infinite sequence $(a_n)_{n\in\mathbb{N}}$ is at most countable.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in X.

We construct recursively a sequence (b_n) such that $\text{Range}(b_n) = \text{Range}(a_n)$ and (b_n) is injective.

Define

$$b_0 := a_0.$$

Having defined $b_0, \dots, b_n$, define k_{n+1} as the least $k \in \mathbb{N}$ such that

$$a_k \neq b_i \quad \forall i \leq n.$$

If such k_{n+1} exists, set

$$b_{n+1} := a_{k_{n+1}}.$$

Otherwise stop the construction.

1	$b_n \in \text{Range}(a_n)$
2	$\quad \mid\ x \in \text{Range}(a_n)$
3	$\quad \mid\ \exists k \in \mathbb{N}\ (a_k = x)$
4	$\quad \mid\ \exists i\ (b_i = x)$
5	$\text{Range}(a_n) \subseteq \{b_i\}$
6	$\{b_i\} \subseteq \text{Range}(a_n)$
7	$\text{Range}(b_n) = \text{Range}(a_n)$
8	$\forall i < j\ (b_i \neq b_j)$
9	(b_n) injective
10	$\text{Range}(a_n)$ is image of an injective sequence
11	$\text{Range}(a_n)$ is at most countable

$\square$

Theorem 9.3.2. *The union of two countable sets is countable.*

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$ and $B = \{b_n : n \in \mathbb{N}\}$.

Define a sequence $(c_n)_{n\in\mathbb{N}}$ by

$$c_{2k} := a_k, \qquad c_{2k+1} := b_k.$$

1	$x \in A \cup B$
2	$\quad x \in A$
3	$\quad \exists k \in \mathbb{N}\ (a_k = x)$
4	$\quad \exists k \in \mathbb{N}\ (c_{2k} = x)$
5	$\quad \exists n \in \mathbb{N}\ (c_n = x)$
6	$\quad x \in B$
7	$\quad \exists k \in \mathbb{N}\ (b_k = x)$
8	$\quad \exists k \in \mathbb{N}\ (c_{2k+1} = x)$
9	$\quad \exists n \in \mathbb{N}\ (c_n = x)$
10	$\forall x \in A \cup B\ \exists n \in \mathbb{N}\ (c_n = x)$
11	$\{c_n : n \in \mathbb{N}\} = A \cup B$
12	$c : \mathbb{N} \xrightarrow[surj]{} A \cup B$
13	$AtMostCountable(A \cup B)$
14	$Infinite(A)$
15	$A \precsim A \cup B$
16	$Infinite(A \cup B)$
17	$Countable(A \cup B)$

□

Corollary 9.3.2. *Finite union of countable sets is countable.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1	$Countable(A_1)$
2	$\quad \forall k < n \quad Countable(A_k) \to Countable(\bigcup_{i \in n} A_i)$
3	$\quad\quad \forall k < s(n) \quad Countable(A_k)$
4	$\quad\quad Countable(A_n)$
5	$\quad\quad Countable(\bigcup_{i \in s(n)} A_i)$
6	$Countable(\bigcup_{i \in s(n)} A_i)$

□

Theorem 9.3.3. *If A and B are countable, then $A \times B$ is countable.*

Proof. It suffices to show $\mathbb{N} \times \mathbb{N}$ is countable.

Define

$$f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

by

$$f(m,n) := \begin{cases} n^2 + m & \text{if } m \leq n, \\ m^2 + 2m - n & \text{if } n < m. \end{cases}$$

1 $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$

2 $f(m_1, n_1) = f(m_2, n_2)$

3 $m \leq n \to n^2 = f(0, n) \leq f(m, n) \leq f(n, n) \leq f(n, m) < f(0, n + 1) = (n + 1)^2$

4 $k = max\{m, n\} \to k^2 \leq f(m, n) < (k + 1)^2$

5 $k' = max\{m', n'\} \to k'^2 \leq f(m', n') < (k' + 1)^2$

6 $f(m, n) = f(m', n')$

7 $k = k'$

> 8 $m \leq n = k \wedge m' \leq n' = k'$
>
> 9 $n = n'$
>
> 10 $n^2 + m = n'^2 + m'$
>
> 11 $n^2 + m = n^2 + m'$
>
> 12 $m = m'$

> 13 $n < m = k \wedge n' < m' = k'$
>
> 14 $m = m'$
>
> 15 $m^2 + 2m - n = m'^2 + 2m' - n'$
>
> 16 $m^2 + 2m - n = m^2 + 2m - n'$
>
> 17 $n = n'$

> 18 $n < m = k \wedge n' = k' > m'$
>
> 19 $m = n'$
>
> 20 $f(m, n) > f(m, m)$
>
> 21 $f(n', n') > f(m', n')$
>
> 22 $f(n', n') = f(m, m)$
>
> 23 $f(m, n) > f(m', n')$
>
> 24 $\perp$

25 $(m_1, n_1) = (m_2, n_2)$

26 $\text{f}isinjective$

> 27 $(m, n) \in \mathbb{N} \times \mathbb{N}$
>
> 28 $\exists k \in \mathbb{N}\ (f(k) = (m, n))$
>
> 29 $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$

30 $Countable(\mathbb{N} \times \mathbb{N})$

□

Corollary 9.3.3. *Finite Cartesian product of countable sets is countable, and A^n is countable for any countable A and $n > 0$.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1 $Countable(A_1)$

2 $\quad\vert\ \forall k < n \quad Countable(A_k) \to Countable(\prod_{i \in n} A_i)$

3 $\quad\vert\quad\vert\ \forall k < s(n) \quad Countable(A_k)$

4 $\quad\vert\quad\vert\ Countable(A_n)$

5 $\quad\vert\quad\vert\ Countable(\prod_{i \in s(n)} A_i)$

6 $Countable(\prod_{i \in s(n)} A_i)$

□

Lemma 9.3.2. *If "$(A_i)_{i\in\mathbb{N}}$" is family of countable sets with a family of sequences "$(a_i)_{i\in\mathbb{N}}$" s.t. "$\forall n \in \mathbb{N} \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} A_n$" then "$\bigcup_{i\in\mathbb{N}} A_i$" is countable.*
Proof Is Left As An Exercise To The Reader.😀

Proof. Define

$$h : \mathbb{N} \times \mathbb{N} \to \bigcup_{i\in\mathbb{N}} A_i \quad \text{by} \quad h(i, n) = a_i(n).$$

1 $x \in \bigcup_{i\in\mathbb{N}} A_i$

2 $\quad\vert\ \exists i \in \mathbb{N}\ (x \in A_i)$

3 $\quad\vert\ \exists n \in \mathbb{N}\ a_i(n) = x$

4 $\quad\vert\ h(i, n) = x$

5 $h : \mathbb{N} \times \mathbb{N} \xrightarrow[surj]{inj} \bigcup_{i\in\mathbb{N}} A_i$

6 $Countable(\mathbb{N} \times \mathbb{N})$

7 $Countable(\bigcup_{i\in\mathbb{N}} A_i)$

□

Theorem 9.3.4. *If A is countable, then $Seq(A)$, the set of all finite sequences of elements of A, is countable.*

Proof. It suffices to prove the theorem for $A = \mathbb{N}$.

Since

$$Seq(\mathbb{N}) = \bigcup_{n\in\mathbb{N}} \mathbb{N}^n,$$

it is enough to construct, for each $n \geq 1$, an enumeration

$$a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n.$$

Let

$$g : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N} \times \mathbb{N}.$$

Define recursively:

$$a_1(i) := (i),$$

and

$$a_{n+1}(i) := (b_0, \ldots, b_{n-1}, i_2),$$

where

$$g(i) = (i_1, i_2)$$

and

$$a_n(i_1) = (b_0, \ldots, b_{n-1}).$$

1 $a_1 : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}$

2 | $a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$

3 | | $(b_0, \ldots, b_n) \in \mathbb{N}^{n+1}$

4 | | $\exists i_1 \in \mathbb{N} \quad a_n(i_1) = (b_0, \ldots, b_{n-1})$

5 | | $\exists i \in \mathbb{N} \quad g(i) = (i_1, b_n)$

6 | | $a_{n+1}(i) = (b_0, \ldots, b_n)$

7 | $a_{n+1} : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^{n+1}$

8 $\forall n \geq 1 \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$

9 $\forall n \in \mathbb{N} \quad \mathbb{N}^n$ is countable

10 $\bigcup_{n \in \mathbb{N}} \mathbb{N}^n$ is countable

11 $Seq(\mathbb{N})$ is countable

Since every countable set is equipotent to a subset of $\mathbb{N}$,

$$Seq(A)$$

is countable.

$\square$

Corollary 9.3.4. *The set of all finite subsets of a countable set is countable.*

Proof. Let A be countable and define

$$\mathcal{F}(A) := \{F \subseteq A : F \text{ finite}\}.$$

For each $n \in \mathbb{N}$ define

$$\mathcal{F}_n(A) := \{F \subseteq A : |F| = n\}.$$

1 A^n is countable

2 $\{(a_1, \ldots, a_n) \in A^n : a_i \text{ distinct}\}$ is countable

3 $\mathcal{F}_n(A)$ is countable

4 | $F \in \mathcal{F}(A)$

5 | $\exists n \in \mathbb{N}\ (F \in \mathcal{F}_n(A))$

6 $\mathcal{F}(A) = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n(A)$

7 $\forall n\ \mathcal{F}_n(A)$ is countable

8 $\bigcup_{n \in \mathbb{N}} \mathcal{F}_n(A)$ is countable

9 $\mathcal{F}(A)$ is countable

$\square$

Corollary 9.3.5. *The set of all finite subsets of a countable set is countable.*

Proof. Let

$$\mathcal{F}(A) := \{X \subseteq A : X \text{ is finite}\}.$$

Define

$$F : Seq(A) \to \mathcal{F}(A)$$

by

$$F((a_0, \ldots, a_{n-1})) = \{a_0, \ldots, a_{n-1}\}.$$

1 $Seq(A)$ is countable

2 $\quad | \; X \in \mathcal{F}(A)$

3 $\quad | \; X = \{a_0, \ldots, a_{n-1}\}$

4 $\quad | \; F((a_0, \ldots, a_{n-1})) = X$

5 $F : Seq(A) \xrightarrow[surj]{} \mathcal{F}(A)$

6 $\mathcal{F}(A)$ is at most countable

7 $\mathcal{F}(A)$ is countable

□

Theorem 9.3.5. *An equivalence relation on a countable set has at most countably many equivalence classes.*

Proof. Let $\sim$ be an equivalence relation on a countable set A.

Define

$$F : A \to A/\sim$$

by

$$F(a) := [a]_\sim$$

1 A is countable

2 $\quad | \; C \in A/\sim$

3 $\quad | \; \exists a \in A \quad C = [a]_E$

4 $\quad | \; F(a) = C$

5 $F : A \xrightarrow[surj]{} A/\sim$

6 $A/\sim$ is at most countable

□

Corollary 9.3.6. *The set of all integers $\mathbb{Z}$ and the set of all rational numbers $\mathbb{Q}$ are countable.*

Proof. **$\mathbb{Z}$ is countable.**

By definition

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N}/\sim .$$

1 $\mathbb{N} \times \mathbb{N}$ is countable

2 $\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim$

3 $\mathbb{Z}$ is at most countable

4 $\mathbb{Z}$ is infinite

5 $\mathbb{Z}$ is countable

$\mathbb{Q}$ is countable.

By definition

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim .$$

1 $\mathbb{Z}$ is countable

2 $\mathbb{Z} - \{0\}$ is countable

3 $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ is countable

4 $F : \mathbb{Z} \times (\mathbb{Z} - \{0\}) \to \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim$

5 $\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} - \{0\}))/\sim$

6 $\mathbb{Q}$ is at most countable

7 $\mathbb{Q}$ is infinite

8 $\mathbb{Q}$ is countable

$\square$

9.4 Uncountable sets

Lemma 9.4.1. $/ \therefore \mathbb{N} \prec 2^{\mathbb{N}}$

Proof. **Claim 0:** $\mathbb{N} \precsim 2^{\mathbb{N}}$

Define

$$f : \mathbb{N} \to 2^{\mathbb{N}}$$

by

$$f(n)(k) = \begin{cases} 1 & k = n, \\ 0 & k \neq n. \end{cases}$$

1	$\quad\vert\; f(m) = f(n)$
2	$\quad\vert\quad\vert\; m \neq n$
3	$\quad\vert\quad\vert\; f(m)(m) = 1 \neq 0 = f(n)(m)$
4	$\quad\vert\quad\vert\; \perp$
5	$\quad\vert\; m = n$
6	$f : \mathbb{N} \xrightarrow{inj} 2^{\mathbb{N}}$
7	$\mathbb{N} \precsim 2^{\mathbb{N}}$

Claim 1: $\mathbb{N} \prec 2^{\mathbb{N}}$

1	$\quad\vert\; g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$
2	$\quad\vert\; d(n) := 1 - g(n)(n)$
3	$\quad\vert\; d \in 2^{\mathbb{N}}$
4	$\quad\vert\; \exists k \in \mathbb{N} \quad g(k) = d$
5	$\quad\vert\; d(k) = g(k)(k)$
6	$\quad\vert\; d(k) = 1 - g(k)(k)$
7	$\quad\vert\; g(k)(k) = 1 - g(k)(k)$
8	$\quad\vert\; \perp$
9	$\nexists g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$
10	$\mathbb{N} \ncong 2^{\mathbb{N}}$
11	$\mathbb{N} \prec 2^{\mathbb{N}}$

□

Lemma 9.4.2. $/ \therefore 2^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N})$
Proof Is Left As An Exercise To The Reader.😃
Theorem 9.4.1.

$$\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$$

Proof. For each

$$S \subseteq \mathbb{N}$$

define its characteristic function

$$\chi_S : \mathbb{N} \to 2$$

by

$$\chi_S(n) := \begin{cases} 1 & n \in S, \\ 0 & n \notin S. \end{cases}$$

Define

$$\chi_{()} : \mathcal{P}(\mathbb{N}) \to 2^{\mathbb{N}}$$

1 | $\chi_S = \chi_T$
2 | $\forall n \in \mathbb{N} \quad (n \in S \leftrightarrow n \in T)$
3 | $S = T$
4 $\chi_{()}$ is injective
5 | $f \in 2^{\mathbb{N}}$
6 | $S := \{n \in \mathbb{N} : f(n) = 1\}$
7 | $\chi_{(S)} = f$
8 $\chi_{()} : \mathcal{P}(\mathbb{N}) \xrightarrow[surj]{inj} 2^{\mathbb{N}}$
9 $\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$

$\square$

Lemma 9.4.3. $/ \therefore 2^{\mathbb{N}} \cong \mathbb{R}$
Proof Is Left As An Exercise To The Reader.😃
Theorem 9.4.2. ***Cantor theorem*** $/ \therefore \forall X \quad Type_0(X) \to X \prec 2^X$
Proof Is Left As An Exercise To The Reader.😃
Theorem 9.4.3. $/ \therefore \forall X \quad Type_0(X) \to 2^X \cong \mathcal{P}(X)$
Proof Is Left As An Exercise To The Reader.😃

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Chapter 10

Ordinals

10.1 Well ordered sets

definition 10.1.1. *if "$(W, <)$" is well ordered and $a \in W$ **"Initial segment"** is*

$$W[a] := \{x \in W \mid x < a\}$$

before get to next lemma Let's have a drfinition. assume "$(W_1, <_1)$" and "$(W_2, <_2)$" are two ordered structures function "f" is increasing when "$f : W_1 \to W_2$" is a (weak) homomorphism.
Lemma 10.1.1. *if "$(W, <)$" is well ordered and $f : W \to W$ is increasing then $\forall x \in W \quad x \leq f(x)$. Proof Is Left As An Exercise To The Reader.*😃
Corollary 10.1.1. *.*

1. *no well ordered structure is isomorph to it's initial segment of itself.*
2. *each well orderd structure has only one automorphism, the identity.*
3. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two isomorphic well ordered structures then the isomorphism is unique.*

Proof Is Left As An Exercise To The Reader.😃
Theorem 10.1.1. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two well ordered structures then exactly one of the following holds.*

1. $(W_1, <_1) \cong (W_2, <_2)$
2. $\exists a \in W_1 \quad (W_1[a], <_1) \cong (W_2, <_2)$
3. $\exists a \in W_2 \quad (W_1, <_1) \cong (W_2[a], <_2)$

Proof Is Left As An Exercise To The Reader.😃

Proof. Let $(W_1, <_1)$ and $(W_2, <_2)$ be well ordered sets.

Define

$$f \subseteq W_1 \times W_2 \quad \text{by} \quad (x, y) \in f \iff W_1[x] \cong W_2[y].$$

Claim 1: f is a function.

1	$(x, y) \in f \wedge (x, y') \in f$	assumption
2	$W_2[y] \cong W_2[y']$	definition of f
3	$y = y'$	uniqueness of isomorphism of well orders

Claim 2: f is injective.

1	$(x, y) \in f \wedge (x', y) \in f$	assumption
2	$W_1[x] \cong W_1[x']$	definition of f
3	$x = x'$	uniqueness of isomorphism of well orders

Claim 3: f is increasing.

1	$x <_1 x'$	assumption
2	$W_1[x'] \cong W_2[f(x')]$	definition of f
3	$W_1[x] \cong W_2[f(x)]$	definition of f
4	$f(x) <_2 f(x')$	comparison of initial segments

Claim 4: f is isomorphism between $S = \operatorname{dom}(f)$ and $T = \operatorname{ran}(f)$.

1	$f : S \to T$ bijective	Claims 1,2
2	$x <_1 x' \Rightarrow f(x) <_2 f(x')$	Claim 3
3	f order isomorphism	definition

Claim 5: S initial segment of W_1.

1	$x \in S \wedge z <_1 x$	assumption
2	$W_1[x] \cong W_2[f(x)]$	definition of S
3	$z \in S$	restriction of initial segments

Claim 6: T initial segment of W_2 or $T = W_2$.

1	$y \in T$	definition of T
2	$T \neq W_2$	assumption of case split
3	$T = W_2[b] \ \vee \ T = W_2$	well-order trichotomy

Claim 7: exclusion of mixed case.

1	$S = W_1[a] \wedge T = W_2[b]$	assumption
2	$f : W_1[a] \cong W_2[b]$	restriction of f
3	$(a, b) \in f$	definition of domain/range
4	$a \in W_1[a]$	initial segment property
5	$a < a$	element characterization
6	$\perp$	irreflexivity

Conclusion:

$$(W_1 \cong W_2) \ \vee \ (W_1[a] \cong W_2) \ \vee \ (W_1 \cong W_2[a]).$$

□

Lemma 10.1.2. *Every element of an ordinal number is an ordinal number.*

10.2 Ordinal numbers

definition 10.2.1. ***Order in sets***

$$<_A := \{(x, y) \in A \times A \mid x \in y\}$$

definition 10.2.2. *A set "α" is a* ***Ordinal*** *if*

1. *α is transitive.*
2. *$(\alpha, <_\alpha)$ is well ordered.*

we denote it by Ord so we have the set of ordinals(Ord).
Lemma 10.2.1. *Every natural number is an ordinal.*

Proof. Let $\mathbb{N}$ be the set of natural numbers.

1	$\forall n \in \mathbb{N}\ (n \subseteq \mathbb{N})$	von Neumann construction
2	$Transitive(\mathbb{N})$	definition of transitive set
3	$<_{\mathbb{N}} = \in \cap (\mathbb{N} \times \mathbb{N})$	definition
4	$(\mathbb{N}, <_{\mathbb{N}})$ is well ordered	standard well-order of $\mathbb{N}$
5	$Ordinal(\mathbb{N})$	definition of ordinal

Hence $\mathbb{N}$ is an ordinal.

□

In ordinal numbers we show "$\mathbb{N}$" with "ω". so it's just a renaming.
Lemma 10.2.2. *if α is an ordinal then $s(\alpha)$ is an ordinal too.*
Proof Is Left As An Exercise To The Reader.😀

Proof.

$$s(\alpha) = \alpha \cup \{\alpha\}.$$

1	α transitive	ordinal assumption
2	$s(\alpha) = \alpha \cup \{\alpha\}$	definition
3	$x \in y \wedge y \in s(\alpha) \Rightarrow x \in s(\alpha)$	case split on $y \in \alpha \cup \{\alpha\}$
4	$Transitive(s(\alpha))$	definition of transitive set
5	$<_\alpha = \in \cap (\alpha \times \alpha)$	definition
6	α is greatest element of $s(\alpha)$	construction
7	$<_{s(\alpha)}$ extends $<_\alpha$	restriction
8	$(s(\alpha), <_{s(\alpha)})$ is well ordered	extension by maximum element
9	$Ordinal(s(\alpha))$	definition of ordinal

Hence $s(\alpha)$ is an ordinal number.

□

Lemma 10.2.3. *prove*

1. $Transitive(Ord)$.
2. $\forall \alpha, \beta \in Ord \quad \alpha \subset \beta \to \alpha \in \beta$.

Proof. **(1):** $Transitive(Ord)$**.**

Claim 1: x **is transitive.**

1	$u \in v \wedge v \in x$	
2	$x \in \alpha$	assumption
3	$v \in \alpha$	transitivity of α
4	$u \in \alpha$	transitivity of α
5	$u, v, x \in \alpha$	closure
6	$u <_\alpha v <_\alpha x$	definition
7	$u <_\alpha x$	linearity of $<_\alpha$

Hence $Transitive(x)$.

Claim 2: $(x, <_x)$ **is well ordered.**

1	$x \subseteq \alpha$	transitivity of α
2	$<_x = <_\alpha \restriction x$	definition
3	$<_\alpha$ well order on α	ordinal assumption
4	$<_x$ well order on x	restriction of well-order

Thus x is transitive and well ordered, hence

$$x \in Ord.$$

1	$\alpha \in Ord$
2	$x \in y \wedge y \in \alpha$
3	α transitive
4	$x \in \alpha$

Hence $Transitive(\alpha)$ for all $\alpha \in Ord$, so $Transitive(Ord)$.

(2): $\alpha \subset \beta \Rightarrow \alpha \in \beta$**.**

Let $\alpha, \beta \in Ord$ and $\alpha \subset \beta$.

1	$\alpha \subset \beta$	assumption
2	$\beta - \alpha \neq \varnothing$	set difference
3	$\exists \gamma \in \beta - \alpha \ \forall \delta \in \beta - \alpha \ (\gamma \leq \delta)$	well-order of β
4	$\gamma \in \beta \wedge \gamma \notin \alpha$	definition of set difference
5	$\forall \delta \in \alpha \ (\delta < \gamma)$	minimality of γ
6	$\alpha \subseteq \gamma$	transitivity of $<$
7	$\gamma \in \beta$	step 8
8	$\alpha = \gamma$	ordinal comparison lemma ($\alpha, \gamma \in Ord$)
9	$\alpha \in \beta$	since $\gamma \in \beta$ and $\alpha = \gamma$

Hence

$$\alpha \subset \beta \Rightarrow \alpha \in \beta.$$

□

Theorem 10.2.1. ***Order of Ordinals*** *prove*

1. $(Ord, <)$ *is lineary ordered.*
2. $(Ord, <)$ *is Well ordered.*

Proof Is Left As An Exercise To The Reader.😃

Theorem 10.2.2. ***Order of Ordinals.*** *prove*

1. *$(Ord, <)$ is linearly ordered.*
2. *$(Ord, <)$ is well ordered.*

Proof. **Claim 1: $(Ord, <)$ is linear.**

1	$\alpha \in \beta$	assumption
2	$\beta \in \gamma$	assumption
3	γ transitive	$\gamma \in Ord$
4	$\alpha \in \gamma$	transitivity of γ

1	$\quad\vert\ \alpha \in \alpha$	assumption
2	$\quad\vert\ (\alpha, <_\alpha)$ is linearly ordered	α is an ordinal
3	$\quad\vert\ x := \alpha$	definition
4	$\quad\vert\ x \in x$	1,3
5	$\quad\vert\ \perp$	asymmetry of $<_\alpha$
6	$\alpha \notin \alpha$	

Let $\alpha, \beta, \gamma \in Ord$.

1	$\alpha \subseteq \beta \wedge \beta \subseteq \gamma$	assumption
2	$\alpha \subseteq \gamma$	transitivity of $\subseteq$
3	$\alpha < \gamma$	Lemma 2.6(a)
4	$\alpha \not< \beta \wedge \beta \not< \alpha \Rightarrow \alpha = \beta$	Lemma 2.6(b),(c)
5	$\alpha < \beta \ \vee \ \alpha = \beta \ \vee \ \beta < \alpha$	Lemma 2.6(c)

Hence $(Ord, <)$ is linearly ordered.

Claim 2: $(Ord, <)$ is well ordered.

Let $X \subseteq Ord$, $X \neq \varnothing$.

1	$X \neq \varnothing$	assumption
2	$\exists \alpha \in X$	nonempty
3	$\exists \beta \in X\ \forall \gamma \in X\ (\beta \leq \gamma)$	Lemma 2.6(d)
4	β is $<$-least element of X	definition

Hence every nonempty set of ordinals has a least element.

Conclusion:

$$(Ord, <) \text{ is linearly ordered and well ordered.}$$

□

Theorem 10.2.3. *The natural numbers are exactly the finite ordinal numbers.*

Proof. **Claim 1: Every natural number is a finite ordinal.**

1	$n \in \mathbb{N}$	
2	$Ordinal(n)$	Theorem 2.3
3	$Finite(n)$	definition of $\mathbb{N}$

Claim 2: Every ordinal that is not a natural number is infinite.

1	$Ordinal(\alpha) \wedge \alpha \notin \mathbb{N}$	assumption
2	$\alpha \neq \omega$	$\alpha \notin \mathbb{N}$
3	$\alpha < \omega \ \vee \ \alpha = \omega \ \vee \ \omega < \alpha$	Theorem 2.6(c)
4	$\omega < \alpha$	Theorem 2.6(b),5,6
5	$\omega \in \alpha$	definition of $<$
6	$\omega \subseteq \alpha$	transitivity of α
7	$Infinite(\omega)$	definition
8	$Infinite(\alpha)$	$\omega \subseteq \alpha$

Hence every finite ordinal is a natural number.

□

Theorem 10.2.4. *Every well ordered structure is isomorphic to a unique ordinal.*

Proof. Let $(W, <)$ be a well ordered structure.
Define

$$A := \{a \in W \mid \exists \alpha \in Ord \ ((W[a], <) \cong (\alpha, <))\}.$$

For each $a \in A$, let α_a be the unique ordinal such that

$$(W[a], <) \cong (\alpha_a, <).$$

Define

$$S := \{\alpha_a \mid a \in A\}.$$

Claim 1: $Transitive(S)$**.**

1	$\alpha_a \in S$	
2	$\gamma \in \alpha_a$	
3	$h : (W[a], <) \cong (\alpha_a, <)$	definition of α_a
4	$c := h^{-1}(\gamma)$	
5	$(W[c], <) \cong (\gamma, <)$	restriction of h
6	$c \in A$	definition of A
7	$\gamma = \alpha_c$	
8	$\gamma \in S$	

Hence $Transitive(S)$.

Claim 2: S **is an ordinal.**

1	$S \subseteq Ord$	definition of S
2	$(S, \in)$ is well ordered	Order of Ordinals
3	$Transitive(S)$	Claim 1
4	$Ordinal(S)$	definition of ordinal

Let

$$S = \alpha.$$

Claim 3: $b < a \wedge a \in A \Rightarrow b \in A$**.**

1	$a \in A \wedge b < a$	
2	$h : (W[a], <) \cong (\alpha_a, <)$	definition of A
3	$h[W[b]]$ is an initial segment of α_a	
4	$\exists \beta \in \alpha_a \quad h[W[b]] = \beta$	initial segments of ordinals
5	$(W[b], <) \cong (\beta, <)$	
6	$b \in A$	definition of A
7	$\alpha_b = \beta$	
8	$\alpha_b \in \alpha_a$	

Claim 4: $A = W$ **or** $A = W[c]$ **for some** $c \in W$**.**

1	$A \subseteq W$	
2	$\forall a \in A\ \forall b < a\ (b \in A)$	Claim 3
3	A is an initial segment of W	
4	$A = W\ \vee\ \exists c \in W\ (A = W[c])$	Lemma 1.2

Define

$$f : A \to \alpha$$

by

$$f(a) := \alpha_a.$$

Claim 5: f is an isomorphism.

1	$a, b \in A$	
2	$a < b \Rightarrow \alpha_a \in \alpha_b$	Claim 3
3	$a < b \Rightarrow f(a) < f(b)$	
4	f injective	uniqueness of ordinals
5	$Range(f) = \alpha$	definition of $\alpha = S$
6	$f : (A, <) \cong (\alpha, <)$	

Claim 6: $A = W$.

1	$\quad\vert\ A = W[c]$	
2	$\quad\vert\ f : (W[c], <) \cong (\alpha, <)$	Claim 5
3	$\quad\vert\ c \in A$	definition of A
4	$\quad\vert\ c \in W[c]$	31
5	$\quad\vert\ \perp$	definition of initial segment
6	$A = W$	

Since $A = W$, Claim 5 yields

$$f : (W, <) \cong (\alpha, <).$$

The ordinal α is unique since isomorphic ordinals are equal.

$\square$

10.3 Transfinite Induction

Theorem 10.3.1 (Second Version of Transfinite Induction)**.** *Let $P(\alpha)$ be a property. Assume*

1. $P(0)$.
2. $\forall\alpha \in Ord \quad P(\alpha) \to P(s(\alpha))$.
3. $\forall\alpha \in Ord \quad \big(Limit(\alpha) \land \alpha \neq 0 \land \forall\beta < \alpha\ P(\beta)\big) \to P(\alpha)$.

Then

$$\forall\alpha \in Ord \quad P(\alpha).$$

Proof. We prove the hypothesis of Transfinite Induction.

1	$\alpha \in Ord$	
2	$\forall\beta < \alpha \quad P(\beta)$	
3	$\alpha = 0$	
4	$P(\alpha)$	assumption (a)
5	$\exists\beta < \alpha \quad \alpha = s(\beta)$	
6	$P(\beta)$	2
7	$P(\alpha)$	assumption (b)
8	$Limit(\alpha) \land \alpha \neq 0$	
9	$P(\alpha)$	assumption (c),2
10	$P(\alpha)$	trichotomy of ordinals
11	$\forall\alpha \in Ord\Big(\forall\beta < \alpha\ P(\beta) \to P(\alpha)\Big)$	
12	$\forall\alpha \in Ord \quad P(\alpha)$	Transfinite Induction

□

Theorem 10.3.2 (Transfinite Induction Principle)**.** *Let $P(\alpha)$ be a property.*

Assume

$$\forall\alpha \in Ord \quad \Big(\forall\beta < \alpha \quad P(\beta) \to P(\alpha)\Big).$$

Then

$$\forall\alpha \in Ord \quad P(\alpha).$$

Proof. Define

$$S := \{\beta \le \gamma \mid \neg P(\beta)\}.$$

1	$\quad \exists \gamma \in Ord \quad \neg P(\gamma)$	assumption
2	$\quad \gamma \in S$	definition of S
3	$\quad S \neq \varnothing$	2
4	$\quad \exists \alpha \in S \quad \forall \delta \in S\ (\alpha \le \delta)$	well ordering of Ord
5	$\quad \alpha \in S$	4
6	$\quad \neg P(\alpha)$	definition of S
7	$\quad \forall \beta < \alpha \quad P(\beta)$	minimality of α
8	$\quad P(\alpha)$	hypothesis
9	$\quad \perp$	6,8
10	$\forall \alpha \in Ord \quad P(\alpha)$	contradiction

□

10.4 Transfinite Recurtion

Theorem 10.4.1 (Transfinite Recursion Theorem)**.** *Let G be an operation. Then there exists a unique operation F such that*

$$F(\alpha) = G(F \restriction \alpha) \quad \textit{for all ordinals } \alpha.$$

Proof. We say t is a **computation of length** α **based on** G if:

$$dom(t) = \alpha + 1 \quad \text{and} \quad \forall \beta \le \alpha\ t(\beta) = G(t \restriction \beta).$$

Define the property:

$$P(x, y)\ :\ \begin{cases} x \in Ord\ \wedge\ y = t(x) \text{ for some computation } t \text{ of length } x, \\ \text{or } x \notin Ord\ \wedge\ y = \varnothing. \end{cases}$$

We show P defines a function.

Claim 1: For every ordinal α there exists a unique computation of length α.

1	$\forall \beta < \alpha\ \exists! t_\beta$ computation of length β	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	Replacement
3	T is a compatible system of functions	uniqueness
4	$t := \bigcup T$	definition
5	$dom(t) = \alpha$	union of domains
6	t is a function	compatibility of , **??**
7	$\forall \beta < \alpha\ t(\beta) = G(t \upharpoonright \beta)$	compatibility
8	$t(\alpha) = G(t \upharpoonright \alpha)$	definition extension step
9	t computation of length α	definition

Claim 2: Uniqueness of computation of length α.

1	t, s computations of length α	
2	$\forall \beta < \alpha\ t(\beta) = s(\beta)$	induction on β
3	$t = s$	extensionality

Hence P defines a function F.

Define
$$F(\alpha) := t(\alpha)$$
where t is the unique computation of length α.

Claim 3: Recursion equation.

1	$\alpha \in Ord$	
2	t computation of length α	
3	$F(\alpha) = t(\alpha)$	definition
4	$t(\alpha) = G(t \upharpoonright \alpha)$	computation rule
5	$F(\alpha) = G(F \upharpoonright \alpha)$	definition of , **??**

Claim 4: Uniqueness of F.

1	F, S satisfy recursion equation	
2	$\forall \beta < \alpha\ F(\beta) = S(\beta)$	induction
3	$F = S$	extensionality

□

Theorem 10.4.2 (Parametric Transfinite Recursion)**.** *Let G be an operation. Then there exists a unique operation F such that*
$$F(z, \alpha) = G\big(z, F(z) \upharpoonright \alpha\big) \quad \textit{for all sets } z \textit{ and ordinals } \alpha.$$

Proof. Fix a set z.

We say t is a **computation of length α based on G and z** if:

$$dom(t) = \alpha + 1 \quad \wedge \quad \forall\beta \leq \alpha\ t(\beta) = G\big(z, t \restriction \beta\big).$$

Define the property

$$Q(z,x,y) : \begin{cases} x \in Ord \ \wedge \ y = t(x) \text{ for some computation } t \text{ of length } x \text{ based on } G, z, \\ \text{or } x \notin Ord \ \wedge \ y = \varnothing. \end{cases}$$

Claim 1: For every ordinal α there exists a unique computation of length α based on G and z.

1	$\forall\beta < \alpha\ \exists! t_\beta$	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	
3	T is compatible	uniqueness
4	$t := \bigcup T$	
5	$dom(t) = \alpha$	
6	t is a function	compatibility
7	$\forall\beta < \alpha\ t(\beta) = G(z, t \restriction \beta)$	
8	$t(\alpha) = G(z, t \restriction \alpha)$	
9	t computation of length α	

Claim 2: Uniqueness.

1	t, s computations of length α
2	$\forall\beta < \alpha\ t(\beta) = s(\beta)$
3	$t = s$

Hence Q defines a function F.

Define

$$F(z,\alpha) := t(\alpha)$$

where t is the unique computation of length α based on G and z.

Claim 3: Recursion equation.

1	$\alpha \in Ord$
2	t computation of length α
3	$F(z,\alpha) = t(\alpha)$
4	$t(\alpha) = G(z, t \restriction \alpha)$
5	$F(z,\alpha) = G(z, F(z) \restriction \alpha)$

Claim 4: Uniqueness of F.

1 F, S satisfy recursion

2 $\forall \beta < \alpha \; F(z, \beta) = S(z, \beta)$

3 $F = S$

□

10.5 Arithmetic of Ordinals

Theorem 10.5.1 (Conway)**.** *For ordinals α, β,*

$$\alpha \oplus \beta = \sup{}' \Big(\{\alpha \oplus \beta' : \beta' < \beta\} \;\cup\; \{\alpha' \oplus \beta : \alpha' < \alpha\} \Big).$$

Proof. Let

$$S_{\alpha,\beta} := \{\alpha \oplus \beta' : \beta' < \beta\} \;\cup\; \{\alpha' \oplus \beta : \alpha' < \alpha\}.$$

Claim 1: $S_{\alpha,\beta}$ is a set of ordinals.

1 $\beta' < \beta \Rightarrow \alpha \oplus \beta' \in Ord$

2 $\alpha' < \alpha \Rightarrow \alpha' \oplus \beta \in Ord$

3 $S_{\alpha,\beta} \subseteq Ord$

Claim 2: $S_{\alpha,\beta}$ is bounded above in Ord.

1 $\forall \gamma \in S_{\alpha,\beta} \; \exists \delta \in Ord \; (\gamma < \delta)$

2 $\exists \sup S_{\alpha,\beta}$

Claim 3: $\sup' S_{\alpha,\beta}$ exists and is an ordinal.

1 $\sup S_{\alpha,\beta} \in Ord$

2 $\sup' S_{\alpha,\beta} \in Ord$

Claim 4: recursion characterization.

1 $\forall \beta' < \beta \quad \alpha \oplus \beta' < \alpha \oplus \beta$

2 $\forall \alpha' < \alpha \quad \alpha' \oplus \beta < \alpha \oplus \beta$

3 $S_{\alpha,\beta} \subseteq \alpha \oplus \beta$

4 $\alpha \oplus \beta$ is upper bound of $S_{\alpha,\beta}$

5 $\sup' S_{\alpha,\beta} \leq \alpha \oplus \beta$

6 $\forall \gamma < \alpha \oplus \beta \; \exists s \in S_{\alpha,\beta} \; (\gamma \leq s)$

7 $\alpha \oplus \beta \leq \sup' S_{\alpha,\beta}$

Hence

$$\alpha \oplus \beta = \sup' S_{\alpha,\beta}.$$

$\square$

Theorem 10.5.2. ***Summation*** *there is a unique function "$+ : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad 0 + \alpha = \alpha = \alpha + 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha + s(\beta) = s(\alpha + \beta) = s(\alpha) + \beta$
3. $\forall \alpha, \beta \in Ord \quad limit(\alpha) \wedge limit(\beta) \to \alpha + \beta = (\bigcup_{\beta' \in \beta} \alpha + \beta') \cup (\bigcup_{\beta' \in \beta} \beta' + \alpha)$

Proof Is Left As An Exercise To The Reader.😃

Theorem 10.5.3. ***Multiplication*** *there is a unique function "$\times : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad \alpha \times 0 = 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha \times s(\beta) = \alpha \times \beta + \alpha$
3. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \alpha < \beta \to \alpha \times \beta = \beta \times \alpha$
4. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \beta \leq \alpha \to \alpha \times \beta = \bigcup_{\beta' \in trunk(\beta)} (\alpha \times \beta') + \alpha \times branch(\beta)$

Proof Is Left As An Exercise To The Reader.😃

we denote also "Multiplication" with "$\alpha \cdot \beta$" and "$\alpha\beta$" too.

10.6 Transfinite Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitary functions and constants and doesn't have meaning we defined before). with "$Limit(x) : \forall y \in N \quad s(y) \neq x$ " and these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x \in N \quad x \cdot 0 = 0$
5. $\forall x, y \in N \quad x + s(y) = s(x + y)$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\forall x, y \in N \quad Limit(y) \to x + y = y + x$
8. $\forall x, y \in N \quad Limit(y) \to xy = yx$
9. $\forall x, y, z \in N \quad Limit(z) \to (x + y) + z = y + (x + z)$
10. $\forall x, y, z \in N \quad Limit(z) \to (xy)z = y(xz)$
11. $\forall x, y, z \in N \quad Limit(z) \to x(y + z) = xy + xz$
12. $\forall x, y, z \in N \quad Limit(z) \to x + z = y + z \to x = y$
13. $\forall x, y \in N \quad Limit(y) \to xy = 0 \to x = 0 \vee y = 0$

14. $\Big(\forall x \in N \quad Limit(x) \to \phi(x)\Big) \wedge \Big(\forall y \in N \quad \big(\phi(y) \to \phi(s(y))\big)\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitary "ϕ")

we call "$\mathfrak{N}$" a model of "Hyper Peano arithmetic".

Lemma 10.6.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in N \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative +)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 10.6.2. *prove for "$(N, +, \cdot)$"*

1. $\forall x, y, z \in N \quad (mn)z = x(nk)$ *(assosiative ×)*
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x, y \in N \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free ×)*
4. $\forall x, y \in N \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in N \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall \alpha, \beta \in Ord \quad \alpha < \beta \leftrightarrow \exists k \neq 0 \quad \alpha + k = \beta$$

Lemma 10.6.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.😃

as you can see "$(Ord, +, \times, s, 0)$" is a model of hyper peano arithmetic. so properties are true for them.

Chapter 11

Cardinals

Chapter 12

Axiom of chioce

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Chapter 13

hyperreal numbers($*\mathbb{N}, *\mathbb{Z}, *\mathbb{Q}, *\mathbb{R}$)

13.1 Filters

Filters are mathematical objects used to formalize the idea of "large subsets" of a set.
definition 13.1.1 (Filter)**.** *Let X be a nonempty set.*

A filter *$\mathcal{F}$ on X is a collection of subsets of X such that:*

1. *$\varnothing \notin \mathcal{F}$.*
2. *$X \in \mathcal{F}$.*
3. *If $A, B \in \mathcal{F}$, then*
$$A \cap B \in \mathcal{F}.$$
4. *If $A \in \mathcal{F}$ and $A \subseteq B \subseteq X$, then*
$$B \in \mathcal{F}.$$

Intuitively:

- Sets in $\mathcal{F}$ are considered "large".
- The intersection of two large sets is still large.
- Any superset of a large set is also large.

Example 13.1.1 (Trivial Filter)**.** *For any set X,*

$$\mathcal{F} = \{X\}$$

is a filter.
Example 13.1.2 (Principal Filter)**.** *Fix $A \subseteq X$ with $A \neq \varnothing$.*

Define
$$\mathcal{F}_A = \{B \subseteq X : A \subseteq B\}.$$

Then $\mathcal{F}_A$ is a filter on X.

This is called the principal filter generated by A.

Example 13.1.3 (Fréchet Filter)**.** *Let* $X = \mathbb{N}$*.*

Define

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} \setminus A \text{ is finite}\}.$$

Then $\mathcal{F}$ *is a filter.*

Its elements are the cofinite subsets of $\mathbb{N}$*.*

This is called the Fréchet filter*.*

13.2 Ultrafilters

Ultrafilters are maximal filters.

definition 13.2.1 (Ultrafilter)**.** *A filter* $\mathcal{U}$ *on* X *is called an* ultrafilter *if it is maximal among filters, that is:*

Whenever $\mathcal{F}$ *is a filter on* X *and*

$$\mathcal{U} \subseteq \mathcal{F},$$

then

$$\mathcal{F} = \mathcal{U}.$$

There is a very important characterization of ultrafilters.

Theorem 13.2.1 (Ultrafilter Characterization)**.** *Let* $\mathcal{U}$ *be a filter on* X*.*

Then $\mathcal{U}$ *is an ultrafilter if and only if for every subset* $A \subseteq X$*, exactly one of the following holds:*

$$A \in \mathcal{U},$$

or

$$X \setminus A \in \mathcal{U}.$$

So an ultrafilter makes a complete decision about every subset.

Example 13.2.1. *Fix* $x \in X$*.*

Define

$$\mathcal{U}_x = \{A \subseteq X : x \in A\}.$$

Then $\mathcal{U}_x$ *is an ultrafilter.*

It is called the principal ultrafilter at x*.*

Proof:

- If $A, B \in \mathcal{U}_x$, then $x \in A \cap B$, so $A \cap B \in \mathcal{U}_x$.
- If $A \in \mathcal{U}_x$ and $A \subseteq B$, then $x \in B$, so $B \in \mathcal{U}_x$.
- For every subset $A \subseteq X$, either $x \in A$ or $x \notin A$.

  Hence either

  $$A \in \mathcal{U}_x$$

  or

  $$X \setminus A \in \mathcal{U}_x.$$

Therefore $\mathcal{U}_x$ is an ultrafilter.

13.3 Nonprincipal Ultrafilters

An ultrafilter on $\mathbb{N}$ is called *nonprincipal* if it contains no finite set.

Equivalently, it extends the Fréchet filter.

Nonprincipal ultrafilters cannot be explicitly constructed in ordinary mathematics; their existence usually requires the Axiom of Choice.

definition 13.3.1 (Extension of a Filter). *Let $\mathcal{F}$ and $\mathcal{G}$ be filters on a set X.*

We say that $\mathcal{G}$ extends $\mathcal{F}$ if

$$\mathcal{F} \subseteq \mathcal{G}.$$

This means every set considered "large" by $\mathcal{F}$ is also considered "large" by $\mathcal{G}$.

Theorem 13.3.1 (Ultrafilter Lemma). *Every filter on a set X can be extended to an ultrafilter.*

In other words, if $\mathcal{F}$ is a filter on X, then there exists an ultrafilter $\mathcal{U}$ such that

$$\mathcal{F} \subseteq \mathcal{U}.$$

So every filter can be enlarged until it becomes maximal.

Example 13.3.1. *Let*

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} \setminus A \text{ is finite}\}.$$

This is the Fréchet filter.

It is not an ultrafilter, because for example neither the set of even numbers

$$E = \{0, 2, 4, 6, \dots\}$$

nor its complement

$$\mathbb{N} \setminus E$$

belongs to $\mathcal{F}$.

By the Ultrafilter Lemma, there exists an ultrafilter $\mathcal{U}$ such that

$$\mathcal{F} \subseteq \mathcal{U}.$$

Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$E \in \mathcal{U}$$

or

$$\mathbb{N} \setminus E \in \mathcal{U}$$

must hold.

The theorem is usually proved using Zorn's Lemma.

Consider the set

$$\mathcal{P} = \{\mathcal{G} : \mathcal{G} \text{ is a filter on } X \text{ and } \mathcal{F} \subseteq \mathcal{G}\}.$$

We order $\mathcal{P}$ by inclusion.

The idea is:

Theorem 13.3.2 (Ultrafilter Lemma). *Let $\mathcal{F}$ be a filter on a set X.*

Then there exists an ultrafilter $\mathcal{U}$ on X such that

$$\mathcal{F} \subseteq \mathcal{U}.$$

Proof. Let

$$\mathcal{P} = \{\mathcal{G} : \mathcal{G} \text{ is a filter on } X \text{ and } \mathcal{F} \subseteq \mathcal{G}\}.$$

We partially order $\mathcal{P}$ by inclusion.

We will apply Zorn's Lemma.

So let

$$\mathcal{C} \subseteq \mathcal{P}$$

be a chain.

Define

$$\mathcal{H} = \bigcup_{\mathcal{G} \in \mathcal{C}} \mathcal{G}.$$

We show that $\mathcal{H}$ is a filter.

First,

$$\varnothing \notin \mathcal{H},$$

because no filter in $\mathcal{C}$ contains $\varnothing$.

Also,

$$X \in \mathcal{H},$$

since every filter contains X.

Now let

$$A, B \in \mathcal{H}.$$

Then there exist

$$\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}$$

such that

$$A \in \mathcal{G}_1, \qquad B \in \mathcal{G}_2.$$

Because $\mathcal{C}$ is a chain, either

$$\mathcal{G}_1 \subseteq \mathcal{G}_2$$

or
$$\mathcal{G}_2 \subseteq \mathcal{G}_1.$$
Without loss of generality, suppose
$$\mathcal{G}_1 \subseteq \mathcal{G}_2.$$
Then
$$A, B \in \mathcal{G}_2.$$
Since $\mathcal{G}_2$ is a filter,
$$A \cap B \in \mathcal{G}_2 \subseteq \mathcal{H}.$$
Thus $\mathcal{H}$ is closed under finite intersections.

Next, suppose
$$A \in \mathcal{H}$$
and
$$A \subseteq B \subseteq X.$$
Then
$$A \in \mathcal{G}$$
for some
$$\mathcal{G} \in \mathcal{C}.$$
Since $\mathcal{G}$ is a filter,
$$B \in \mathcal{G} \subseteq \mathcal{H}.$$
Therefore $\mathcal{H}$ is a filter.

Also, every element of $\mathcal{C}$ contains $\mathcal{F}$, so
$$\mathcal{F} \subseteq \mathcal{H}.$$
Hence
$$\mathcal{H} \in \mathcal{P}.$$
Thus every chain in $\mathcal{P}$ has an upper bound.

By Zorn's Lemma, $\mathcal{P}$ has a maximal element, say $\mathcal{U}$.

We claim that $\mathcal{U}$ is an ultrafilter.

Suppose $\mathcal{U} \subseteq \mathcal{V}$, where $\mathcal{V}$ is a filter on X.

Since
$$\mathcal{F} \subseteq \mathcal{U} \subseteq \mathcal{V},$$
we have
$$\mathcal{V} \in \mathcal{P}.$$
But $\mathcal{U}$ is maximal in $\mathcal{P}$, so
$$\mathcal{V} = \mathcal{U}.$$
Therefore $\mathcal{U}$ is an ultrafilter extending $\mathcal{F}$. □

Theorem 13.3.3. *Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$.*

Then

$$\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}.$$

That is, every cofinite subset of $\mathbb{N}$ belongs to $\mathcal{U}$.

Proof. Let

$$A \in \mathcal{F}_{\mathrm{Fr}}.$$

Then

$$\mathbb{N} \setminus A$$

is finite.

Suppose, toward a contradiction, that

$$A \notin \mathcal{U}.$$

Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$A \in \mathcal{U}$$

or

$$\mathbb{N} \setminus A \in \mathcal{U}$$

holds.

Therefore

$$\mathbb{N} \setminus A \in \mathcal{U}.$$

But

$$\mathbb{N} \setminus A$$

is finite.

Hence $\mathcal{U}$ contains a finite set.

We show this implies that $\mathcal{U}$ is principal.

Let

$$F = \{n_1, \ldots, n_k\} \in \mathcal{U}.$$

Since

$$F = n_1 \cup \cdots \cup n_k,$$

if no singleton $\{n_i\}$ belonged to $\mathcal{U}$, then for every i we would have

$$\mathbb{N} \setminus \{n_i\} \in \mathcal{U},$$

because $\mathcal{U}$ is an ultrafilter.

Since filters are closed under finite intersections,

$$\bigcap_{i=1}^{k} (\mathbb{N} \setminus \{n_i\}) \in \mathcal{U}.$$

But

$$\bigcap_{i=1}^{k} (\mathbb{N} \setminus \{n_i\}) = \mathbb{N} \setminus F.$$

Thus

$$\mathbb{N} \setminus F \in \mathcal{U}.$$

Since also

$$F \in \mathcal{U},$$

we obtain

$$F \cap (\mathbb{N} \setminus F) = \varnothing \in \mathcal{U},$$

contradicting the definition of a filter.

Therefore

$$\{n_i\} \in \mathcal{U}$$

for some i.

Hence $\mathcal{U}$ is principal, contradiction.

Therefore

$$A \in \mathcal{U}.$$

Since A was arbitrary,

$$\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}.$$

□

13.4 Definition of $(*\mathbb{N}, *\mathbb{Z}, *\mathbb{Q}, *\mathbb{R})$

definition 13.4.1 (Ultrapower Construction)**.** *Let X be a set and let $\mathcal{U}$ be a nonprincipal ultrafilter on ω.*

Consider the set

$$X^{\omega}$$

of all functions

$$a : \omega \to X.$$

We usually write such a function as

$$a = (a_i)_{i \in \omega},$$

where

$$a_i \in X$$

for every

$$i \in \omega.$$

definition 13.4.2. *Define a relation $\sim$ on X^ω by*

$$a \sim b$$

if

$$\{i \in \omega : a_i = b_i\} \in \mathcal{U}.$$

In words, two sequences are equivalent if they agree on a set of indices belonging to the ultrafilter.
Theorem 13.4.1. *The relation $\sim$ is an equivalence relation on X^ω.*

Proof. We verify reflexivity, symmetry, and transitivity.

Reflexivity. Let

$$a \in X^\omega.$$

Then

$$\{i \in \omega : a_i = a_i\} = \omega.$$

Since every ultrafilter contains ω,

$$a \sim a.$$

Symmetry. Suppose

$$a \sim b.$$

Then

$$\{i \in \omega : a_i = b_i\} \in \mathcal{U}.$$

But

$$\{i \in \omega : a_i = b_i\} = \{i \in \omega : b_i = a_i\},$$

hence

$$b \sim a.$$

Transitivity. Suppose

$$a \sim b$$

and

$$b \sim c.$$

Then

$$A = \{i \in \omega : a_i = b_i\} \in \mathcal{U},$$

and

$$B = \{i \in \omega : b_i = c_i\} \in \mathcal{U}.$$

Since $\mathcal{U}$ is a filter,

$$A \cap B \in \mathcal{U}.$$

If
$$i \in A \cap B,$$
then
$$a_i = b_i \quad \text{and} \quad b_i = c_i,$$
so
$$a_i = c_i.$$

Thus
$$A \cap B \subseteq \{i \in \omega : a_i = c_i\}.$$

Since ultrafilters are closed upward,
$$\{i \in \omega : a_i = c_i\} \in \mathcal{U}.$$

Therefore
$$a \sim c.$$

□

definition 13.4.3. *The quotient*
$$X^\omega / \sim$$
is called the ultrapower *of X modulo $\mathcal{U}$.*

The equivalence class of
$$a \in X^\omega$$
is denoted by
$$[a].$$

We now apply this construction to the standard number systems.

definition 13.4.4. *The set of hypernatural numbers is*
$$^*\mathbb{N} = \mathbb{N}^\omega / \sim .$$

definition 13.4.5. *The set of hyperintegers is*
$$^*\mathbb{Z} = \mathbb{Z}^\omega / \sim .$$

definition 13.4.6. *The set of hyperrational numbers is*
$$^*\mathbb{Q} = \mathbb{Q}^\omega / \sim .$$

definition 13.4.7. *The set of hyperreal numbers is*
$$^*\mathbb{R} = \mathbb{R}^\omega / \sim .$$

The algebraic operations are defined componentwise.

For example, if

$$[a], [b] \in {}^*\mathbb{R},$$

then

$$[a] + [b] = [(a_i + b_i)_{i\in\omega}],$$

and

$$[a] \cdot [b] = [(a_i b_i)_{i\in\omega}].$$

Similarly, the order relation is defined by

$$[a] < [b]$$

if

$$\{i \in \omega : a_i < b_i\} \in \mathcal{U}.$$

Theorem 13.4.2. *The standard structures*

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R}$$

embed into

$${}^*\mathbb{N}, \quad {}^*\mathbb{Z}, \quad {}^*\mathbb{Q}, \quad {}^*\mathbb{R},$$

respectively.

Proof. We prove the statement for $\mathbb{R}$.

Define

$$\iota : \mathbb{R} \to {}^*\mathbb{R}$$

by

$$\iota(r) = [(r, r, r, \dots)].$$

Suppose

$$\iota(r) = \iota(s).$$

Then

$$\{i \in \omega : r = s\} \in \mathcal{U}.$$

If

$$r \neq s,$$

then

$$\{i \in \omega : r = s\} = \varnothing,$$

which cannot belong to a filter.

Hence

$$r = s.$$

Therefore ι is injective. □

Example 13.4.1. *Define*

$$a_i = i$$

for every

$$i \in \omega.$$

Then

$$[a] \in {}^*\mathbb{N}$$

is larger than every standard natural number.

Indeed, if

$$n \in \mathbb{N},$$

then

$$\{i \in \omega : n < a_i\} = \{i \in \omega : n < i\}$$

is cofinite, hence belongs to $\mathcal{U}$.

Therefore

$$n < [a].$$

13.5 Hyperreals: Great and Small

Throughout this chapter we work inside the hyperreal field ${}^*\mathbb{R}$ constructed as an ultrapower of $\mathbb{R}$ with respect to a nonprincipal ultrafilter on ω.

Elements of ${}^*\mathbb{R}$ are written as

$$[a], \quad a = (a_i)_{i \in \omega}, \quad a_i \in \mathbb{R}.$$

13.6 (Un)limited, Infinitesimal, and Appreciable Numbers

definition 13.6.1 (Limited and Unlimited)**.** *Let $x \in {}^*\mathbb{R}$.*

- *x is* limited *if there exists $r \in \mathbb{R}$ such that*

$$|x| < r.$$

- *x is* unlimited *if it is not limited.*

definition 13.6.2 (Infinitesimal)**.** *An element $x \in {}^*\mathbb{R}$ is infinitesimal if for every $r > 0$ in $\mathbb{R}$,*

$$|x| < r.$$

definition 13.6.3 (Appreciable)**.** *An element $x \in {}^*\mathbb{R}$ is appreciable if it is limited and not infinitesimal.*

Example 13.6.1. *Let*

$$\varepsilon = \left[\left(\frac{1}{i+1} \right)_{i \in \omega} \right].$$

Then ε is infinitesimal.

13.7 Arithmetic of Hyperreals

Let $x = [a]$ and $y = [b]$ in $^*\mathbb{R}$.

Operations are defined componentwise:

$$x + y = [a_i + b_i]_{i\in\omega}, \qquad x \cdot y = [a_i b_i]_{i\in\omega}.$$

Lemma 13.7.1. *The sum of two infinitesimals is infinitesimal.*

Proof. Let x, y be infinitesimal and $r > 0$. Then choose $r/2 > 0$. We have

$$|x| < \frac{r}{2}, \quad |y| < \frac{r}{2}.$$

Hence

$$|x + y| \leq |x| + |y| < r.$$

So $x + y$ is infinitesimal. □

Lemma 13.7.2. *The product of a limited number and an infinitesimal is infinitesimal.*

Proof. Let x be infinitesimal and y limited. Then $|y| < M$ for some $M \in \mathbb{R}$.

For any $r > 0$, choose $\delta = \frac{r}{M+1}$. Since x is infinitesimal, $|x| < \delta$, hence

$$|xy| < r.$$

So xy is infinitesimal. □

13.8 On the Use of "Finite" and "Infinite"

In $^*\mathbb{R}$ the terminology is interpreted as follows:

- **Finite** means limited (bounded by a real number).
- **Infinite** means unlimited (larger in magnitude than any real).

This distinction is relative to $\mathbb{R}$ inside $^*\mathbb{R}$: a number may be "finite" in this sense while still being nonstandard.

13.9 Halos, Galaxies, and Real Comparisons

Let $x, y \in {}^*\mathbb{R}$.

definition 13.9.1 (Infinitesimal Closeness)**.** *We say that x is* infinitely close *to y, and write*

$$x \approx y,$$

if $x - y$ is infinitesimal.

definition 13.9.2 (Halo)**.** *The* halo *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{hal}(x) = \{y \in {}^*\mathbb{R} : y \approx x\}.$$

definition 13.9.3 (Galaxy). *The* galaxy *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{gal}(x) = \{y \in {}^*\mathbb{R} : x - y \textit{ is limited}\}.$$

Lemma 13.9.1. *Every halo is contained in a galaxy.*

Proof. If $y \in \mathrm{hal}(x)$, then $x - y$ is infinitesimal, hence limited. Therefore $y \in \mathrm{gal}(x)$. □

Lemma 13.9.2. *For standard $r, s \in \mathbb{R}$,*

$$r \approx s \quad \iff \quad r = s.$$

Proof. If $r \approx s$, then $r - s$ is a real infinitesimal, hence 0. So $r = s$. □

13.10 Exercises on Halos and Galaxies

1. Show that $x \approx y$ defines an equivalence relation.
2. Prove that halos partition ${}^*\mathbb{R}$.
3. Show that each galaxy is a union of halos.

13.11 Shadows

definition 13.11.1 (Standard Part). *If $x \in {}^*\mathbb{R}$ is limited, then there exists a unique real number $r \in \mathbb{R}$ such that*

$$x \approx r.$$

This r is called the standard part *of x, denoted*

$$\mathrm{st}(x) = r.$$

Lemma 13.11.1. *The standard part map is well-defined on limited hyperreals.*

Proof. If $x \approx r$ and $x \approx s$ with $r, s \in \mathbb{R}$, then $r - s$ is infinitesimal, hence $r = s$. □

13.12 Exercises on Infinite Closeness

1. Show that if $x \approx y$ and $y \approx z$, then $x \approx z$.
2. Prove that $x \approx y$ implies $x + z \approx y + z$.
3. If $x \approx y$ and z is limited, show $xz \approx yz$.

13.13 Shadows and Completeness

Lemma 13.13.1. *Every limited $x \in {}^*\mathbb{R}$ is infinitely close to a unique real number.*

Proof. This follows from the construction of ${}^*\mathbb{R}$ as an ultrapower and the fact that bounded sequences in $\mathbb{R}$ have a well-defined ultralimit modulo a nonprincipal ultrafilter. □

13.14 Exercise on Dedekind Completeness

1. Show that $\mathbb{R}$ is Dedekind complete.
2. Use the standard part map to explain why every limited hyperreal has a shadow in $\mathbb{R}$.
3. Compare this with completeness of $^*\mathbb{R}$ (is it Dedekind complete?).

Chapter 14

fantacy numbers($\mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord), \mathbb{C}(Ord)$)

14.1 fantacy Integers“$\mathbb{Z}(Ord)$”

we learned about “Ord” before and we constructed structure “$(Ord, +, \cdot, <)$”. if you assume “$Ord \times Ord$” we can define an equivalence relation “$(\alpha_1, \beta_1) \sim (\alpha_2, \beta_2) : \alpha_1 + \beta_2 = \alpha_2 + m_1$”(check it's an equivalence relation).
definition 14.1.1. ***fantacy Integers“$\mathbb{Z}(Ord)$”*** *is a set*

$$\mathbb{Z}(Ord) := Ord \times Ord / \sim$$

we denote it's members with “$z_0, z_1, ...$”

in this system “$0 := [(0,0)]$” and “$\forall \beta \in Ord \quad \beta := [(\beta, 0)]$” and “$\forall \beta \in Ord \quad -\beta := [(0, \beta)]$”. in general“$\forall \alpha, \beta \in Ord \quad \alpha - \beta := [(\alpha, \beta)]$”. so we can say “$Ord \subseteq \mathbb{Z}(Ord)$”.
Theorem 14.1.1. ***Summation*** *there is a unique function “$+_{\mathbb{Z}}(Ord) : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] +_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1 + \alpha_2, \beta_1 + \beta_2)]$$

Proof Is Left As An Exercise To The Reader.😃
Theorem 14.1.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Z}(Ord)} : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] \times_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1\alpha_2 + \beta_1\beta_2, \alpha_1\beta_2 + \alpha_2\beta_1)]$$

Proof Is Left As An Exercise To The Reader.😃
definition 14.1.2. ***Order in*** $\mathbb{Z}(Ord)$

$$[(\alpha_1, \beta_1)] <_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] : \alpha_1 + \beta_2 < \alpha_2 + \beta_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}(Ord)}$", "$+$" instead of "$+_{\mathbb{Z}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{Z}(Ord)}$".

Lemma 14.1.1. *prove.*

1. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha <_{\mathbb{Z}(Ord)} \beta \leftrightarrow \alpha < \beta$
2. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha +_{\mathbb{Z}(Ord)} \beta = \alpha + \beta$
3. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha \times_{\mathbb{Z}(Ord)} \beta = \alpha \times \beta$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 14.1.2. *prove for "$(\mathbb{Z}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 14.1.3. *prove for "$(\mathbb{Z}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Z}(Ord) \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.1.3. ***Order of fantacy Integers($\mathbb{Z}(Ord)$)*** *prove for "$(\mathbb{Z}(Ord), +, \times, <_{\mathbb{Z}(Ord)})$"*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \to x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \wedge 0 < z \to xz < y\gamma$

Proof Is Left As An Exercise To The Reader.😃

14.2 fantacy Rational numbers "$\mathbb{Q}(Ord)$"

we learned about "$\mathbb{Z}(Ord)$" before and we constructed structure "$(\mathbb{Z}(Ord), +, \cdot, <)$". if you assume "$\mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})$" we can define an equivalence relation "$(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).

definition 14.2.1. ***fantacy Rational numbers*"$\mathbb{Q}(Ord)$"** *is a set*

$$\mathbb{Q}(Ord) := \mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})/\sim$$

we denote it's members with "$q_0, q_1, ...$"

in this system "$0 := [(0,1)]$" and "$1 := [(1,1)]$" and "$\forall x \in \mathbb{Z}(Ord) \quad x := [(x,1)]$" and "$\forall x \in \mathbb{Z}(Ord) \quad \frac{1}{x} := [(1,x)]$". in general "$\forall x, y \in \mathbb{Z}(Ord) \quad \frac{x}{y} := [(x,y)]$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord)$".

Theorem 14.2.1. ***Summation*** *there is a unique function "$+_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] +_{\mathbb{Q}(Ord)} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.2.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] \times_{\mathbb{Q}(Ord)} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

definition 14.2.2. ***Order in*** $\mathbb{Q}(Ord)$.
if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Q}(Ord)}$", "$+$" instead of "$+_{\mathbb{Q}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{Q}(Ord)}$".

Lemma 14.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z <_{\mathbb{Q}(Ord)} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z +_{\mathbb{Q}(Ord)} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z \times_{\mathbb{Q}(Ord)} y = z \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 14.2.2. *prove for "$(\mathbb{Q}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 14.2.3. *prove for "$(\mathbb{Q}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{Q}(Ord) - \{0\} \quad \exists y \in \mathbb{Q}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.2.3. ***Order of fantacy Rational numbers(**$\mathbb{Q}(Ord)$**)*** *prove for "$(\mathbb{Q}(Ord), <_{\mathbb{Q}(Ord)})$"*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q}(Ord) \quad \exists z \in \mathbb{Q}(Ord) \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \wedge 0 < z \to xz < y\gamma$

Proof Is Left As An Exercise To The Reader.😃

14.3 fantacy Real numbers "$\mathbb{R}(Ord)$"

before we we get to lesson define function "$|\_| : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not fimiliar to this notation it means "$|\_| := \{(x, y) \in \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \mid (0 \leq x \to y = x) \wedge (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}(Ord)$" before and we constructed structure "$(\mathbb{Q}(Ord), +, \cdot, <)$". now assume $\mathbb{Q}(Ord)^{Ord}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}(Ord)$). we don't want it all we just want a subset.

definition 14.3.1. ***Cauchy sequence*** *is a sequence like "$(a_i)_{i \in Ord} \in \mathbb{Q}(Ord)^{Ord}$" s.t. elements of it get infinitly close to each other i.e.*

$$Cauchy(a_i)_{i \in Ord} : \forall \epsilon > 0 \quad \exists N \quad \forall n, m \quad N < n, m < roof(N) \to |a_n - a_m| < \epsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in Ord} \sim (b_i)_{i \in Ord} : \forall \epsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to |a_n - b_n| < \epsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in Ord}]$ with $[a_i]_{i \in Ord}$.

definition 14.3.2. ***fantacy Real numbers "$\mathbb{R}(Ord)$"*** *is a set*

$$\mathbb{R}(Ord) := Cauchy/\sim$$

we denote it's members with "$r_0, r_1, \ldots$"

in this system "$0 := [0]_{i\in Ord}$" and "$1 := [1]_{i\in Ord}$" and "$\forall q \in \mathbb{Q}(Ord) \quad q := [q]_{i\in Ord}$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord)$".
Theorem 14.3.1. ***Summation*** *there is a unique function "$+_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$" s.t.*

$$[a_i]_{i\in Ord} +_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i + b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.😃
Theorem 14.3.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$" s.t.*

$$[a_i]_{i\in Ord} \times_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.😃
definition 14.3.3. ***Order in*** $\mathbb{R}(Ord)$ *.*

$$[a_i]_{i\in Ord} <_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} : \exists \epsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to b_n - a_n > \epsilon$$

this lemma let us use "$<$" instead of "$<_{\mathbb{R}(Ord)}$", "$+$" instead of "$+_{\mathbb{R}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{R}(Ord)}$".
Lemma 14.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x <_{\mathbb{R}(Ord)} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x +_{\mathbb{R}(Ord)} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x \times_{\mathbb{R}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$x \cdot y$" and "xy" too.
Lemma 14.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in \mathbb{R}(Ord) \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃
Lemma 14.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall x \in \mathbb{R}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*

3. $\forall x \in \mathbb{R}(Ord) - \{0\} \quad \exists y \in \mathbb{R}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.3.3. ***Linear order of fantacy Real numbers(*$\mathbb{R}(Ord)$*)*** *prove "*$(\mathbb{R}(Ord), <_{\mathbb{R}(Ord)}$
$)$

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{R}(Ord) \quad \exists z \in \mathbb{R}(Ord) \quad x \neq y \rightarrow x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R}(Ord) \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R}(Ord) \quad Uppreb(X, y)) \rightarrow (\exists y \in \mathbb{R}(Ord) \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \rightarrow x + z < y + z$
6. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader.😃

14.4 fantacy Complex numbers"$\mathbb{C}(Ord)$"

we learned about "$\mathbb{R}(Ord)$" before and we constructed structure "$(\mathbb{R}(Ord), +, \cdot, <)$".

definition 14.4.1. ***fantacy Complex numbers"*$\mathbb{C}(Ord)$*"*** *is a set*

$$\mathbb{C}(Ord) := \mathbb{R}(Ord) \times \mathbb{R}(Ord)$$

we denote it's members with "$z_0, x, ...$"

in this system "$0 := (0, 0)$" and "$\forall n \in \mathbb{R}(Ord) \quad r := (r, 0)$" . in general "$\forall x, y \in \mathbb{R}(Ord) \quad x + iy := (x, y)$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord) \subseteq \mathbb{C}(Ord)$".

Theorem 14.4.1. ***Summation*** *there is a unique function "$+_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \rightarrow \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) +_{\mathbb{C}(Ord)} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.4.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \rightarrow \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) \times_{\mathbb{C}(Ord)} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.😃

this lemma let us use "$+$" instead of "$+_{\mathbb{C}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{C}(Ord)}$".

Lemma 14.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R}(Ord) \quad x +_{\mathbb{C}(Ord)} y = x + y$
2. $/ \therefore \forall x, y \in \mathbb{R}(Ord) \quad x \times_{\mathbb{C}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 14.4.2. *prove for "$(\mathbb{C}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall z \in \mathbb{C}(Ord) \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{C}(Ord) \quad \exists y \in \mathbb{C}(Ord) \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{C}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 14.4.3. *prove for "$(\mathbb{C}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall z \in \mathbb{C}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{C}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
4. $\forall x, y, z \in \mathbb{C}(Ord) \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.4.3. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall n \in Ord \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.😃

Lemma 14.4.4. *prove for "$(\mathbb{C}(Ord), +, \cdot, (\_)^{(\_)})$"*

1. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.4.4. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall m, n \in Ord \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.😃

Lemma 14.4.5. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.😃

Theorem 14.4.5. ***Big Sum*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\sum_{i=0}^{(\_)}(\_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😃

Lemma 14.4.6. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in Ord \quad \forall a \quad k\sum_{i=0}^{n} a_n = \sum_{i=0}^{n} ka_n$

Proof Is Left As An Exercise To The Reader. 😃

Theorem 14.4.6. ***Big Product*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\prod_{i=0}^{(\_)}(\_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😃

Lemma 14.4.7. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n}(a_n b_n)$
2. $\forall n \in Ord \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n}(a_n)^k$

Proof Is Left As An Exercise To The Reader. 😃

Chapter 15

New beginning for calculus

15.1 Derivatives

first we have to fix a "Main variable" it's almost all the time "x". (somtimes it's "t").
definition 15.1.1. *assume $f : \mathbb{R}(Ord) \to \mathbb{R}(Ord)$ we define **Difrential** $(d_\alpha(f))$*

$$d_\alpha(f) := f(x+\alpha) - f(x)$$

Main reason we choose "x" in defintion becuase it's "Main variable". you can prove $\forall x \in \mathbb{R}(Ord) \quad d_\alpha(x) = \alpha(x)$.
definition 15.1.2. *we define **Derivative***

$$\frac{d}{dx} := \{\big(f, sh(\frac{d_\alpha(f)}{d_\alpha(x)})\big) \mid f : \mathbb{R}(Ord) \to \mathbb{R}(Ord)\}$$

there is a biggest set like A s.t. $\frac{d}{dx}A$ is a function (you can prove it as an exercise). we call that set "Derivatible funsions" and that function "Derivative".

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Chapter 16

Lebesgue Intergral

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Chapter 17

New beginning for Probability theory